

5G RAN

Uplink Scheduling Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
Added descriptions of the mutually exclusive relationship between uplink DCI extension-based scheduling and combining noise suppression. For details, see [6.14.3.2.2 Mutually Exclusive Functions](#).
Added descriptions of cell requirements for the uplink non-consecutive scheduling switch. For details, see [6.5.3.4 Cells](#).
Revised descriptions of the impact relationship between uplink non-consecutive scheduling and flexible experience differentiation. For details, see [6.5.3.2.3 Function Impacts](#).
Added descriptions of the impact relationships between FSD transmission mode and the following functions: uplink basic preallocation and uplink smart preallocation. For details, see [5.10.3.2.3 Function Impacts](#).

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 05 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added the mutually exclusive relationship between uplink scheduling on a per RB basis and uplink precise MCS optimization. For details, see 6.5.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationships of uplink target IBLER adaptation with uplink precise MCS optimization and probability-based uplink MU-MIMO MCS selection. For details, see 7.1.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationships of uplink PAMC with uplink precise MCS optimization and probability-based uplink MU-MIMO MCS selection. For details, see 7.8.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the mutually exclusive relationship between time-domain power saving BWP and uplink DCI extension-based scheduling. For details, see 6.14.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added the mutually exclusive relationship between DMRS configuration type and DCI-based dynamic waveform switching. For details, see 8.3.2.2 Mutually Exclusive Functions .	None	High - frequency TDD	5900 series base stations (macro base stations)
Added the impact relationship between uplink PAMC and uplink 4-layer transmission for a single FWA user. For details, see 7.8.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added uplink scheduling gain optimization. For details, see: • 6.5.1 Principles • 6.5.2.1 Benefits • 6.5.4.1.1 Data Preparation • 6.5.4.1.2 Using MML Commands	Modified parameter: Added the UL_SCH_GAIN_OPT_SW option to the NRDUCellPusch.PuschAlgoExtSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial

features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-010011	Scheduling	4 UE Scheduling Type Determining
FBFD-010011	Scheduling	5 UE Scheduling
FBFD-010011	Scheduling	6 PUSCH Time-Frequency Resource Allocation
FBFD-010011	Scheduling	7 MCS Selection
FBFD-010011	Scheduling	8 DMRS Resource Allocation

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
UE scheduling type determining	None	4 UE Scheduling Type Determining
UE scheduling	None	5 UE Scheduling

Function Name	Difference	Chapter/Section
PUSCH time-frequency resource allocation	None	6 PUSCH Time-Frequency Resource Allocation
MCS selection	None	7 MCS Selection
DMRS resource allocation	None	8 DMRS Resource Allocation

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
UE scheduling type determining	None	4 UE Scheduling Type Determining
UE scheduling	None	5 UE Scheduling
PUSCH time-frequency resource allocation	None	6 PUSCH Time-Frequency Resource Allocation
MCS selection	None	7 MCS Selection
DMRS resource allocation	None	8 DMRS Resource Allocation

3 Overview

3.1 Basic Concepts and Definitions

The New Radio (NR) system uses shared channels for transmission, with time-frequency resources dynamically allocated among UEs. The Scheduling feature allows the gNodeB to achieve time-frequency resource allocation in the uplink and downlink and maximizes system capacity while guaranteeing user fairness.

Scheduler

Schedulers are located at the Media Access Control (MAC) layer of the NR system. The basic function of a scheduler is to allocate appropriate system resources in each slot to UEs for transmission and reception. Schedulers are classified by direction into downlink schedulers and uplink schedulers. This document describes uplink schedulers. For details about downlink schedulers, see *Downlink Scheduling*.

An uplink scheduler allocates physical uplink shared channel (PUSCH) resources to UEs and selects an appropriate MCS index for transmitting UE data. It performs the following functions:

- Allocates PUSCH time-frequency resources to UEs.
- Allocates DMRS resources to UEs for the gNodeB to demodulate the PUSCH.
- Selects an appropriate MCS index for UEs to transmit UE data.

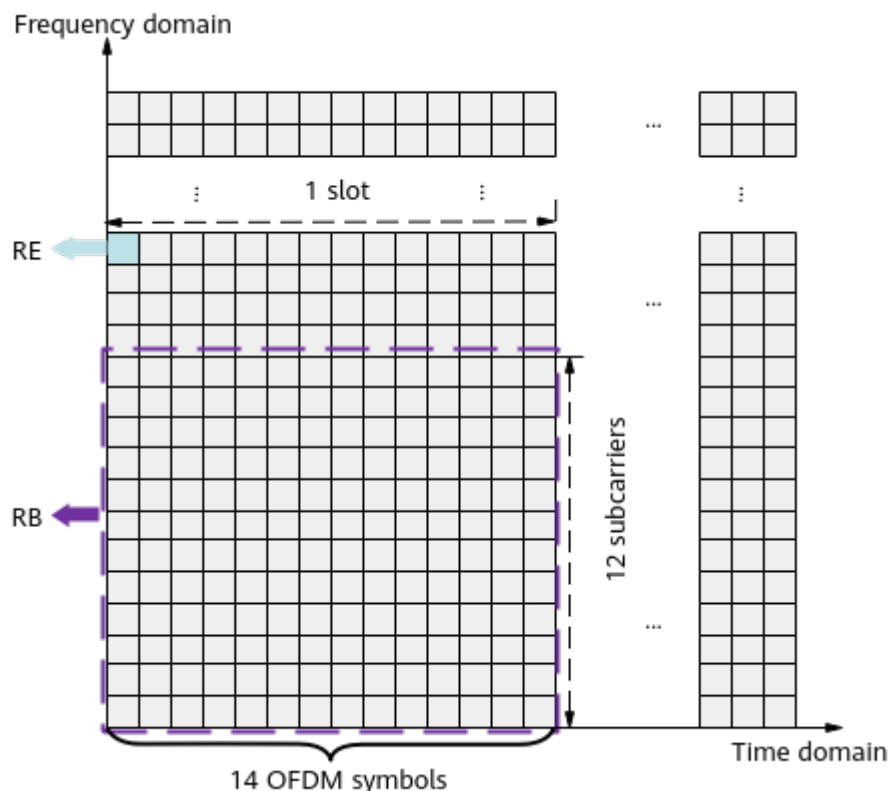
RE and RB

A resource element (RE) is the smallest time-frequency resource unit in the NR system, comprising one subcarrier in the frequency domain and one OFDM symbol in the time domain.

An RB is the smallest unit of resources that can be allocated for data transmission at the physical layer in the NR system, composed of 12 subcarriers in the frequency domain and any number of symbols in the time domain.

Figure 3-1 illustrates an RE and an RB (in one slot in the time domain as an example).

Figure 3-1 RE and RB



A UE can be allocated multiple RBs in a slot. The maximum number of RBs that can be allocated correlates with the system bandwidth and subcarrier spacing. For example: Up to 273 RBs can be allocated in NR TDD when the bandwidth is 100 MHz and the subcarrier spacing is 30 kHz.

MCS

A modulation and coding scheme (MCS) is a combination of a modulation scheme and code rate. It defines the number of valid bits that can be carried by a resource element (RE). Different modulation schemes use different channel code rates. If the channel quality is good, a gNodeB selects an MCS with a more efficient modulation scheme and a higher code rate (with fewer redundant bits inserted). If the channel quality is poor, the gNodeB selects one with a less efficient modulation scheme and a lower code rate (with more redundant bits inserted). For details, see section 5.1 "Modulation mapper" in 3GPP TS 38.211 V15.5.0.

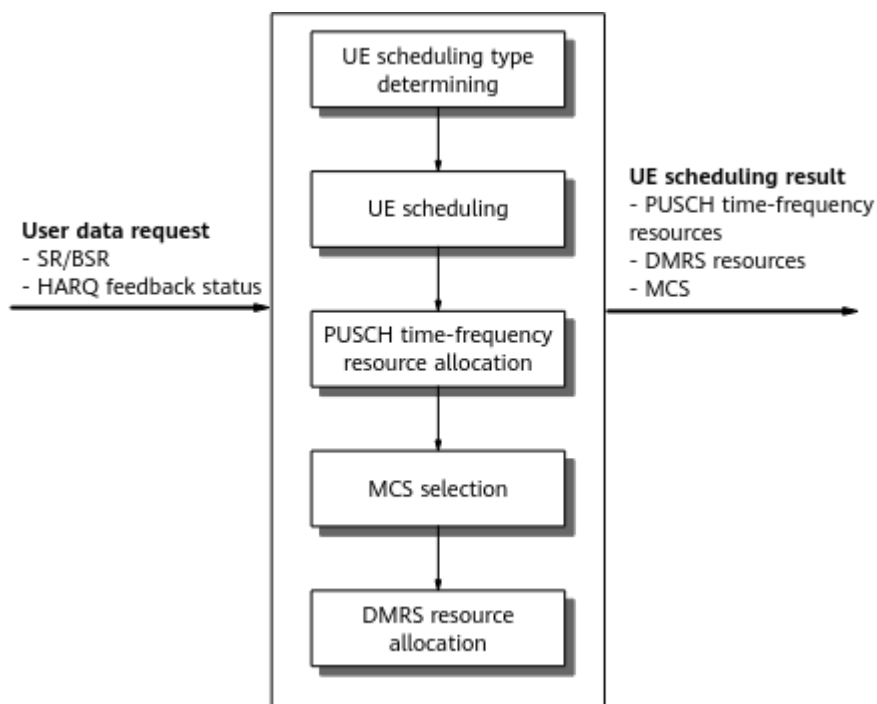
TBS

A transport block size (TBS) is the size of a transport block transmitted from the MAC layer to the PHY layer. Peak traffic volume calculation requires that the TBS of all slots be obtained within a second. The TBS depends on the MCS and the number of resources allocated for data transmission.

3.2 Basic Procedure of Uplink Scheduling

Figure 3-2 shows the basic procedure of uplink scheduling. A scheduler uses UE data requests as the inputs and UE scheduling results as the outputs. For details, see chapter 6 "Physical uplink shared channel related procedure" in 3GPP TS 38.214 V15.4.0.

Figure 3-2 Basic procedure of uplink scheduling



The following describes the inputs and outputs of an uplink scheduler.

- UE data request
 - Scheduling request (SR): Indicates the information that a UE sends to the gNodeB over the physical uplink control channel (PUCCH) to request resources for uplink data transmission. Based on the SR, the scheduler allocates PUSCH time-frequency resources to the UE.
 - Buffer status report (BSR): Indicates the amount of data to be sent by a UE to the scheduler in the uplink buffer. A UE periodically sends its BSR based on the periodic BSR timer (periodicBSR-Timer), the length of which is specified by the **NRDUCellPusch.PeriodicBsrTimer** parameter. After this timer expires, the UE triggers BSR reporting and sends the BSR to the scheduler. The timer then restarts. The timer length can be configured only in low frequency bands, not in high frequency bands. For details about periodicBSR-Timer, see section 5.4.5 "Buffer Status Reporting" in 3GPP TS 38.321 V15.9.0 and section 6.3.2 "Radio resource control information elements" in 3GPP TS 38.331 V15.9.0.
 - HARQ feedback status: Indicates whether UE data has been correctly transmitted or retransmitted. HARQ feedback can be ACK or NACK.

- UE scheduling result
 - PUSCH time-frequency resources: Refer to the PUSCH time-frequency resources allocated by the scheduler to a UE, including both time-domain and frequency-domain resources.
 - DMRS resources: Refer to the DMRS resources allocated by the scheduler to a UE.
 - MCS: Refers to the modulation and coding scheme signaled by the scheduler to each successfully scheduled UE.

 NOTE

Information such as the UE capability and uplink power also affects the scheduling result.

An uplink scheduler performs scheduling in each slot as follows:

1. Before scheduling data, the uplink scheduler needs to determine the scheduling content and type. For details, see [4 UE Scheduling Type Determining](#).
2. Based on inputs, the uplink scheduler determines the scheduling priorities of bearers and selects the UEs to be scheduled. This satisfies scheduling fairness requirements and maximizes system throughput. For details, see [5 UE Scheduling](#).
3. The uplink scheduler indicates to the UE the slot for scheduling and the start and end positions of the scheduled orthogonal frequency division multiplexing (OFDM) symbols in that slot to complete PUSCH time-frequency resource allocation for system information or UE data transmission. For details, see [6 PUSCH Time-Frequency Resource Allocation](#).
4. The uplink scheduler selects an MCS for uplink scheduling based on the SINR. MCS selection for uplink scheduling involves SINR adjustment, initial MCS selection, and MCS adjustment. For details, see [7 MCS Selection](#).
5. After PUSCH resources are allocated, the uplink scheduler allocates DMRS resources to UEs for PUSCH demodulation. Section 6.4.1 "Reference signals" in 3GPP TS 38.211 V15.4.0 defines PUSCH DMRS, in terms of DMRS symbols in the time domain and DMRS configuration types in the frequency domain. For details, see [8 DMRS Resource Allocation](#).

4 UE Scheduling Type Determining

4.1 Principles

Before scheduling data, an uplink scheduler needs to determine the scheduling type of a UE, which can either be initial transmission or retransmission.

Initial transmission scheduling refers to the initial scheduling of data blocks, and retransmission scheduling refers to the re-scheduling of data blocks.

The scheduler uses either of the following methods to trigger uplink initial transmission scheduling:

- The gNodeB receives the SR of a UE. In this case, the scheduler considers that new data is to be transmitted. To ensure UE scheduling performance, the following optimization functions are provided:
 - **SR-based smart scheduling**

SR-based smart scheduling is introduced to preferentially guarantee services of UEs that are about to experience service drops and UEs that are accessing the network. This function is controlled by the **SR_SMART_SCH_SW** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter. After this function is enabled, the gNodeB adjusts the SR-related scheduling priorities to preferentially guarantee services of UEs that are about to experience service drops, then UEs that are accessing the cell, and finally other UEs that have sent SRs.
 - **SR-based scheduling optimization**

SR-based scheduling optimization is introduced to promptly schedule UEs whose buffer is not empty after SR-based smart scheduling takes effect. This function is controlled by the **SR_SCH_OPT_SW** option of the **NRDUCellPusch.PuschPerformanceSwitch** parameter. After SR-based scheduling optimization is enabled, the gNodeB lowers the scheduling priorities for UEs whose buffer is empty and preferentially schedules UEs whose buffer is not empty; in addition, to ensure that the uplink data of the latter UEs can be scheduled at a time in best effort mode, the gNodeB adjusts the volume of the first packet based on the historical data packet transmission status of these UEs. This function is supported in both low and high frequency bands. This document describes only SR-based scheduling optimization in high frequency bands. For details about

SR-based scheduling optimization in low frequency bands, see *Turbo Uplink (Low-Frequency TDD)*.

- The gNodeB actively schedules uplink data of UEs in preallocation mode. For details about preallocation, see [5.10 Uplink Preallocation](#).

After an initial transmission scheduling procedure is complete:

- The gNodeB receives uplink data from a UE. If the data is correctly demodulated, the HARQ feedback status is ACK and is sent to the UE to indicate successful scheduling.
 - If there is still UE data to be scheduled in the BSR carried in the data block transmitted last time, initial transmission scheduling continues.
 - If there is no data to be scheduled in the BSR carried in the data block transmitted last time, the scheduling procedure ends.
- If the HARQ feedback status of uplink UE data reception by the gNodeB is NACK, the scheduling fails. The scheduler needs to re-schedule the data block by performing retransmission scheduling.

4.2 Network Analysis

4.2.1 Benefits

The uplink scheduler first needs to determine whether the scheduling type is initial transmission or retransmission scheduling, and ensures preferential scheduling of important information so as to maximize system capacity while ensuring user fairness.

Other sub-functions have the following impacts on the network:

- **SR-based smart scheduling**
After this function takes effect, UEs about to experience service drops and newly admitted UEs can be preferentially guaranteed. This ensures service experience of these UEs.
- **SR-based scheduling optimization**
After this function takes effect in high-frequency scenarios, the average uplink UE throughput (**User Uplink Average Throughput (DU)**) and average uplink cell throughput may increase. The benefits increase with a larger number of RRC_CONNECTED UEs or a higher uplink PRB usage.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

UE scheduling type determining is a basic function and has no impact on the network.

Other sub-functions have the following impacts on the network:

- **SR-based smart scheduling**
None

- **SR-based scheduling optimization**

This function may have the following impacts on network performance:

The values of the following indicators increase:

- **Uplink Resource Block Utilizing Rate (DU)**
- Average uplink MCS index

The values of the following indicators decrease:

- **Cell Downlink Average Throughput (DU)**
- **User Downlink Average Throughput (DU)**
- **N.ThpVol.DL**
- Average downlink MCS index

The values of the following indicators fluctuate, depending on the proportions of UEs near the cell center, at a medium distance from the cell center, and at the cell edge:

- PDCCH CCE usage
- Average uplink rank

4.3 Requirements

4.3.1 Licenses

There are no license requirements for basic functions.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	SR-based smart scheduling	SR_SMART_SCH_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	<i>Uplink Scheduling</i>	SR-based scheduling optimization takes effect only when both SR-based smart scheduling and SR-based scheduling optimization are enabled (by selecting the SR_SMART_SCH_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter and the SR_SCH_OPT_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter).
Low-frequency TDD	None	None	None	None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.3.4 Others

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

This function is enabled by default and does not need activation.

Table 4-1 describes the parameters used for function optimization. This table does not describe parameters related to cell establishment.

Table 4-1 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
Uplink PUSCH Algorithm Switch	NRDUCellPusch.UlPuschAlgoSwitch	SR_SMART_SCH_SW	You are advised to select this option.
PUSCH Performance Switch	NRDUCellPusch.PuschPerformanceSwitch	SR_SCH_OPT_SW	You are advised to select this option for a high-frequency cell where the uplink PRB usage is greater than 30% and the number of RRC_CONNECTED UEs is greater than 70. You are advised to deselect this option in other scenarios.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

MML commands are not required for activating this function.

Optimization Command Examples

```
//Turning on the switch for SR-based smart scheduling  
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPuschAlgoSwitch=SR_SMART_SCH_SW-1;  
//Turning on the switch for SR-based scheduling optimization  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=SR_SCH_OPT_SW-1;
```

Deactivation Command Examples

MML commands are not required for deactivating this function. You can determine whether to restore the settings of other parameters based on actual network conditions.

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

UE scheduling type determining is a basic function and takes effect by default. Activation verification is not involved.

The activation verification methods for other sub-functions are as follows:

- **SR-based smart scheduling**
None
- **SR-based scheduling optimization**
If the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

4.4.3 Network Monitoring

UE scheduling type determining is a basic function that ensures basic service performance. Separate monitoring is not required.

The indicators listed in [4.2 Network Analysis](#) can be used to monitor the benefits and impacts of other sub-functions.

5 UE Scheduling

After determining the scheduling type of a UE, the gNodeB performs scheduling based on the result. The basic procedure and optimization functions for UE scheduling are as follows:

- Basic procedure for UE scheduling: For details, see [5.1 Basic Procedure for UE Scheduling](#).
- Optimization functions for UE scheduling: In addition to the basic procedure, scheduling also needs to optimize policies (for example, adjust scheduling priorities for UEs or apply packet bundled scheduling) based on the actual data transmission process in different service scenarios to ensure better transmission performance. [Table 5-1](#) lists optimization functions for UE scheduling.

Table 5-1 Optimization functions for UE scheduling

Function Name	Low-Frequency NR TDD	High-Frequency NR TDD	Reference
Uplink balanced scheduling	√	-	5.2 Uplink Balanced Scheduling
Stopping uplink scheduling for uplink-abnormal UEs	√	√	5.3 Stopping Uplink Scheduling for Uplink-Abnormal UEs

Function Name	Low-Frequency NR TDD	High-Frequency NR TDD	Reference
Scheduling optimization for UEs with aperiodic uplink CSIs/Uplink enhanced scheduling of aperiodic CSI reports	√	-	5.4 Scheduling Optimization for UEs with Aperiodic Uplink CSIs/Uplink Enhanced Scheduling of Aperiodic CSI Reports
To-be-scheduled UE optimization	√	-	5.5 To-Be-Scheduled UE Optimization
Uplink scheduling based on beam priorities defined by the Hybrid EPF algorithm	-	√	5.6 Uplink Scheduling Based on Beam Priorities Defined by the Hybrid EPF Algorithm
Uplink aperiodic CSI reporting optimization	-	√	5.7 Uplink Aperiodic CSI Reporting Optimization
Adaptive extension of the uplink signaling reporting interval	√	-	5.8 Adaptive Extension of the Uplink Signaling Reporting Interval

Function Name	Low-Frequency NR TDD	High-Frequency NR TDD	Reference
Doppler measurement based on the maximum RB segment	√	-	5.9 Doppler Measurement based on the Maximum RB Segment
Uplink preallocation	√	√	5.10 Uplink Preallocation
Proactive scheduling for uplink signaling	-	√	5.11 Proactive Scheduling for Uplink Signaling
MPS-based preferential access and scheduling	√	-	5.12 MPS-based Preferential Access and Scheduling
Aperiodic CSI-RS optimization in heavy-load scenarios	√	-	5.13 Aperiodic CSI-RS Optimization in Heavy-Load Scenarios

 NOTE

In the preceding table, "√" indicates that the function is supported in the specified RAT, while "-" indicates that the function is not supported.

5.1 Basic Procedure for UE Scheduling

5.1.1 Principles

After determining the scheduling type of a UE, the gNodeB performs scheduling based on the result.

- Initial transmission scheduling

Uplink initial transmission scheduling uses the EPF algorithm to prioritize multiple UEs with uplink data transmission requests and schedules them in descending order of priority.

The scheduling priority in EPF is calculated as follows:

$$\text{Priority} = \frac{\text{eff}}{r} \times \gamma(QCI), \text{ where:}$$

- **eff**: Indicates the channel quality of a UE.
- **r**: Indicates the historical transmission rate of a UE.
- **$\gamma(QCI)$** : For details, see *QoS Management*.
- Retransmission scheduling
Uplink retransmission scheduling uses asynchronous adaptive HARQ. During HARQ retransmissions, the system can adaptively select an MCS and use the same TBS as that for initial transmissions. When the number of uplink HARQ transmissions reaches the maximum, uplink HARQ retransmission stops.
Uplink scheduling supports up to four HARQ redundancy versions. For details, see "Table 7.3.1.1.1-2: Redundancy version" in 3GPP TS 38.212 V15.4.0.

5.1.2 Network Analysis

5.1.2.1 Benefits

With the basic procedure for UE scheduling, the scheduler determines whether to perform initial transmission scheduling or retransmission scheduling based on the HARQ feedback status of uplink data transmission. In addition, the scheduler determines the scheduling priorities of UEs in different scheduling types to ensure UE fairness.

5.1.2.2 Impacts

The impacts between this function and other functions are described in [5.1.3.2.3 Function Impacts](#).

Basic procedure for UE scheduling has no impact on the network.

5.1.3 Requirements

5.1.3.1 Licenses

There are no license requirements for basic functions.

5.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.1.3.2.1 Prerequisite Functions

None

5.1.3.2.2 Mutually Exclusive Functions

None

5.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	NRDUCell.NrD uCellNetworki ngMode set to HYPER_CELL	<i>Hyper Cell</i>	The uplink common control information (such as the PUCCH, PRACH, and SRS information) for each TRP within a hyper cell is scheduled in a unified manner. The gNodeB allocates the same time-frequency resources to each TRP. The uplink UE-dedicated signaling and data for each TRP within a hyper cell are independently scheduled and allocated by the serving TRP.
Low-frequency TDD	Cell Combination	NRDUCell.NrD uCellNetworki ngMode set to HYPER_CELL_C OMBINE_MOD E	<i>Cell Combination</i>	The PUCCH, PRACH, and SRS for each TRP for a combined cell are scheduled in a unified manner. The gNodeB allocates the same time-frequency resources to each TRP.

5.1.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.1.3.4 Others

None

5.1.4 Operation and Maintenance

5.1.4.1 Data Configuration

5.1.4.1.1 Data Preparation

This function is enabled by default and does not need activation.

5.1.4.1.2 Using MML Commands

Before using MML commands, refer to [5.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual

network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

MML commands are not required for activating this function.

Deactivation Command Examples

Deactivation command examples are not involved.

5.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.1.4.2 Activation Verification

This function is a basic function and takes effect by default. Activation verification is not involved.

5.1.4.3 Network Monitoring

This basic function ensures basic service performance. Separate monitoring is not required.

5.2 Uplink Balanced Scheduling

5.2.1 Principles

Uplink balanced scheduling is introduced to solve the problem of low uplink PRB usage caused by insufficient PDCCH CCEs in uplink heavy-load scenarios. When uplink PDCCH CCEs are limited and the number of RRC_CONNECTED UEs in a cell is greater than the related threshold (specified by the **NRDUCellUISch.UlBalancedSchUeNumThld** parameter) for the uplink balanced scheduling function, this function takes effect for UEs whose data volume to be scheduled is larger than the relevant data volume threshold of large-packet-service UEs (specified by the **NRDUCellUISch.UlLargePktUeDataVolumeThld** parameter), to increase the uplink PRB usage. This function is supported only in low frequency bands, not in high frequency bands.

5.2.2 Network Analysis

5.2.2.1 Benefits

After this function takes effect in uplink low-frequency heavy-load scenarios, the uplink PRB usage, uplink cell throughput, and average uplink user-perceived rate increase.

5.2.2.2 Impacts

The impacts between this function and other functions are described in [5.2.3.2.3 Function Impacts](#).

After this function takes effect in low-frequency heavy-load scenarios, the proportion of UEs with low uplink throughput increases and the downlink cell throughput may decrease.

5.2.3 Requirements

5.2.3.1 Licenses

There are no license requirements for basic functions.

5.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.2.3.2.1 Prerequisite Functions

None

5.2.3.2.2 Mutually Exclusive Functions

None

5.2.3.2.3 Function Impacts

None

5.2.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.2.3.4 Others

None

5.2.4 Operation and Maintenance

5.2.4.1 Data Configuration

5.2.4.1.1 Data Preparation

Table 5-2 describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

Table 5-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
UL Balanced Schedule UE Num Threshold	NRDUCellULSch. <i>ULBalancedSchUeNumThld</i>	None	The value 20 is recommended.
UL Large-Packet UE Data Volume Threshold	NRDUCellULSch. <i>ULLargePktUeDataVolumeThld</i>	None	Use the default value.

5.2.4.1.2 Using MML Commands

Before using MML commands, refer to **5.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Configuring the threshold of the number of UEs for uplink balanced scheduling to take effect  
MOD NRDUCELLULSCH: NrDuCellId=0, ULBalancedSchUeNumThld=20;  
//Configuring the data volume threshold of large-packet-service UEs in uplink balanced scheduling  
MOD NRDUCELLULSCH: NrDuCellId=0, ULLargePktUeDataVolumeThld=50;
```

Deactivation Command Examples

Deactivation command examples are not involved.

5.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.2.4.2 Activation Verification

If the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

5.2.4.3 Network Monitoring

The indicators listed in [5.2.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.3 Stopping Uplink Scheduling for Uplink-Abnormal UEs

5.3.1 Principles

The function of stopping uplink scheduling for uplink-abnormal UEs is introduced to effectively control the uplink RBLER in a cell. After this function is enabled, the gNodeB stops uplink scheduling for UEs with a consecutively high uplink BLER. This reduces the uplink RBLER, improves spectral efficiency, and increases the uplink cell throughput in heavy-load scenarios. This function is enabled by selecting the **ABN_UE_STOP_ULSCH_SW** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter.

5.3.2 Network Analysis

5.3.2.1 Benefits

After this function is enabled for a low-frequency cell serving UEs with a consecutively high uplink BLER, the uplink RBLER of the cell decreases and the average uplink cell throughput increases.

Uplink RBLER of a cell (%) =
$$\frac{(\text{N.UL.SCH.HalfPiBPSK.ErrTB.Rbler} + \text{N.UL.SCH.QPSK.ErrTB.Rbler} + \text{N.UL.SCH.16QAM.ErrTB.Rbler} + \text{N.UL.SCH.64QAM.ErrTB.Rbler} + \text{N.UL.SCH.256QAM.ErrTB.Rbler})}{(\text{N.UL.SCH.HalfPiBPSK.TB} + \text{N.UL.SCH.QPSK.TB} + \text{N.UL.SCH.16QAM.TB} + \text{N.UL.SCH.64QAM.TB} + \text{N.UL.SCH.256QAM.TB})} \times 100\%$$

5.3.2.2 Impacts

The impacts between this function and other functions are described in [5.3.3.2.3 Function Impacts](#).

After this function is enabled in low-frequency scenarios, the gNodeB stops uplink scheduling for abnormal UEs. As a result, the uplink PRB usage of the cell may decrease.

5.3.3 Requirements

5.3.3.1 Licenses

There are no license requirements for basic functions.

5.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.3.2.1 Prerequisite Functions

None

5.3.3.2.2 Mutually Exclusive Functions

None

5.3.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	UE link fault detection	NRDUCellAlgoSwitch.UeLinkAbnormalDetectionSwitch	None	After the function of stopping uplink scheduling for uplink-abnormal UEs is enabled, link fault detection may fail for UEs for which scheduling has been stopped. As a result, these UEs cannot be released, but air interface resources can be saved.

5.3.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.3.3.4 Others

None

5.3.4 Operation and Maintenance

5.3.4.1 Data Configuration

5.3.4.1.1 Data Preparation

Table 5-3 describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

Table 5-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink PUSCH Algorithm Switch	NRDUCellPusch. UIPuschAlgoSwitch	ABN_UE_STOP_UL SCH_SW	You are advised to select this option.

5.3.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for stopping uplink scheduling for uplink-abnormal UEs  
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPuschAlgoSwitch=ABN_UE_STOP_ULSCH_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch for stopping uplink scheduling for uplink-abnormal UEs  
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPuschAlgoSwitch=ABN_UE_STOP_ULSCH_SW-0;
```

5.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.3.4.2 Activation Verification

When a cell serves UEs with a consecutively high uplink BLER, if the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

5.3.4.3 Network Monitoring

The indicators listed in [5.3.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.4 Scheduling Optimization for UEs with Aperiodic Uplink CSIs/Uplink Enhanced Scheduling of Aperiodic CSI Reports

5.4.1 Principles

When uplink periodic CSI resources are insufficient or some features (such as intra-base-station DL CoMPDPSCH type II codebook or frequency selective scheduling) take effect, UEs need to aperiodically report CSI measurement results over the PUSCH. These UEs are referred to as UEs with aperiodic uplink CSIs and they occupy PUSCH resources. As such, the following optimization functions are introduced:

- Scheduling optimization for UEs with aperiodic uplink CSIs is introduced to increase available PUSCH resources when the uplink load is heavy and uplink performance needs to be preferentially guaranteed. This function is controlled by the **UL_APERIODIC_CSI_SCH_OPT_SW** option of the **NRDUCellPusch.PuschPerformanceSwitch** parameter. After this function takes effect, the capacity of scheduling UEs with aperiodic uplink CSIs decreases and the PUSCH resources occupied by these UEs decrease, thereby increasing the average uplink user-perceived rate. This function is supported only in TDD in low frequency bands, not in high frequency bands.
- In low-frequency NR TDD scenarios, downlink transmission strongly depends on CSI reports from UEs. Therefore, uplink enhanced scheduling of aperiodic CSI reports is introduced to increase the probability of scheduling UEs with aperiodic uplink CSIs when the downlink load is heavy and downlink performance needs to be preferentially guaranteed. This function is controlled by the **APER_CSI_RPT_ENHANCE_SW** option of the **NRDUCellCsirs.CsiSwitch** parameter. After this function takes effect, the gNodeB can schedule UEs with aperiodic uplink CSIs in more subframes, increasing the probability of scheduling these UEs and the average downlink UE throughput.

5.4.2 Network Analysis

5.4.2.1 Benefits

Scheduling Optimization for UEs with Aperiodic Uplink CSIs

After this function takes effect in low-frequency NR TDD scenarios with heavy load, available PUSCH resources and the average uplink user-perceived rate increase.

Uplink Enhanced Scheduling of Aperiodic CSI Reports

After this function takes effect in low-frequency NR TDD scenarios with downlink heavy load, the probability of scheduling UEs with aperiodic uplink CSIs increases

and the average downlink UE throughput (**User Downlink Average Throughput (DU)**) increases.

5.4.2.2 Impacts

The impacts between this function and other functions are described in [5.4.3.2.3 Function Impacts](#).

Scheduling Optimization for UEs with Aperiodic Uplink CSIs

In low-frequency NR TDD scenarios with heavy load, this function reduces the probability of CSI measurement reporting, probably affecting the CQI measurement accuracy. If this occurs, the MCS index for downlink scheduling and the **Cell Downlink Average Throughput (DU)** decrease.

Uplink Enhanced Scheduling of Aperiodic CSI Reports

In low-frequency scenarios, this function increases the number of scheduling times for UEs with aperiodic uplink CSIs. This results in increase in PUSCH resource usage, uplink PRB usage, uplink CCE usage, and service drop rate. In addition, the user-plane CPU usage of gNodeB boards (indicated by **VS.NRBoard.UPlane.CPULoad.Avg**, **VS.NRBoard.UPlane.CPULoad.Max**, and **VS.NRBoard.UPlane.CumulativeHighloadCount**) may slightly increase.

5.4.3 Requirements

5.4.3.1 Licenses

There are no license requirements for basic functions.

5.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.4.3.2.1 Prerequisite Functions

None

5.4.3.2.2 Mutually Exclusive Functions

Scheduling optimization for UEs with aperiodic uplink CSIs

None

Uplink enhanced scheduling of aperiodic CSI reports

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Aperiodic CSI CCE resources	APERIODIC_CSI_CCE_RES option of the NRDUCellPdcch.PDCCHAdjRefResType parameter	<i>Channel Management</i>	The switch for enhanced scheduling of aperiodic CSI reports cannot be turned on (by selecting the APER_CSI_RPT_ENHANCE_SW option of the NRDUCellCsirs.CsiSwitch parameter) when the switch for aperiodic CSI CCE resources is turned on (by selecting the APERIODIC_CSI_CCE_RES option of the NRDUCellPdcch.PDCCHAdjRefResType parameter). Reason: When the switch for aperiodic CSI CCE resources is on, the PDCCH scheduling resource adjustment algorithm obtains the scheduling resource adjustment amount based on aperiodic CSI reporting. In this case, if enhanced scheduling of aperiodic CSI reports is in effect, the number of scheduling times of UEs with aperiodic uplink CSIs increases. This may lead to more CCEs allocated for uplink scheduling but fewer for downlink scheduling in PDCCH CCE adjustment, affecting the downlink throughput. Therefore, the switch for enhanced scheduling of aperiodic CSI reports cannot be turned on together with the switch for aperiodic CSI CCE resources.

5.4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	The status (enabled or disabled) of uplink enhanced scheduling of aperiodic CSI reports must be consistent between the PCell and all SCells of a CA UE. Status inconsistency affects the downlink throughput of the UE.
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FRIINTER_BAND_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	The status (enabled or disabled) of uplink enhanced scheduling of aperiodic CSI reports must be consistent between the PCell and all SCells of a CA UE. Status inconsistency affects the downlink throughput of the UE.

5.4.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.4.3.4 Others

None

5.4.4 Operation and Maintenance

5.4.4.1 Data Configuration

5.4.4.1.1 Data Preparation

[Table 5-4](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-4 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
PUSCH Performance Switch	NRDUCellPusch.PuschPerformanceSwitch	UL_APERIODIC_CSI_SCH_OPT_SW	You are advised to select this option when the uplink load is heavy (the PDCCH CCEs are limited or the uplink PRB usage is greater than or equal to 90%) and uplink performance needs to be preferentially ensured. You are advised to deselect this option in other scenarios.
CSI Switch	NRDUCellCsirs.CsirsSwitch	APER_CSI_RPT_ENHANCE_SW	You are advised to select this option in low-frequency NR TDD scenarios with downlink heavy load and deselect this option in other scenarios.

5.4.4.1.2 Using MML Commands

Before using MML commands, refer to [5.4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling scheduling optimization for UEs with aperiodic uplink CSIs
MOD NRDUCCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL_APERIODIC_CSI_SCH_OPT_SW-1;
//Enabling uplink enhanced scheduling of aperiodic CSI reports
MOD NRDUCCELLCSIRS: NrDuCellId=0, CsirsSwitch=APER_CSI_RPT_ENHANCE_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling scheduling optimization for UEs with aperiodic uplink CSIs  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL_APERIODIC_CSI_SCH_OPT_SW-0;  
//Disabling uplink enhanced scheduling of aperiodic CSI reports  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=APER_CSI_RPT_ENHANCE_SW-0;
```

5.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.4.2 Activation Verification

Scheduling Optimization for UEs with Aperiodic Uplink CSIs

In low-frequency NR TDD scenarios with heavy load, if the average uplink UE throughput (**User Uplink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

Uplink Enhanced Scheduling of Aperiodic CSI Reports

In low-frequency NR TDD scenarios with downlink heavy load, if the average downlink cell throughput (**User Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

5.4.4.3 Network Monitoring

The indicators listed in [5.4.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.5 To-Be-Scheduled UE Optimization

5.5.1 Principles

When there is a relatively large number of RRC_CONNECTED UEs, the uplink scheduler, having to balance the CPU load and uplink performance, periodically updates the scheduling information of some UEs to be scheduled to reduce the CPU load. If there is buffered data for a UE to be scheduled, the scheduling information sometimes is not promptly updated, prolonging the scheduling delay and affecting the average uplink UE throughput. The to-be-scheduled UE optimization function can be enabled using the **TO_BE_SCHEDULED_UE_OPT_SW** option of the **NRDUCellServExp.ServiceExpAlgoSwitch** parameter to more promptly update scheduling information of to-be scheduled UEs with buffered data, thereby improving the average uplink UE throughput. This function applies only to low-frequency NR TDD cells and SUL cells.

5.5.2 Network Analysis

5.5.2.1 Benefits

When there is a relatively large number of RRC_CONNECTED UEs, the number of UEs scheduled in the uplink may increase after this function takes effect, leading to the following impacts on the network:

- The uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**), uplink interference (**N.UL.NI.Avg**), and average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) increase.
- If the channel conditions of new UEs in uplink scheduling differ greatly from those of existing UEs, the average uplink UE throughput (**User Uplink Average Throughput (DU)**), average uplink MCS index, average uplink rank, average number of paired layers for uplink MU pairing, average proportion of PRBs for uplink MU pairing, and PDCCH CCE usage fluctuate.

When the proportion of RRC_CONNECTED UEs in the downlink DRX state is high, the number of UEs scheduled in the downlink may also increase, leading to the following additional impacts on the network:

- The downlink PRB usage (**Downlink Resource Block Utilizing Rate (DU)**), average downlink UE throughput (**User Downlink Average Throughput (DU)**), and average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increase.
- The average downlink MCS index, average downlink rank, average number of paired layers for downlink MU pairing, and average proportion of PRBs for downlink MU pairing fluctuate.
- When TCP-based services (such as web browsing, email, chatting, FTP, P2P file sharing, and progressive video) are used, the number of uplink TCP response packets increases after the number of UEs scheduled in the downlink increases. The network impacts are the same as those of the increase in the number of UEs scheduled in the uplink.

5.5.2.2 Impacts

The impacts between this function and other functions are described in [5.5.3.2.3 Function Impacts](#).

The network impacts produced by to-be-scheduled UE optimization are described in benefit analysis. For details, see [5.5.2.1 Benefits](#).

5.5.3 Requirements

5.5.3.1 Licenses

There are no license requirements for basic functions.

5.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.5.3.2.1 Prerequisite Functions

None

5.5.3.2.2 Mutually Exclusive Functions

None

5.5.3.2.3 Function Impacts

None

5.5.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.5.3.4 Others

None

5.5.4 Operation and Maintenance

5.5.4.1 Data Configuration

5.5.4.1.1 Data Preparation

[Table 5-5](#) describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

Table 5-5 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Service Experience Algorithm Switch	NRDUCellServExp.ServiceExpAlgoSwitch	TO_BE_SCHEDULED_UE_OPT_SW	You are advised to select this option when there is a large number of RRC_CONNECTED UEs.

5.5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the to-be-scheduled UE optimization switch (You are advised to turn on this switch when there is a relatively large number of RRC_CONNECTED UEs.)  
MOD NRDUCELLSERVEXP: NrDuCellId=0, ServiceExpAlgoSwitch=TO_BE_SCHEDULED_UE_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the to-be-scheduled UE optimization switch  
MOD NRDUCELLSERVEXP: NrDuCellId=0, ServiceExpAlgoSwitch=TO_BE_SCHEDULED_UE_OPT_SW-0;
```

5.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.5.4.2 Activation Verification

In scenarios with a large number of RRC_CONNECTED UEs, if the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

5.5.4.3 Network Monitoring

The indicators listed in [5.5.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.6 Uplink Scheduling Based on Beam Priorities Defined by the Hybrid EPF Algorithm

5.6.1 Principles

In high frequency bands, NR TDD uses the analog-digital hybrid beamforming architecture. The scheduler preferentially allocates analog beam resources to UEs

and then allocates digital beam resources. Due to the restriction that RB resources of one analog beam cannot be used by UEs of another beam, when a UE performing small-packet services exclusively occupies an analog beam, remaining RB resources of the beam cannot be used by UEs of other beams, resulting in a waste of resources. To address this issue, uplink scheduling based on beam priorities defined by the Hybrid EPF algorithm is introduced. This function is controlled by the **NRDUCellUlsch.UlBeamPriBasedSch** parameter. When this parameter is set to **HEPF**, this function is enabled to calculate the priority of each beam based on the EPF priority and required RB quantity of UEs in each beam. Then, scheduling among beams is performed by beam priority, but within a beam, the EPF priority of each UE applies. This raises the scheduling priorities of the beams serving UEs with medium- or large-packet services, increasing the downlink PRB usage, average downlink UE throughput, and average downlink cell throughput. This function is supported only in high-frequency NR TDD scenarios.

5.6.2 Network Analysis

5.6.2.1 Benefits

After this function takes effect in high-frequency NR TDD with medium or heavy load (with the uplink PRB usage being about 50%–70%) where the small-packet and large-packet hybrid burst traffic model is used, the **Cell Uplink Average Throughput (DU)** and **User Uplink Average Throughput (DU)** increase. In other scenarios, this function may not deliver any benefits.

5.6.2.2 Impacts

The impacts between this function and other functions are described in [5.6.3.2.3 Function Impacts](#).

In high-frequency NR TDD, this function has the following impacts on the network:

- Processing delay at the MAC layer and RLC layer in the gNodeB may decrease.
- The values of the following indicators fluctuate, depending on the proportions of UEs near the cell center, at a medium distance from the cell center, and at the cell edge as well as the inbound traffic models of UEs:
 - Average uplink MCS index and average uplink rank
 - Uplink IBLER, uplink RBLER, and uplink retransmission proportion
 - Proportion of using the 64QAM MCS index table in the uplink, proportion of using 64QAM, 16QAM, or QPSK in uplink scheduling, uplink CCE aggregation level, and PDCCH CCE usage
 - Number of active UEs in the uplink and number of scheduled UEs in the uplink
 - **Cell Downlink Average Throughput (DU)**, **User Downlink Average Throughput (DU)**, average downlink MCS index, and proportion of using each modulation scheme in downlink scheduling

5.6.3 Requirements

5.6.3.1 Licenses

There are no license requirements for basic functions.

5.6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.6.3.2.1 Prerequisite Functions

None

5.6.3.2.2 Mutually Exclusive Functions

None

5.6.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Downlink scheduling based on beam priorities defined by the Hybrid EPF algorithm	NRDUCellDlsc h.DlBeamPriBasedSch set to HEPF	<i>Downlink Scheduling</i>	Enabling "uplink scheduling based on beam priorities defined by the Hybrid EPF algorithm" increases the average uplink cell throughput. However, if "downlink scheduling based on beam priorities defined by the Hybrid EPF algorithm" is enabled afterwards and most UEs with downlink large-packet services are located at the cell edge, the number of uplink response packets increases and the uplink data volume at the RLC layer, average uplink cell throughput, and average uplink UE throughput may decrease.

5.6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

5.6.3.4 Others

None

5.6.4 Operation and Maintenance

5.6.4.1 Data Configuration

5.6.4.1.1 Data Preparation

[Table 5-6](#) describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

Table 5-6 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink Beam Priority-based Schedule	NRDUCellUlsch.UlBeamPriBasedSch	None	NR TDD in high frequency bands: In scenarios where beam resources are insufficient prior to RB resources (when the TTI duty cycle is close to 100% and the uplink PRB usage is less than 70%), you are advised to set this parameter to HEPF . In other scenarios, you are advised to set this parameter to OFF .

5.6.4.1.2 Using MML Commands

Before using MML commands, refer to [5.6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink scheduling based on beam priorities defined by the Hybrid EPF algorithm (only for high-frequency TDD scenarios)  
MOD NRDUCELLULSCH: NrDuCellId=0, ULBeamPriBasedSch=HEPF;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink scheduling based on beam priorities defined by the Hybrid EPF algorithm (only for high-frequency TDD scenarios)  
MOD NRDUCELLULSCH: NrDuCellId=0, ULBeamPriBasedSch=OFF;
```

5.6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.6.4.2 Activation Verification

In medium- and heavy-load scenarios (with the uplink PRB usage being about 50%–70%) where the small-packet and large-packet hybrid burst traffic model is used, if the **Cell Uplink Average Throughput (DU)** increases after this function is enabled, this function has taken effect.

5.6.4.3 Network Monitoring

The indicators listed in [5.6.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.7 Uplink Aperiodic CSI Reporting Optimization

5.7.1 Principles

To reduce the impact of aperiodic CSI reporting on uplink scheduling, uplink aperiodic CSI reporting optimization is introduced. The optimization includes

aperiodic CSI reporting interval extension and combined reporting of beam information, which are controlled by the **APER_CSI_RPT_INTVL_INCREASE_SW** and **BEAM_INFO_COMBINED_RPT_SW** options of the **NRDUCellCsirs.CsiSwitch** parameter, respectively. After either of the preceding functions takes effect, the number of uplink aperiodic CSI reports decreases, reducing uplink resource usage and increasing average uplink cell throughput. This function is supported only in high-frequency NR TDD scenarios.

5.7.2 Network Analysis

5.7.2.1 Benefits

In high-frequency NR TDD scenarios with medium or heavy load (for example, with the uplink PRB usage being greater than 60%) where the proportion of UEs at the cell edge is less than 5%, after this function (either aperiodic CSI reporting interval extension or combined reporting of beam information) takes effect, aperiodic CSI reporting decreases, increasing the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) and average uplink UE throughput (**User Uplink Average Throughput (DU)**). Proportion of UEs at the cell edge in a cell = $\frac{\text{SUM}(\text{N.ChMeas.PUSCH.MCS.0 to N.ChMeas.PUSCH.MCS.9})}{\text{SUM}(\text{N.ChMeas.PUSCH.MCS.0 to N.ChMeas.PUSCH.MCS.28})}$.

5.7.2.2 Impacts

The impacts between this function and other functions are described in [5.7.3.2.3 Function Impacts](#).

In high-frequency NR TDD, this function (aperiodic CSI reporting interval extension or combined reporting of beam information) has the following impacts on the network:

- The average uplink MCS index fluctuates, depending on the inbound traffic model and the proportions of UEs near the cell center, at a medium distance from the cell center, and at the cell edge.
- In an uplink light-load cell (for example, with the PRB usage being less than 30%), the overall network is not congested and service requirements of all UEs can be met. In this case, the **Uplink Resource Block Utilizing Rate (DU)** and **Cell Uplink Average Throughput (DU)** may slightly decrease because the reduction in uplink aperiodic CSI reports decreases the number of padding packets.

5.7.3 Requirements

5.7.3.1 Licenses

There are no license requirements for basic functions.

5.7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.7.3.2.1 Prerequisite Functions

None

5.7.3.2.2 Mutually Exclusive Functions

None

5.7.3.2.3 Function Impacts

None

5.7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

5.7.3.4 Others

None

5.7.4 Operation and Maintenance

5.7.4.1 Data Configuration

5.7.4.1.1 Data Preparation

[Table 5-7](#) describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

Table 5-7 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
CSI Switch	NRDUCellCsirs.Cs <i>iSwitch</i>	APER_CSI_RPT_IN TVL_INCREASE_S W	In high-frequency NR TDD FWA scenarios, you are advised to select this option.
CSI Switch	NRDUCellCsirs.Cs <i>iSwitch</i>	BEAM_INFO_COM BINED_RPT_SW	In high-frequency NR TDD FWA scenarios, you are advised to select this option.

5.7.4.1.2 Using MML Commands

Before using MML commands, refer to [5.7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink aperiodic CSI reporting optimization  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=APER_CSI_RPT_INTVL_INCREASE_SW-1;  
//Enabling combined reporting of beam information  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=BEAM_INFO_COMBINED_RPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink aperiodic CSI reporting optimization  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=APER_CSI_RPT_INTVL_INCREASE_SW-0;  
//Disabling combined reporting of beam information  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=BEAM_INFO_COMBINED_RPT_SW-0;
```

5.7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.7.4.2 Activation Verification

In high-frequency NR TDD scenarios with medium or heavy load (for example, with the uplink PRB usage being greater than 60%) where the proportion of UEs at the cell edge is less than 5%, if the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect. Proportion of UEs at the cell edge in a cell = $\frac{\text{SUM}(\text{N.ChMeas.PUSCH.MCS.0 to N.ChMeas.PUSCH.MCS.9})}{\text{SUM}(\text{N.ChMeas.PUSCH.MCS.0 to N.ChMeas.PUSCH.MCS.28})}$.

5.7.4.3 Network Monitoring

The indicators listed in [5.7.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.8 Adaptive Extension of the Uplink Signaling Reporting Interval

5.8.1 Principles

In uplink heavy-load scenarios (with a large number of UEs or uplink CCE congestion), frequent A3 measurement reports triggered by coordination-based features (such as intra-base-station DL CoMP, intra-base-station UL CoMP, and event A3 for intra-frequency neighboring cell measurement) occupy a large number of uplink CCEs and PUSCH RBs, severely affecting the average uplink UE throughput. To address this issue, adaptive extension of the uplink signaling reporting interval is introduced. This function is controlled by the **MEAS_OVERHEAD_REDUCE_SW** option of the **NRCellMeasConfig.MeasSwitch** parameter. After this function is enabled, the event A3 reporting interval is prolonged and the reporting frequency is reduced when uplink CCE congestion occurs or there is a large number of UEs. This decreases occupied CCEs and PUSCH RBs and increases the average uplink UE throughput. This function is supported only in low frequency bands, not in high frequency bands.

5.8.2 Network Analysis

5.8.2.1 Benefits

After this function takes effect in low-frequency scenarios with heavy load in the uplink (with a relatively large number of UEs or uplink CCE congestion), available uplink PUSCH resources and the average uplink UE throughput (**User Uplink Average Throughput (DU)**) increase.

5.8.2.2 Impacts

The impacts between this function and other functions are described in [5.8.3.2.3 Function Impacts](#).

In low-frequency cells with heavy load in the uplink (there is a relatively large number of UEs or uplink CCE congestion), this function may have the following impacts on the network:

- The uplink throughput at the MAC layer decreases because the reduction in the number of A3-relevant uplink scheduling times decreases the number of padding packets.
- If downlink performance is limited by uplink TCP ACK transmission, the downlink throughput increases.
- The number of UEs reporting event A3 decreases, and the number of active UEs in the uplink as well as the uplink PDCCH usage decrease.

5.8.3 Requirements

5.8.3.1 Licenses

There are no license requirements for basic functions.

5.8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.8.3.2.1 Prerequisite Functions

None

5.8.3.2.2 Mutually Exclusive Functions

None

5.8.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station UL CoMP	INTRA_GNB_UL_COMP_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	When both "adaptive extension of the uplink signaling reporting interval" and intra-base-station UL CoMP are enabled, the proportion of intra-base-station UL CoMP in effect decreases.
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	When both "adaptive extension of the uplink signaling reporting interval" and intra-base-station DL CoMP are enabled, the proportion of intra-base-station DL CoMP in effect decreases.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Coordinated interference management	INTRA_GNB_DL_CS_CBF_SW and INTER_GNB_DL_CS_CBF_SW options of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	When both "adaptive extension of the uplink signaling reporting interval" and coordinated interference management are enabled, the proportion of coordinated interference management in effect decreases.
Low-frequency TDD	Event A3-related intra-frequency neighboring cell measurement	INTRA_FREQ_NCELL_STAT_SW option of the NRDUCellCollabServ.NeighborNrCellMgmtSw parameter	None	In cells with heavy load in the uplink, enabling "adaptive extension of the uplink signaling reporting interval" prolongs the event A3 reporting interval for intra-frequency neighboring cell measurement.

5.8.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.8.3.4 Others

None

5.8.4 Operation and Maintenance

5.8.4.1 Data Configuration

5.8.4.1.1 Data Preparation

[Table 5-8](#) describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

Table 5-8 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Measurement Switch	NRCellMeasConfig.MeasSwitch	MEAS_OVERHEAD_REDUCE_SW	In uplink heavy-load scenarios (with uplink CCE congestion or a relatively large number of UEs), you are advised to select this option. In other scenarios, you are advised to deselect this option.

5.8.4.1.2 Using MML Commands

Before using MML commands, refer to [5.8.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling adaptive extension of the uplink signaling reporting interval  
MOD NRCCELLMEASCONFIG: NrCellId=0, MeasSwitch=MEAS_OVERHEAD_REDUCE_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling adaptive extension of the uplink signaling reporting interval  
MOD NRCCELLMEASCONFIG: NrCellId=0, MeasSwitch=MEAS_OVERHEAD_REDUCE_SW-0;
```

5.8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.8.4.2 Activation Verification

In uplink low-frequency heavy-load scenarios (with a relatively large number of UEs or uplink CCE congestion), if the average uplink UE throughput (**User Uplink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

5.8.4.3 Network Monitoring

The indicators listed in [5.8.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.9 Doppler Measurement based on the Maximum RB Segment

5.9.1 Principles

When there are small RB segments, Doppler measurement results sometimes are inaccurate, affecting the uplink user-perceived rate. To resolve this issue, Doppler measurement based on the maximum RB segment is introduced. This function is controlled by the **MAX_RB_SEG_DOPPLER_MEAS_SW** option of the **NRDUCellPusch.PuschMeasSw** parameter. When this option is selected, Doppler measurement is performed based on the maximum RB segment, improving Doppler measurement accuracy.

5.9.2 Network Analysis

5.9.2.1 Benefits

This function may improve the Doppler measurement accuracy and therefore increase the average uplink cell rate (**Cell Uplink Average Throughput (DU)**).

5.9.2.2 Impacts

The impacts between this function and other functions are described in [5.9.3.2.3 Function Impacts](#).

5.9.3 Requirements

5.9.3.1 Licenses

There are no license requirements for basic functions.

5.9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.9.3.2.1 Prerequisite Functions

None

5.9.3.2.2 Mutually Exclusive Functions

None

5.9.3.2.3 Function Impacts

None

5.9.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.9.3.4 Others

None

5.9.4 Operation and Maintenance

5.9.4.1 Data Configuration

5.9.4.1.1 Data Preparation

Table 5-9 describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

Table 5-9 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
PUSCH Measurement Sw	NRDUCellPusch.<i>PuschMeasSw</i>	MAX_RB_SEG_DO PPLER_MEAS_SW	Select this option.

5.9.4.1.2 Using MML Commands

Before using MML commands, refer to [5.9.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling Doppler measurement based on the maximum RB segment
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschMeasSw=MAX_RB_SEG_DOPPLER_MEAS_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling Doppler measurement based on the maximum RB segment
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschMeasSw=MAX_RB_SEG_DOPPLER_MEAS_SW-0;
```

5.9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.9.4.2 Activation Verification

If the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

5.9.4.3 Network Monitoring

The indicators listed in [5.9.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.10 Uplink Preallocation

5.10.1 Principles

Preallocation refers to a mechanism whereby the gNodeB actively schedules a UE at a specified interval regardless of whether the UE has sent an SR to the gNodeB. This shortens the gap from when the UE sends an SR to when the UE obtains the uplink scheduling grant. The minimum preallocation interval (interval between two times that the same UE is scheduled) is specified by the **NRDUCellPusch.MinPreallocationPeriod** parameter. **NRDUCellPusch.PreschUeNumUpperLimit** specifies the maximum number of UEs in a slot's preallocation queue, and **NRDUCellPusch.PreallocationSize** specifies the amount of UE data to be scheduled for uplink preallocation.

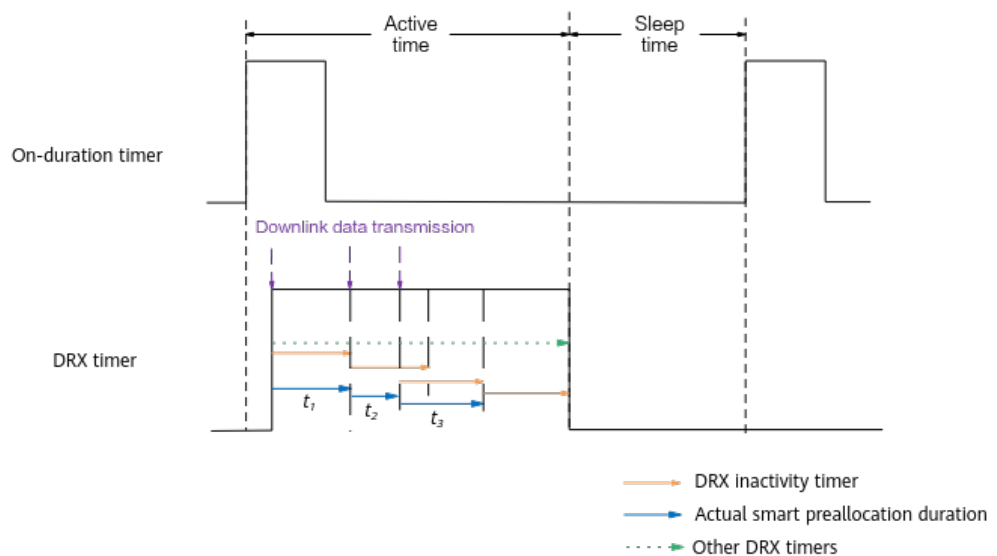
Uplink preallocation is classified into uplink basic preallocation and uplink smart preallocation.

- Uplink basic preallocation is controlled by the **UL_BASIC_PREALLOCATION_SWITCH** option of the **NRDUCellPusch.UlPreallocationSwitch** parameter and takes effect only for newly admitted UEs, UEs involved in incoming handovers, or UEs for which RRC connections are reconfigured. When basic preallocation takes effect, a UE

is scheduled as long as there are available resources, regardless of whether the UE has service requests. Basic preallocation consumes air interface resources, and the UE standby time is short.

- Uplink smart preallocation is controlled by the **UL_SMART_PREALLOCATION_SWITCH** option of the **NRDUCellPusch.UlPreallocationSwitch** parameter. In smart preallocation mode, a UE with no service requests enters the DRX sleep state when all DRX-related timer expires. This saves air interface resources and prolongs the UE's standby duration. After the **UL_SMART_PREALLOCATION_SWITCH** option is selected, smart preallocation is triggered each time the core network sends data to the gNodeB. If not all data is transmitted within the smart preallocation duration specified by **NRDUCellPusch.UlSmartPreallocationDur**, smart preallocation is automatically extended to allow all remaining data to be transmitted. During smart preallocation, a UE remains in the DRX active period. It enters the DRX sleep state when (1) it has no service requests and (2) all DRX-related timer expires. This reduces power consumption. **Figure 5-1** illustrates smart preallocation. In the figure, t_1 , t_2 , and t_3 indicate three actual smart preallocation durations, separately.

Figure 5-1 Smart preallocation principles



For details about the on-duration timer, DRX inactivity timer, and other DRX timers, see *DRX*.

In addition, the gNodeB supports uplink preallocation for sparse-packet services (namely ping services, for the moment) in low frequency bands. This function is collectively controlled by the **SPARSE_DIFF_PROC_SW** option of the **gNodeBParam.ServiceDiffProcSwitch** parameter and the **SPARSE_DIFF_SCH_SW** option of the **NRDUCellAlgoSwitch.ServiceDiffSwitch** parameter. This function takes effect only for UEs performing sparse-packet services. The evaluation period for uplink preallocation for sparse-packet services is specified by **NRDUCellServExp.SparseServEvalPrd**. This parameter takes effect only when the **SPARSE_DIFF_SCH_SW** option of the **NRDUCellAlgoSwitch.ServiceDiffSwitch**

parameter is selected. It is recommended that this parameter be set to the same value as the expected period to be guaranteed for sparse-packet services.

- A smaller value of this parameter results in a shorter time for entry into and exit from uplink preallocation for sparse-packet services, reducing resource consumption by sparse-packet services.
- A larger value of this parameter results in a longer time for entry into and exit from uplink preallocation for sparse-packet services, increasing resource consumption by sparse-packet services.

In low-frequency scenarios, the **SPARSE_SERV_IDENT_BEARER_SW** option of the **NRDUCellServExp.ServiceExpAlgoSwitch** parameter specifies the bearer policy for identifying UEs running sparse-packet services.

- If this option is deselected, only UEs with a single bearer can be identified as UEs running sparse-packet services.
- If this option is selected, a UE with two bearers (one with a QCI of 5) can be identified as a UE running sparse-packet services.

In high-frequency scenarios, if the **SPARSE_DIFF_PROC_SW** option of the **gNodeBParam.ServiceDiffProcSwitch** parameter and the **SPARSE_DIFF_SCH_SW** option of the **NRDUCellAlgoSwitch.ServiceDiffSwitch** parameter are both selected, only UEs with a single bearer can be identified as UEs performing sparse-packet services and uplink preallocation for sparse-packet services is not supported.

5.10.2 Network Analysis

5.10.2.1 Benefits

- Uplink basic preallocation is a basic function that serves as the gain baseline of uplink smart preallocation. Uplink smart preallocation results in less UE power consumption and lower uplink PRB usage than basic preallocation.

The E2E ping delay with only uplink preallocation for sparse-packet services in effect is shorter than that with only uplink smart preallocation in effect.

- When uplink preallocation for sparse-packet services takes effect, a smaller value of the **NRDUCellServExp.SparseServEvalPrd** parameter results in a shorter time for entry into and exit from uplink preallocation for sparse-packet services, reducing resource consumption by sparse-packet services. When air interface resources are insufficient, the data rate of UEs not running sparse-packet services increases.

5.10.2.2 Impacts

The impacts between this function and other functions are described in [5.10.3.2.3 Function Impacts](#).

When uplink preallocation for sparse-packet services is in effect, UEs running sparse-packet services always occupy uplink resources for preallocation. As a result, the uplink peak rate of other UEs cannot be reached.

When uplink preallocation for sparse-packet services is in effect, a larger value of the **NRDUCellServExp.SparseServEvalPrd** parameter results in longer periods for

entry into and exit from uplink preallocation for sparse-packet services, as well as increased resource consumption by sparse-packet services. When air interface resources are insufficient, the data rate of UEs not running sparse-packet services decreases.

After uplink preallocation is enabled, the following may increase:

- Number of uplink scheduling times in each modulation scheme:
N.UL.SCH.QPSK.TB, N.UL.SCH.16QAM.TB, N.UL.SCH.64QAM.TB, and N.UL.SCH.256QAM.TB
- Total volume of uplink data received at the MAC layer in a cell:
N.ThpVol.UL.Cell
- Maximum total volume of uplink data received at the MAC layer in a cell:
N.ThpVol.UL.Cell.Max
- Average number of PRBs used by UEs in uplink preallocation:
N.PR.B.UL.Prealloc.Used.Avg
- Total duration of uplink data transmission at the MAC layer in a cell:
N.ThpTime.UL.Cell

The values of the following indicators fluctuate depending on traffic model:

- Average number of used uplink PRBs: **N.PR.B.UL.Used.Avg**
- Average uplink cell throughput: **Cell Uplink Average Throughput (DU)**

5.10.3 Requirements

5.10.3.1 Licenses

There are no license requirements for basic functions.

5.10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.10.3.2.1 Prerequisite Functions

None

5.10.3.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	gNodeB MAC padding test	STRGNBMACPADDINGTEST	None	Uplink preallocation for sparse-packet services does not take effect when a padding test is started at the MAC layer of the gNodeB.
Low-frequency TDD High-frequency TDD	Uplink preallocation	• UL_BASIC_PREALLOCATION_SWITCH option of the NRDUCellPusch.UlPreallocationSwitch parameter • UL_SMART_PREALLOCATION_SWITCH option of the NRDUCellPusch.UlPreallocationSwitch parameter	<i>Uplink Scheduling</i>	Uplink basic preallocation (controlled by the UL_BASIC_PREALLOCATION_SWITCH option of the NRDUCellPusch.UlPreallocationSwitch parameter) and uplink smart preallocation (controlled by the UL_SMART_PREALLOCATION_SWITCH option of the NRDUCellPusch.UlPreallocationSwitch parameter) are mutually exclusive, indicating that they cannot be both enabled.

5.10.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	DRX	BASIC_DRX_SWITCH option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter	<i>DRX</i>	- After uplink basic preallocation is enabled (by selecting the UL_BASIC_PREALLOCATION_SWITCH option of the NRDUCellPusch.UlPreallocationSwitch parameter), UEs cannot enter the sleep state. - After uplink smart preallocation is enabled (by selecting the UL_SMART_PREALLOCATION_SWITCH option of the NRDUCellPusch.UlPreallocationSwitch parameter), UEs cannot enter the sleep state within the uplink smart preallocation duration. - After uplink preallocation for sparse-packet services is enabled (by selecting both the SPARSE_DIFF_PROC_SW option of the gNodeBParam.ServiceDiffProcSwitch parameter and the SPARSE_DIFF_SCH_SW option of the NRDUCellAlgoSwitch.ServiceDiffSwitch parameter), UEs performing sparse-packet services cannot enter the sleep state.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-specific scheduling based on the target IBLER	• UL_IBLER_QCI_SW option of the gNBDUSchParamGroup.ScheduleOptSwitch parameter • DL_IBLER_QCI_SW option of the gNBDUSchParamGroup.ScheduleOptSwitch parameter	<i>Service-differentiated Experience Guarantee</i>	QCI-specific scheduling based on the target IBLER does not take effect for UEs enabled with uplink preallocation for sparse-packet services.
Low-frequency TDD	QCI-based uplink basic preallocation	UL_PREALLOCATION_QCI_SW option of the gNBDUSchParamGroup.ScheduleOptSwitch parameter	<i>Service-differentiated Experience Guarantee</i>	For QCI-specific services, when both uplink basic preallocation and QCI-based uplink basic preallocation are enabled, the following rules apply: • If the SERV_DIFF_POLICY_SW option of the NRDUCellAlgoSwitch.LowLatencySwitch parameter is selected, uplink basic preallocation does not take effect. • If the SERV_DIFF_POLICY_SW option of the NRDUCellAlgoSwitch.LowLatencySwitch parameter is deselected, both functions can take effect but are mutually exclusive at the TTI level. Within a single TTI, QCI-based uplink basic preallocation preferentially takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-based uplink smart preallocation	UL_SMART_PREALLOC_QCI_SW option of the gNB-DUSchParamGroup.ScheduleOptSwitch parameter	<i>Service-differentiated Experience Guarantee</i>	For QCI-specific services, when both uplink smart preallocation and QCI-based uplink smart preallocation are enabled, the following rules apply: • If the SERV_DIFF_POLICY_SW option of the NR-DUCellAlgoSwitch.LowLateNcySwitch parameter is selected, uplink smart preallocation does not take effect. • If the SERV_DIFF_POLICY_SW option of the NR-DUCellAlgoSwitch.LowLateNcySwitch parameter is deselected, both functions can take effect but are mutually exclusive at the TTI level. Within a single TTI, QCI-based uplink smart preallocation preferentially takes effect.
Low-frequency TDD	PUCCH channel management	SR_PERIOD_ADAPT_SWITCH option of the NR-DUCellPucch.SrResourceAllocationSwitch parameter SR_PERIOD_OPT_SWITCH option of the NR-DUCellPucch.SrResourceAllocationSwitch parameter NR-DUCellPucch.SrPeriod	<i>Channel Management</i>	If the SR period is shortened due to reconfiguration or SR period adaptation, the number of times basic and smart preallocations are performed decreases. This reduces the average number of PRBs used by uplink preallocation UEs (N.PRB.UL.Prealloc.Used.Avg). If there is a large volume of data configured for preallocation, the number of uplink padding bits decreases, reducing the average number of PRBs used by DRBs on the PUSCH (N.PRB.UL.DrbUsed.Avg) and the average uplink cell throughput (Cell Uplink Average Throughput (DU)).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	- The low latency and high reliability function does not apply to UEs enabled with uplink preallocation for sparse-packet services. - Uplink basic preallocation does not take effect when the low latency and high reliability function is enabled for a specific QCI. - Uplink smart preallocation does not take effect when the low latency and high reliability function is enabled for a specific QCI.
Low-frequency TDD High-frequency TDD	NSA uplink preallocation	NR: NSA_DC_PREALLOCATION_SW option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter LTE: NSA_DC_PREALLOCATION_SW option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Basics</i>	In NSA networking, if NSA uplink preallocation is enabled (by selecting the NSA_DC_PREALLOCATION_SW option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter on the NR side or the NSA_DC_PREALLOCATION_SW option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter on the LTE side) and the SPARSE_DIFF_SCH_SW option of the NRDUCellAlgoSwitch.ServiceDiffSwitch parameter is selected, uplink preallocation for sparse-packet services can take effect.
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapBasedFunSw parameter	<i>RedCap</i>	When RedCap Introduction is enabled, uplink basic preallocation does not take effect on RedCap UEs. The UL_REDCAP_PREALLOCATION_SW option of the NRDUCellPusch.UlPreallocationSwitch parameter must also be selected so that uplink basic preallocation takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapBasedFunSw parameter	<i>RedCap</i>	When RedCap Introduction is enabled, uplink smart preallocation does not take effect on RedCap UEs. The UL_REDCAP_PREALLOCATION_SW option of the NRDUCellPusch.UlPreallocationSwitch parameter must also be selected so that uplink smart preallocation takes effect.
Low-frequency TDD	Device-pipe synergy preallocation	UL_PREFERENTIAL_SCH_SW option of the NRDUCellQciBearer.ServiceDiffAlgoSwitch parameter	<i>FWA</i>	Uplink basic preallocation does not take effect for UEs with device-pipe synergy preallocation in effect.
Low-frequency TDD	Device-pipe synergy preallocation	UL_PREFERENTIAL_SCH_SW option of the NRDUCellQciBearer.ServiceDiffAlgoSwitch parameter	<i>FWA</i>	Uplink smart preallocation does not take effect for UEs with device-pipe synergy preallocation in effect.
Low-frequency TDD	FSD transmission mode	• NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL • NRDUCellMltiTrp.DataTransMode set to FSD_MODE	<i>Hyper Cell</i>	• If the FSD transmission mode is used for a hyper cell and uplink basic preallocation is enabled, the number of preallocation times and the average uplink MCS index may decrease due to staggering of overlapping areas. • If the FSD transmission mode is used for a hyper cell and uplink smart preallocation is enabled, the number of preallocation times and the average uplink MCS index may decrease due to staggering of overlapping areas.

5.10.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.10.3.4 Others

None

5.10.4 Operation and Maintenance

5.10.4.1 Data Configuration

5.10.4.1.1 Data Preparation

[Table 5-10](#) and [Table 5-11](#) describe the parameters used for function activation and optimization, respectively.

Table 5-10 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink Preallocation Switch	NRDUCellPusch.UlPreallocationSwitch	UL_BASIC_PREALLOCATION_SWITCH	Use the default value.
Uplink Preallocation Switch	NRDUCellPusch.UlPreallocationSwitch	UL_SMART_PREALLOCATION_SWITCH	Select this option.
Service Differentiation Process Switch	gNodeBParam.ServiceDiffProcSwitch	SPARSE_DIFF_PROC_SW	Use the default value.
Service Differentiation Switch	NRDUCellAlgoSwitch.ServiceDiffSwitch	SPARSE_DIFF_SCH_SW	Use the default value.
Service Experience Algorithm Switch	NRDUCellServExp.ServiceExpAlgoSwitch	SPARSE_SERV_IDENTITY_BEARER_SW	Use the default value.
Sparse Service Evaluation Period	NRDUCellServExp.SparseServEvalPrd	N/A	Use the default value.

Table 5-11 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
Minimum Preallocation Period	NRDUCellPusch.MinPreallocationPeriod	Use the default value.
Prescheduled UE Num Upper Limit	NRDUCellPusch.PreschUeNumUpperLimit	Use the default value.
Preallocation Size	NRDUCellPusch.PreallocationSize	Use the recommended value.
Uplink Smart Preallocation Duration	NRDUCellPusch.UISmartPreallocationDur	Use the recommended value.

5.10.4.1.2 Using MML Commands

Before using MML commands, refer to [5.10.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling the uplink basic preallocation function (When this function is enabled, the user-perceived delay decreases, but the uplink interference and UE power consumption increase. This function is mutually exclusive with the uplink smart preallocation function. Therefore, it is recommended that this function be disabled and the uplink smart preallocation function be enabled.)
MOD NRDUCELLPUSCH: NrDuCellId=0,
UIPreallocationSwitch=UL_BASIC_PREALLOCATION_SWITCH-1&UL_SMART_PREALLOCATION_SWITCH-0;
//Enabling uplink smart preallocation
MOD NRDUCELLPUSCH: NrDuCellId=0,
UIPreallocationSwitch=UL_SMART_PREALLOCATION_SWITCH-1&UL_BASIC_PREALLOCATION_SWITCH-0;
//When there are UEs for which two bearers (one with a QCI of 5) are set up by default in a cell, the bearer policy switch for identifying UEs running sparse-packet services must be turned on so that these UEs can be identified as UEs running sparse-packet services.
MOD NRDUCELLSERVEXP: NrDuCellId=0, ServiceExpAlgoSwitch=SPARSE_SERV_IDENT_BEARER_SW-1;
//Selecting both the SPARSE_DIFF_PROC_SW and SPARSE_DIFF_SCH_SW options to enable uplink preallocation for sparse-packet services (When this function is enabled, UEs running sparse-packet services always occupy uplink resources for the purpose of preallocation, making the uplink rates of other UEs fail to reach the peak values. Therefore, it is recommended that this function be disabled.)
MOD GNODEBPARAM: ServiceDiffProcSwitch=SPARSE_DIFF_PROC_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, ServiceDiffSwitch=SPARSE_DIFF_SCH_SW-1;
//Setting the evaluation period for uplink preallocation for sparse-packet services
MOD NRDUCELLSERVEXP: NrDuCellId=0, SparseServEvalPrd=10;
```

Optimization Command Examples

```
//Configuring the minimum preallocation period
MOD NRDUCELLPUSCH: NrDuCellId=0, MinPreallocationPeriod=5;
//Configuring the maximum number of UEs in the preallocation queue
MOD NRDUCELLPUSCH: NrDuCellId=0, PreschUeNumUpperLimit=2;
//Configuring the amount of UE data to be scheduled for uplink preallocation
MOD NRDUCELLPUSCH: NrDuCellId=0, PreallocationSize=100;
//Configuring uplink smart preallocation duration
MOD NRDUCELLPUSCH: NrDuCellId=0, UISmartPreallocationDur=50;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink basic preallocation
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPreallocationSwitch=UL_BASIC_PREALLOCATION_SWITCH-0;
//Disabling uplink smart preallocation
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPreallocationSwitch=UL_SMART_PREALLOCATION_SWITCH-0;
//Turning off the bearer policy switch for identifying UEs running sparse-packet services
MOD NRDUCELLSERVEXP: NrDuCellId=0, ServiceExpAlgoSwitch=SPARSE_SERV_IDENT_BEARER_SW-0;
//Disabling uplink preallocation for sparse-packet services
MOD GNODEBPARAM: ServiceDiffProcSwitch=SPARSE_DIFF_PROC_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, ServiceDiffSwitch=SPARSE_DIFF_SCH_SW-0;
```

5.10.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.10.4.2 Activation Verification

If the average number of PRBs used by UEs in uplink preallocation (measured by **N.PRB.UL.Prealloc.Used.Avg**) changes from 0 to a value greater than 0 after the uplink basic preallocation function is enabled, this function has taken effect.

If the average number of PRBs used by UEs in uplink preallocation (measured by **N.PRB.UL.Prealloc.Used.Avg**) changes from 0 to a value greater than 0 after the uplink smart preallocation function is enabled, this function has taken effect.

Uplink preallocation for sparse-packet services is a basic function and does not require separate activation verification.

5.10.4.3 Network Monitoring

- Observe **N.UL.SCH.QPSK.TB** to **N.UL.SCH.256QAM.TB** to obtain the number of uplink scheduling times when a modulation scheme is used.
- Observe **N.ThpVol.UL.Cell** to obtain the total volume of uplink data received at the MAC layer in a cell.
- Observe **N.ThpVol.UL.Cell.Max** to obtain the maximum total volume of uplink data received at the MAC layer in a cell.
- Observe **N.PRB.UL.Prealloc.Used.Avg** to obtain the average number of PRBs used by UEs in uplink preallocation.

5.11 Proactive Scheduling for Uplink Signaling

5.11.1 Principles

Currently, the gNodeB cannot receive scheduling requests from some UEs for a long time during RRC or service setup in high-frequency NR TDD scenarios. This causes UE access failures or service drops especially in the event of heavy traffic, and then a low RRC setup success rate. As such, proactive scheduling for uplink signaling is introduced. This function includes uplink RRC preallocation and uplink QoS flow preallocation.

Uplink RRC Preallocation

During contention-based random access initiated by a UE, the UE needs to respond with an RRCSetupComplete message (Msg5) to an RRCSetup message (Msg4) from the gNodeB. If the gNodeB fails to receive a scheduling request (SR) from the UE, the UE cannot obtain scheduling resources, in which case the RRC response signaling of the UE cannot be scheduled and sent to the gNodeB.

In this regard, the uplink RRC preallocation function is introduced. This function is controlled by the **UL_RRC_PREALLOCATION_SW** option of the **NRDUCellPusch.UlPreallocationSwitch** parameter. If this option is selected, the gNodeB proactively schedules a UE if the gNodeB does not receive an SR from the UE within 500 ms after sending RRC signaling to the UE. This allows the UE to send RRC response signaling to the gNodeB.

Uplink QoS Flow Preallocation

After a UE completes initial access, the service setup request phase starts, where services are mapped to QCI-specific bearers of the gNodeB. After the gNodeB initiates a service setup procedure, if it fails to receive a SR from the UE, the UE cannot obtain scheduling resources. As a result, the service setup response signaling of the UE cannot be scheduled and sent to the gNodeB.

In this regard, the uplink QoS flow preallocation function is introduced. This function is controlled by the **UL_QOS_FLOW_PREALLOCATION_SW** option of the **NRDUCellPusch.UlPreallocationSwitch** parameter. If this option is selected, the gNodeB proactively schedules a UE if the gNodeB does not receive an SR from the UE within 500 ms after sending service setup signaling to the UE. This allows the UE to send service setup response signaling to the gNodeB.

5.11.2 Network Analysis

5.11.2.1 Benefits

Uplink RRC preallocation: If the function switch is turned on, more resources are allocated for scheduling of uplink signaling, increasing the RRC setup success rate. Higher interference to SR transmission by UEs indicates larger benefits.

Uplink QoS flow preallocation: If the function switch is turned on, more resources are allocated for scheduling of uplink signaling, increasing the QoS flow setup

success rate. Higher interference to SR transmission by UEs indicates larger benefits.

5.11.2.2 Impacts

The impacts between this function and other functions are described in [5.11.3.2.3 Function Impacts](#).

Enabling either uplink RRC preallocation or uplink QoS flow preallocation results in a decrease in the **Cell Uplink Average Throughput (DU)**.

5.11.3 Requirements

5.11.3.1 Licenses

There are no license requirements for basic functions.

5.11.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.11.3.2.1 Prerequisite Functions

None

5.11.3.2.2 Mutually Exclusive Functions

None

5.11.3.2.3 Function Impacts

None

5.11.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

5.11.3.4 Networking

Uplink RRC preallocation and uplink QoS flow preallocation apply to SA networking and NSA and SA hybrid networking.

5.11.4 Operation and Maintenance

5.11.4.1 Data Configuration

5.11.4.1.1 Data Preparation

Table 5-12 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-12 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink Preallocation Switch	NRDUCellPusch.UlPreallocationSwitch	UL_RRC_PREALLOCATION_SW	Select this option.
Uplink Preallocation Switch	NRDUCellPusch.UlPreallocationSwitch	UL_QOS_FLOW_PREALLOCATION_SW	Select this option.

5.11.4.1.2 Using MML Commands

Before using MML commands, refer to **5.11.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink RRC preallocation  
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPreallocationSwitch=UL_RRC_PREALLOCATION_SW-1;  
//Enabling uplink QoS flow preallocation  
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPreallocationSwitch=UL_QOS_FLOW_PREALLOCATION_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink RRC preallocation  
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPreallocationSwitch=UL_RRC_PREALLOCATION_SW-0;  
//Disabling uplink QoS flow preallocation  
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPreallocationSwitch=UL_QOS_FLOW_PREALLOCATION_SW-0;
```

5.11.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.11.4.2 Activation Verification

Uplink RRC preallocation is a basic function and does not require separate activation verification.

Uplink QoS flow preallocation is a basic function and does not require separate activation verification.

5.11.4.3 Network Monitoring

Monitor **Cell Uplink Average Throughput (DU)** to obtain the average uplink cell throughput.

Monitor **User Uplink Average Throughput (DU)** to obtain the average uplink UE throughput.

5.12 MPS-based Preferential Access and Scheduling

5.12.1 Principles

In heavy-traffic scenarios, the increasing number of UEs may lead to delays in uplink scheduling during RRC connection setup or service setup. As a result, UEs fail to access the network. To address this issue, media priority service (MPS)-based preferential access and scheduling is introduced. For UEs whose RRC connection setup requests contain the cause value "mps-PriorityAccess", this function enhances the scheduling efficiency during RRC connection setup and service request setup. In this way, these UEs can preferentially access the network. This function is controlled by the **MPS_PRIORITY_ACCESS_SCH_SW** option of the

NRDUCellPusch.PuschAlgoExtSwitch parameter. This function applies only to low-frequency TDD and FDD cells.

5.12.2 Network Analysis

5.12.2.1 Benefits

On a heavy-load network, MPS-based preferential access and scheduling allows UEs whose RRC connection setup requests contain the cause value "mps-PriorityAccess" to preferentially access the network.

5.12.2.2 Impacts

This function may have the following impacts:

- These indicators fluctuate: RRC connection setup success rate, QoS flow setup success rate, radio access success rate, and service drop rate. The **Cell Uplink Average Throughput (DU)** decreases.
- The access success rate and service drop rate of UEs other than those with the cause value "mps-PriorityAccess" may fluctuate.

5.12.3 Requirements

5.12.3.1 Licenses

There are no license requirements for basic functions.

5.12.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.12.3.2.1 Prerequisite Functions

None

5.12.3.2.2 Mutually Exclusive Functions

None

5.12.3.2.3 Function Impacts

None

5.12.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.12.3.4 Others

None

5.12.4 Operation and Maintenance

5.12.4.1 Data Configuration

5.12.4.1.1 Data Preparation

Table 5-13 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-13 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
PUSCH Algorithm Extension Switch	NRDUCellPusch.PuschA lgoExtSwitch	MPS_PRIORITY_A CCESS_SCH_SW	Use the default value.

5.12.4.1.2 Using MML Commands

Before using MML commands, refer to [5.12.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling MPS-based preferential access and scheduling  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschAlgoExtSwitch=MPS_PRIORITY_ACCESS_SCH_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling MPS-based preferential access and scheduling  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschAlgoExtSwitch=MPS_PRIORITY_ACCESS_SCH_SW-0;
```

5.12.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.12.4.2 Activation Verification

MPS-based preferential access and scheduling is a basic function and does not require activation verification.

5.12.4.3 Network Monitoring

The indicators listed in [5.12.2.2 Impacts](#) can be used to monitor the benefits and impacts of this function.

5.13 Aperiodic CSI-RS Optimization in Heavy-Load Scenarios

5.13.1 Principles

Aperiodic CSI-RS reporting enables the gNodeB to flexibly obtain channel state information as required. Given this, "aperiodic CSI-RS optimization in heavy-load scenarios" is introduced. With this function, aperiodic CSI-RS reporting remains

effective even under heavy load, fully utilizing aperiodic CSI-RS capabilities. This function is controlled by the **APER_CSIRS_FLEX_OPT_SW** option of the **NRDUCellCsirs.CsiRsExtSwitch** parameter. This function applies only to low-frequency TDD and FDD cells.

5.13.2 Network Analysis

5.13.2.1 Benefits

After this function takes effect, aperiodic CSI-RS capabilities can be fully used, and more features that depend on aperiodic CSI-RS can take effect. The benefits are as follows:

- The **User Downlink Average Throughput (DU)** increases.
- The service drop rate decreases.
- The uplink and downlink CCE aggregation levels decrease.

5.13.2.2 Impacts

The impacts between this function and other functions are described in [5.13.3.2.3 Function Impacts](#).

This function has the following impacts on the network:

Aperiodic CSI-RS reporting over the PUSCH may decrease the **User Uplink Average Throughput (DU)**.

5.13.3 Requirements

5.13.3.1 Licenses

There are no license requirements for basic functions.

5.13.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.13.3.2.1 Prerequisite Functions

None

5.13.3.2.2 Mutually Exclusive Functions

None

5.13.3.2.3 Function Impacts

None

5.13.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.13.3.4 Others

None

5.13.4 Operation and Maintenance

5.13.4.1 Data Configuration

5.13.4.1.1 Data Preparation

[Table 5-14](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-14 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
CSI-RS Extend Switch	NRDUCellCsirs.Cs iRsExtSwitch	APER_CSIRS_FLEX _OPT_SW	You are advised to select this option when the system is heavily loaded.

5.13.4.1.2 Using MML Commands

Before using MML commands, refer to [5.13.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling aperiodic CSI-RS optimization in heavy-load scenarios  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsRsExtSwitch=APER_CSIRS_FLEX_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling aperiodic CSI-RS optimization in heavy-load scenarios  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiRsExtSwitch=APER_CSIRS_FLEX_OPT_SW-0;
```

5.13.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.13.4.2 Activation Verification

If the **User Downlink Average Throughput (DU)** increases or the service drop rate decreases after this function is enabled, this function has taken effect.

5.13.4.3 Network Monitoring

The indicators listed in [5.13.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

6 PUSCH Time-Frequency Resource Allocation

PUSCH time-frequency resources include both time- and frequency-domain resources. PUSCH resource allocation is to allocate resources of appropriate sizes and positions to UEs.

PUSCH Frequency-Domain Resource Allocation

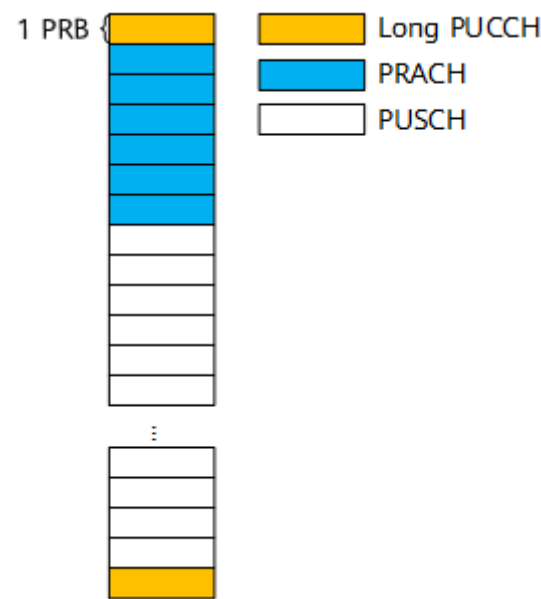
- Determining the resource size

An uplink scheduler determines the RB resource size required by a UE in a slot based on information such as the BSR, measured SINR, and power headroom status reported by the UE. Given the same data volume requested by multiple UEs, the scheduler allocates fewer RBs to the UEs with better channel quality. For details about the power headroom, see *Power Control*.

Using the long PUCCH format

PUSCH frequency-domain resources include the RBs that remain available after RBs for the long PUCCH and the PRACH have been allocated, as shown in [Figure 6-1](#).

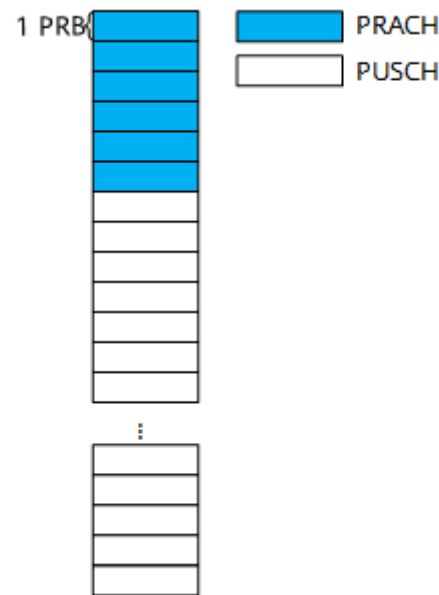
Figure 6-1 PUSCH frequency-domain resources



Using the short PUCCH format

PUSCH frequency-domain resources are the RBs that remain available after RBs for the PRACH have been allocated, as shown in [Figure 6-2](#).

Figure 6-2 Frequency-domain resources



- Determining the resource position
The scheduled RB positions are determined based on the resource allocation result. After RBs are allocated to other UEs that have higher scheduling priorities, the RBs that remain available are selected.

In addition to the preceding basic procedures, uplink scheduling also needs to optimize PUSCH frequency-domain resource allocation policies based on the actual data transmission process in different service scenarios to ensure better

transmission performance. [Table 6-1](#) lists optimization functions for PUSCH frequency-domain resource allocation.

Table 6-1 Optimization functions for PUSCH frequency-domain resource allocation

Function Name	Low-Frequency NR TDD	High-Frequency NR TDD	Reference
Uplink RB estimation optimization	√	√	6.1 Uplink RB Estimation Optimization
Even uplink RB division optimization	√	-	6.2 Even Uplink RB Division Optimization
Uplink retransmission UE scheduling optimization	√	-	6.3 Uplink Retransmission UE Scheduling Optimization
PUSCH and PRACH FDM	-	√	6.4 PUSCH and PRACH FDM
Uplink consecutive scheduling and non-consecutive scheduling	√	√	6.5 Uplink Consecutive Scheduling and Non-Consecutive Scheduling
Uplink frequency selective scheduling and non-frequency-selective scheduling	√	√	6.6 Uplink Frequency Selective Scheduling and Non-Frequency-Selective Scheduling
Uplink interference randomization-based scheduling	-	√	6.7 Uplink Interference Randomization-based Scheduling

Function Name	Low-Frequency NR TDD	High-Frequency NR TDD	Reference
Adaptive HARQ-ACK feedback on downlink transmission (high-frequency TDD)	-	√	6.9 Adaptive HARQ-ACK Feedback on Downlink Transmission (High-Frequency TDD)
Uplink scheduling enhancement	√	-	6.10 Uplink Scheduling Enhancement
PUSCH and PUCCH RB adaptation	√	-	6.12 PUSCH and PUCCH RB Adaptation

NOTE

In the preceding table, "√" indicates that the function is supported in the specified RAT, while "-" indicates that the function is not supported.

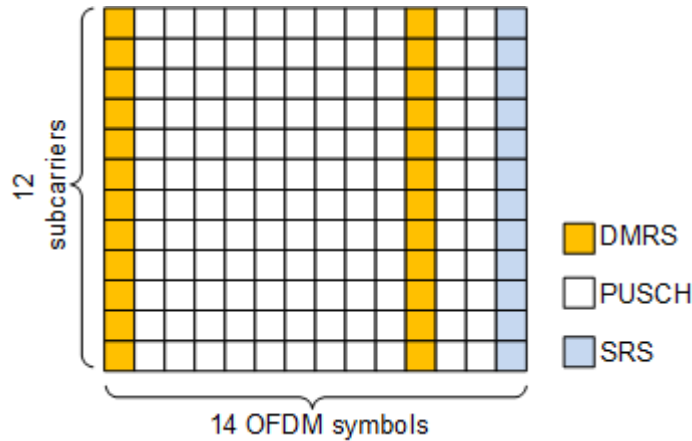
PUSCH Time-Domain Resource Allocation

The scheduler needs to indicate the scheduled slot and both the start and end positions of the scheduled OFDM symbols in the slot to a UE. PUSCH time-domain resource allocation is classified into type A and type B, depending on the combination of the start symbol position and scheduling length. For details, see section 6.1.2.1 "Resource allocation in time domain" in 3GPP TS 38.214 V15.4.0.

Using the long PUCCH format

PUSCH time-domain resources include the OFDM symbols that remain available after the SRS and DMRS have been scheduled, as shown in [Figure 6-3](#).

Figure 6-3 PUSCH time-domain resources



Using the short PUCCH format

Figure 6-4 shows the PUSCH time-domain resources using the one-symbol short PUCCH as an example in low-frequency scenarios.

Figure 6-4 Time-domain resources (low frequency bands)

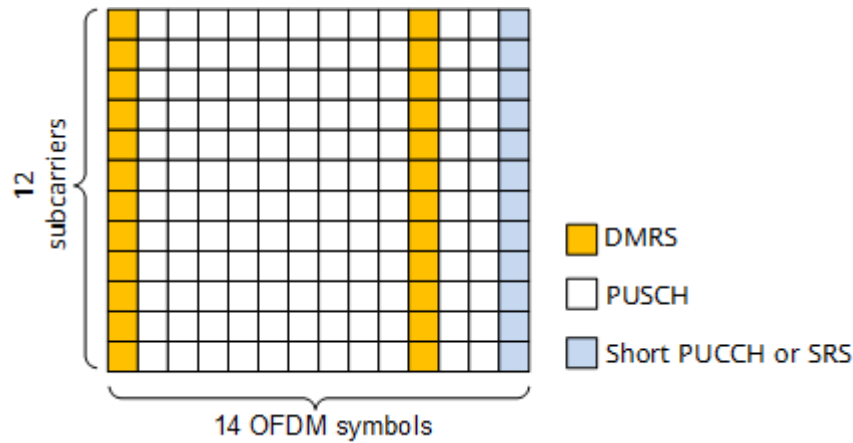
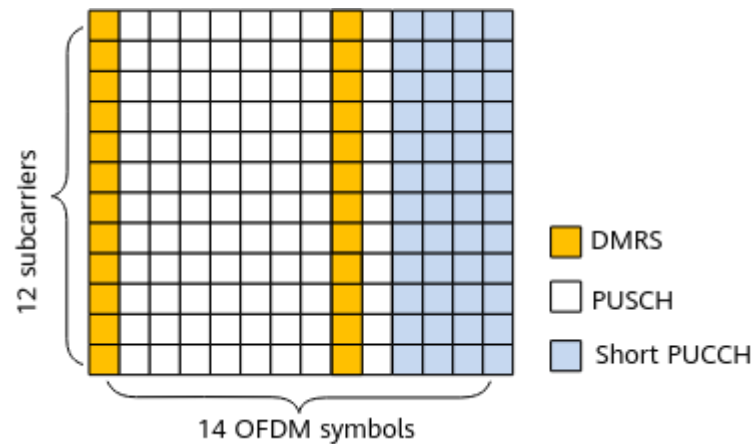


Figure 6-5 shows the PUSCH time-domain resources using the uplink slot as an example in high-frequency scenarios when the short PUCCH occupies five symbols (one symbol, symbol 12 by default, in the self-contained slot, and four symbols in the uplink slot).

Figure 6-5 Time-domain resources (high frequency bands)



In addition to the preceding basic procedures, uplink scheduling also needs to optimize PUSCH time-domain resource allocation policies based on the actual data transmission process in different service scenarios to ensure better transmission performance. Table 6-2 lists optimization functions for PUSCH time-domain resource allocation.

Table 6-2 Optimization functions for PUSCH time-domain resource allocation

Function Name	Low - Frequency NR TDD	High-Frequency NR TDD	Reference	
PUCCH symbol sharing with the PUSCH	-	√	-	6.11 PUCCH Symbol Sharing with the PUSCH
Uplink scheduling of multi-SLIV UEs in the same slot	√	√	√	6.13 Uplink Scheduling of Multi-SLIV UEs in the Same Slot

Function Name	Low - Frequency NR TDD	High-Frequency NR TDD	Reference
Uplink DCI extension-based scheduling (low-frequency TDD)	√	-	6.14 Uplink DCI Extension-based Scheduling (Low-Frequency TDD)

 NOTE

In the preceding table, "√" indicates that the function is supported in the specified RAT, while "-" indicates that the function is not supported.

6.1 Uplink RB Estimation Optimization

6.1.1 Principles

When the power of a UE is insufficient, the uplink RB estimation optimization function can be used to adjust the estimated number of RBs for the UE, increasing the average uplink user-perceived rate. This function is controlled by the **UL_RB_ESTIMATE_OPT_SW** option of the **NRDUCellPusch.PuschPerformanceSwitch** parameter and is recommended in light- and medium-load scenarios.

6.1.2 Network Analysis

6.1.2.1 Benefits

If the function switch is turned on, the average uplink UE throughput (**User Uplink Average Throughput (DU)**) and the RB usage of the cell increase.

6.1.2.2 Impacts

The impacts between this function and other functions are described in [6.1.3.2.3 Function Impacts](#).

For cells with light or medium load, this function can be enabled to prevent data fragmentation caused by UE power insufficiency. This function increases the average uplink user-perceived rate but decreases the average MCS index for uplink scheduling.

6.1.3 Requirements

6.1.3.1 Licenses

There are no license requirements for basic functions.

6.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.1.3.2.1 Prerequisite Functions

None

6.1.3.2.2 Mutually Exclusive Functions

None

6.1.3.2.3 Function Impacts

None

6.1.3.3 Hardware

For details, see [4.3.3 Hardware](#).

6.1.3.4 Others

None

6.1.4 Operation and Maintenance

6.1.4.1 Data Configuration

6.1.4.1.1 Data Preparation

Table 6-3 describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

Table 6-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
PUSCH Performance Switch	NRDUCellPusch.PuschPerformanceSwitch	UL_RB_ESTIMATE_OPT_SW	You are advised to select this option in light- or medium-load scenarios (with the uplink PRB usage being less than 60%) and deselect this option in other scenarios.

6.1.4.1.2 Using MML Commands

Before using MML commands, refer to **6.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for uplink RB estimation optimization
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL_RB_ESTIMATE_OPT_SW-1;
```

Deactivation Command Examples

```
//Turning off the switch for uplink RB estimation optimization
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL_RB_ESTIMATE_OPT_SW-0;
```

6.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.1.4.2 Activation Verification

In light-or medium-load scenarios where the uplink PRB usage is less than 60%, if the average uplink UE throughput (**User Uplink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

6.1.4.3 Network Monitoring

The indicators listed in [6.1.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

6.2 Even Uplink RB Division Optimization

6.2.1 Principles

On the base station side, RB resources can be evenly allocated to some UEs, improving resource allocation fairness and cell throughput. When the **RB_DIVISION_OPT_SW** option of the **NRDUCellPusch.PuschPerformanceSwitch** parameter is selected, the maximum number of UEs for which the scheduler can evenly allocate RBs is specified by the **NRDUCellPusch.RbDividedUserUpperLimit** parameter. When the **RB_DIVISION_OPT_SW** option of the **NRDUCellPusch.PuschPerformanceSwitch** parameter is deselected, the scheduler can select a maximum of four UEs in the current slot for even RB division. This function is supported only in low frequency bands, not in high frequency bands.

6.2.2 Network Analysis

6.2.2.1 Benefits

After this optimization takes effect:

- When the cell is heavily loaded, a larger number of UEs involved in uplink even RB division results in a larger average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) and a lower average uplink UE throughput (**User Uplink Average Throughput (DU)**).
- When the cell is heavily loaded, a smaller number of UEs involved in uplink even RB division results in a lower average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) and a larger average uplink UE throughput (**User Uplink Average Throughput (DU)**).

6.2.2.2 Impacts

The impacts between this function and other functions are described in [6.2.3.2.3 Function Impacts](#).

6.2.3 Requirements

6.2.3.1 Licenses

There are no license requirements for basic functions.

6.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.2.3.2.1 Prerequisite Functions

None

6.2.3.2.2 Mutually Exclusive Functions

None

6.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink experience-based scheduling	UL_EXP_BASED_SCH_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	When both even uplink RB division optimization and uplink experience-based scheduling are enabled, benefits delivered by uplink experience-based scheduling decrease as the number of UEs to which RBs are evenly allocated increases. This is because even uplink RB division optimization affects RB resource allocation.

6.2.3.3 Hardware

For details, see [4.3.3 Hardware](#).

6.2.3.4 Others

None

6.2.4 Operation and Maintenance

6.2.4.1 Data Configuration

6.2.4.1.1 Data Preparation

[Table 6-4](#) describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

Table 6-4 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
PUSCH Performance Switch	NRDUCellPusch.PuschPerformanceSwitch	RB_DIVISION_OPT_SW	You are advised to deselect this option in low-frequency scenarios.
RB Divided User Upper Limit	NRDUCellPusch.RbDividedUserUpperLimit	None	Use the recommended value.

6.2.4.1.2 Using MML Commands

Before using MML commands, refer to [6.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning off the switch for even uplink RB division optimization in low-frequency scenarios
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=RB_DIVISION_OPT_SW-0;
//Configuring the maximum number of UEs involved in the optimization of even RB division
MOD NRDUCELLPUSCH: NrDuCellId=0, RbDividedUserUpperLimit=4;
```

Deactivation Command Examples

Deactivation command examples are not involved.

6.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.2.4.2 Activation Verification

None

6.2.4.3 Network Monitoring

None

6.3 Uplink Retransmission UE Scheduling Optimization

6.3.1 Principles

For UEs requiring uplink retransmission — uplink retransmission UEs for short, a certain retransmission combining gain margin is reserved during retransmission resource allocation. This wastes allocated RB resources to a certain extent. To save on uplink RB resources, the uplink retransmission UE scheduling optimization function is introduced. This function reduces the number of RBs used for the first retransmission of a UE and allocates the saved RBs to other UEs, increasing the average uplink UE throughput (**User Uplink Average Throughput (DU)**). The retransmission-specific RB reduction proportion is configured using the **NRDUCellULSch.UlFirstRetransMinRbPct** parameter. If this parameter is set to **100**, the uplink retransmission UE scheduling optimization function does not take effect. This function applies to low-frequency NR TDD cells and SUL cells.

6.3.2 Network Analysis

6.3.2.1 Benefits

After this function takes effect in low-frequency scenarios, the average uplink UE throughput (**User Uplink Average Throughput (DU)**) increases.

6.3.2.2 Impacts

The impacts between this function and other functions are described in [6.3.3.2.3 Function Impacts](#).

After this function takes effect in low-frequency scenarios, the proportion of uplink retransmissions may increase.

6.3.3 Requirements

6.3.3.1 Licenses

There are no license requirements for basic functions.

6.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.3.2.1 Prerequisite Functions

None

6.3.3.2 Mutually Exclusive Functions

None

6.3.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	None	None	None	None

6.3.3.3 Hardware

For details, see [4.3.3 Hardware](#).

6.3.3.4 Others

None

6.3.4 Operation and Maintenance

6.3.4.1 Data Configuration

6.3.4.1.1 Data Preparation

[Table 6-5](#) describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

Table 6-5 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
UL First Retrans Minimum RB Percentage	NRDUCellULSch. <i>ULFirstRetransMinRbPct</i>	None	If uplink RB resources are limited, you are advised to set this parameter to 50 . In other scenarios, use the default value.

6.3.4.1.2 Using MML Commands

Before using MML commands, refer to [6.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual

network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink retransmission UE scheduling optimization  
MOD NRDUCELLULSCH: NrDuCellId=0, UIFirstRetransMinRbPct=50;
```

Deactivation Command Examples

```
//Disabling uplink retransmission UE scheduling optimization  
MOD NRDUCELLULSCH: NrDuCellId=0, UIFirstRetransMinRbPct=100;
```

6.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.3.4.2 Activation Verification

If the average uplink UE throughput (**User Uplink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

6.3.4.3 Network Monitoring

The indicators listed in [6.3.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

6.4 PUSCH and PRACH FDM

6.4.1 Principles

In high-frequency scenarios, PUSCH and PRACH FDM is introduced to increase the number of available PUSCH RBs. This function is controlled by the **NRDUCellPrach.PuschPrachFdmSwitch** parameter. When this function takes effect, the remaining RBs in the PRACH slots can also be used by the PUSCH, thereby increasing the uplink cell traffic volume.

6.4.2 Network Analysis

6.4.2.1 Benefits

After this function is enabled in high-frequency FWA scenarios, the number of available PUSCH RBs and the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) increase.

6.4.2.2 Impacts

The impacts between this function and other functions are described in [6.4.3.2.3 Function Impacts](#).

In high-frequency FWA scenarios, this function decreases the number of antennas for receiving PRACH signals on the base station. As a result, the PRACH coverage shrinks and the number of UEs that can access the cell may decrease.

6.4.3 Requirements

6.4.3.1 Licenses

There are no license requirements for basic functions.

6.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.4.3.2.1 Prerequisite Functions

None

6.4.3.2.2 Mutually Exclusive Functions

None

6.4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Uplink UE SINR optimization	UL_USER_SINR_OPT_SWITCH option of the NRDUCellPusch .UlPuschAlgoSwitch parameter	None	When both uplink UE SINR optimization and PUSCH and PRACH FDM are enabled, the uplink cell throughput fluctuation caused by PUSCH and PRACH FDM decreases.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Optimal selection of narrow beams for access channels	ACC_CHN_OPT_NARROW_BEAM_SEL_SW option of the NRDUCellTrpMmwavBeam.BeamPerformanceSw parameter	<i>mmWave Beam Management (High-Frequency TDD)</i>	For 4T4R modules, optimal selection of narrow beams for access channels cannot take effect if PUSCH/PRACH FDM Switch is set to BASIC or if PUSCH/PRACH FDM Switch is set to FULL_BUFFER and there are speed turbo UEs or MNT UEs in the sector.

6.4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

6.4.3.4 Others

None

6.4.4 Operation and Maintenance

6.4.4.1 Data Configuration

6.4.4.1.1 Data Preparation

Table 6-6 describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

Table 6-6 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
PUSCH/PRACH FDM Switch	NRDUCellPrach.PuschPrachFdmSwitch	None	In high-frequency scenarios, enabling this function will decrease the number of antennas for receiving PRACH signals on the base station side and causes PRACH coverage to shrink. As such, it is good practice to set this parameter to OFF . If the parameter value differs between co-sector cells, the effective value for this function is unpredictable. Therefore, it is good practice to configure the same parameter value for co-sector cells.

6.4.4.1.2 Using MML Commands

Before using MML commands, refer to [6.4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling PUSCH and PRACH FDM  
MOD NRDUCELLPRACH: NrDuCellId=0, PuschPrachFdmSwitch=BASIC;
```

Deactivation Command Examples

```
//Disabling PUSCH and PRACH FDM  
MOD NRDUCELLPRACH: NrDuCellId=0, PuschPrachFdmSwitch=OFF;
```

6.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.4.2 Activation Verification

If the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

6.4.4.3 Network Monitoring

The indicators listed in [6.4.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

6.5 Uplink Consecutive Scheduling and Non-Consecutive Scheduling

6.5.1 Principles

Section 6.1.2.2 "Resource allocation in frequency domain" in 3GPP TS 38.214 V15.4.0 defines two resource allocation schemes: type 0 and type 1. In type 0, contiguous or non-contiguous resource block groups (RBGs) can be allocated. In type 1, only contiguous RBs can be allocated. Uplink scheduling can use type 1 or type 0 for resource allocation.

- Uplink non-consecutive scheduling: The uplink scheduler searches the available bandwidth range for a group of contiguous RBs or several groups of non-contiguous RBs to be allocated to UEs. This helps fully utilize frequency-domain resources. With uplink non-consecutive scheduling, resources are allocated at the RBG level. For details, see section 6.1.2.2 "Resource allocation in frequency domain" in 3GPP TS 38.214 V15.4.0.
- Uplink consecutive scheduling: The uplink scheduler searches for and allocates a group of contiguous RBs to the UEs from the start position of available bandwidth. Resources in uplink consecutive scheduling are allocated at the RB or RBG level. For details, see section 6.1.2.2 "Resource allocation in frequency domain" in 3GPP TS 38.214 V15.4.0.

NOTE

The number of RBs contained in an RBG varies with the bandwidth. For details, see Table 5.3.2-1 Transmission bandwidth configuration NRB for FR1 in 3GPP TS 38.104 V15.5.0 and Table 6.1.2.2.1-1 Nominal RBG size P in 3GPP TS 38.214 V15.4.0. 3GPP specifications define two RBG size configurations: configuration 1 and configuration 2. Huawei uses configuration 1.

Table 6-7 describes how to select uplink consecutive or non-consecutive scheduling.

Table 6-7 Uplink scheduling mode

Scenario	Scheduling Mode
Low frequency bands	The UL_NON_CON_SCH_SW option of the NRDUCellAlgoSwitch.<i>UlinconsecutiveSchSwitch</i> parameter specifies the resource allocation mode for uplink scheduling. • If this option is deselected, uplink consecutive scheduling is used. • If this option is selected, uplink non-consecutive scheduling is used. In low frequency bands, uplink non-consecutive scheduling or consecutive scheduling is adaptively selected in scenarios where UEs support uplink non-contiguous resource allocation and dynamic switching between contiguous and non-contiguous allocation. (That is, UEs are required to report <i>almostContiguousCP-OFDM-UL</i>, <i>ra-Type0-PUSCH</i>, and <i>dynamicSwitchRA-Type0-1-PUSCH</i> IEs.) Otherwise, uplink consecutive scheduling is used. When the scheduling mode is adapted to uplink consecutive scheduling, if the UL_NONCONSEC_SCH_OPT_SW option of the NRDUCellPusch.<i>PuschPerformanceSwitch</i> parameter is also selected, scheduling is performed at the RB level; if this option is deselected, scheduling is performed at the RBG level. Regardless of whether uplink continuous or non-continuous scheduling is used, if uplink scheduling gain optimization is enabled (by selecting the UL_SCH_GAIN_OPT_SW option of the NRDUCellPusch.<i>PuschAlgoExtSwitch</i> parameter), the gNodeB adjusts the UE resource allocation mode to increase the average uplink UE throughput.
High frequency bands	Only uplink consecutive scheduling is supported.

Table 6-8 describes the uplink scheduling granularity when uplink consecutive scheduling is performed at the RB level.

Table 6-8 Uplink scheduling granularity

Scenario	Scheduling Granularity
Low frequency bands	The scheduling granularity is related to the DMRS configuration type (specified by the NRDUCellPusch.UIDmrsType parameter). - When NRDUCellPusch.UIDmrsType is set to TYPE1: - If uplink scheduling on a per RB basis is enabled, scheduling is performed at a granularity of one RB with a minimum of one RB allocated. The UL_1RB_SCH_SWITCH option of the NRDUCellPusch.UlPuschAlgoSwitch parameter specifies whether to enable uplink scheduling on a per RB basis^a. - If enhancement of uplink scheduling on a per RB basis is enabled together with uplink scheduling on a per RB basis, scheduling is performed at a granularity of one RB with a minimum of two RBs allocated. This enhancement is controlled by the UL_1RB_SCH_ENH_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter^b. - If enhancement of uplink scheduling on a per RB basis is disabled together with uplink scheduling on a per RB basis, scheduling is performed at a granularity of four RBs. - If minimum RB control for cell edge UEs and uplink scheduling on a per RB basis are both enabled, cell edge UEs are scheduled at a granularity of one RB with a minimum of four RBs allocated; non-cell-edge UEs are scheduled at a granularity of one RB with a minimum of one RB allocated. Minimum RB control for cell edge UEs is controlled by the FAR_UE_MIN_RB_CONTROL_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter.^b - When NRDUCellPusch.UIDmrsType is set to TYPE2, scheduling is performed at a granularity of four RBs. For details about DMRS configuration types, see 8 DMRS Resource Allocation.
High frequency bands	Scheduling is performed at a granularity of four RBs.

Scenario	Scheduling Granularity
	a: When uplink scheduling on a per RB basis is enabled together with FDM between PUSCH data and DMRS (controlled by NRDUCellPusch.PuschDmrsFdmSwitch), the PUSCH_DMRS_FDM_AND_1RB_OPT_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter specifies the sequence in which the two functions take effect. • When this option is selected, the scheduler preferentially determines whether uplink scheduling on a per RB basis takes effect: – If so, uplink scheduling is performed on a per RB basis, with a minimum of one RB allocated. In this case, FDM between PUSCH data and DMRS can take effect only for UEs allocated four or more RBs, not for UEs allocated fewer than four RBs. – If not, uplink scheduling is performed on a per four-RB basis, and FDM between PUSCH data and DMRS can take effect. • When this option is deselected, the scheduler preferentially determines whether FDM between PUSCH data and DMRS takes effect: – If so, when the estimated number of required RBs is less than four, scheduling is performed on a per four-RB basis. When the estimated number of required RBs is greater than or equal to 4, scheduling on a per RB basis can take effect, whereby the estimated number of RBs can be allocated. – If not, a decision is made on whether scheduling on a per RB basis takes effect. If it does, uplink scheduling is performed on a per RB basis, with a minimum of one RB allocated. If it does not, scheduling is performed on a per four-RB basis. For details about the FDM between PUSCH data and DMRS function, see 8 DMRS Resource Allocation. b: When enhancement of uplink scheduling on a per RB basis is enabled together with FDM between PUSCH data and DMRS, both functions can take effect. That is, scheduling is performed at a granularity of one RB with a minimum of four RBs allocated and frequency-domain multiplexing is allowed between PUSCH data and DMRS.

6.5.2 Network Analysis

6.5.2.1 Benefits

Uplink Non-Consecutive Scheduling

After this function takes effect, more RBs can be allocated in non-consecutive mode to UEs supporting uplink non-consecutive resource allocation. This helps improve the average uplink UE throughput (**User Uplink Average Throughput (DU)**).

Uplink Scheduling Gain Optimization

After this function takes effect, better RBs can be allocated to uplink UEs, increasing the average uplink UE throughput.

Uplink Non-Consecutive Scheduling Optimization

After this function takes effect, the number of scheduled RBs decreases, the PRB usage decreases, the uplink throughput at the MAC layer decreases, and the interference to neighboring cells decreases.

Uplink Scheduling on a per RB Basis

After this function takes effect, the uplink PRB usage decreases and the uplink spectral efficiency increases.

- Uplink PRB usage = $N.PR.B.UL.Used.Avg/N.PR.B.UL.Avail.Avg$
- Uplink spectral efficiency = $N.ThpVol.UL/N.PR.B.UL.Used.Avg$

Enhancement of Uplink Scheduling on a per RB Basis

After this function takes effect, the uplink PRB usage of the cell decreases in scenarios with a light cell load, and the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) increases in scenarios with a heavy cell load.

Minimum RB control for CEUs

After this function takes effect, the uplink PRB usage increases, the average uplink UE throughput (**User Uplink Average Throughput (DU)**) increases, and the average uplink MCS index decreases.

6.5.2.2 Impacts

The impacts between this function and other functions are described in [6.5.3.2.3 Function Impacts](#).

Uplink non-consecutive scheduling

Uplink non-consecutive scheduling increases the number of available RBs. Since the interference varies across RBs, scheduling a UE on RBs experiencing strong interference may decrease the MCS index. If this occurs, the uplink user-perceived rate fluctuates.

Uplink non-consecutive scheduling optimization

The network impacts produced by uplink non-consecutive scheduling optimization are described in benefit analysis. For details, see [6.5.2.1 Benefits](#).

Uplink scheduling on a per RB basis

Given the same traffic volume, this function has the following impacts:

- The **User Uplink Average Throughput (DU)** decreases.
- The **N.UL.NI.Avg** decreases.
- The number of padding bits at the MAC layer decreases, causing the **VS.FEGE.RxMinSpeed**, **VS.FEGE.TxMinSpeed**, **N.ThpVol.UL.Cell**, and **N.PR.B.UL.DrbUsed.Avg** to decrease. For details about padding bits, see section 5.4.5 "Buffer Status Reporting" in 3GPP TS 38.321 V15.5.0.
- The following impacts arise if uplink control information (UCI) is transmitted on the PUSCH:

- When the probability of UCI reporting on the PUSCH is relatively high, the **N.ChMeas.PUSCH.MCS.0** to **N.ChMeas.PUSCH.MCS.28** may change, and the average MCS index and the BLER slightly decrease.
- When the probability of UCI reporting on the PUSCH is relatively low, the **N.ChMeas.PUSCH.MCS.0** to **N.ChMeas.PUSCH.MCS.28** may change, and the average MCS index and the BLER slightly increase.
- For downlink small-packet services with certain packet lengths, the proportion of small packets with MCS index reduction decreases, leading to the following impacts (insignificant on the whole):
 - The **N.ChMeas.PDSCH.MCS.0** to **N.ChMeas.PDSCH.MCS.27** and **N.ThpVol.UL.Cell** may change.
 - The average downlink MCS index, throughput, and PRB usage fluctuate.
 - The BLER and delay fluctuate.
- In cells with heavy load in the downlink, uplink scheduling on a per RB basis enables UEs to provide more timely feedback on downlink data transmission. This may increase downlink traffic at the PDCP layer and improve the downlink cell throughput. However, the downlink user-perceived rate may decrease due to limited downlink air interface capabilities.
- In a cell with heavy load in the uplink, if the amount of padding data added by CCUs is low, uplink scheduling on a per RB basis may increase the number of RBs occupied by CEUs, which may decrease the uplink spectral efficiency.
- In uplink small-packet scenarios, uplink scheduling on a per RB basis decreases the PRB usage in preallocation. Therefore, UEs transmit some data in preallocation mode, causing the uplink user-perceived rate to decrease.
- The uplink BLER increases. In this case, the **UL_1RB_MCS_SEL_SW** option of the **NRDUCellUIMc.UIMcPerformanceSw** parameter can be selected to mitigate the increase.
- If uplink scheduling on a per RB basis is enabled, a CEU with poor signal quality can send only a small amount of data in each scheduling occasion. Consequently, the UE cannot report its capability information to the base station within the specified timer. This causes an access failure, and the uplink QoS flow setup success rate and RRC connection setup success rate on the base station may decrease. To address this issue, the **FAR_UE_MIN_RB_CONTROL_SW** option of the **NRDUCellPusch.PuschPerformanceSwitch** parameter can be selected. This prevents uplink scheduling on a per RB basis from working on a CEU.

Enhancement of uplink scheduling on a per RB basis

None

Minimum RB control for CEUs

The network impacts produced by minimum RB control for cell edge UEs are described in benefit analysis. For details, see [6.5.2.1 Benefits](#).

6.5.3 Requirements

6.5.3.1 Licenses

There are no license requirements for basic functions.

6.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.5.3.2.1 Prerequisite Functions

None

6.5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink precise MCS optimization	UL_PRECISE_MCS_OPT_SW option of the NRDUCellAiAlgo.AiAmcAlgoSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	Uplink scheduling on a per RB basis and uplink precise MCS optimization cannot be both enabled.

6.5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Flexible experience differentiation	FLEX_EXP_DIFF_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>Service-differentiated Experience Guarantee</i>	When the UL_NON_CON_SCH_SW option of the NRDUCellAlgoSwitch.UllInconsecutiveSchSwitch parameter is selected, and the UL_UNLIMITED_RATE_SW option of the gNBFlexExpDiffCfg.EquityLevelAlgoSw parameter is selected for UEs at the associated privilege level, the estimated maximum uplink target rate may be lower than expected for uplink UEs with uplink non-consecutive scheduling in effect. If this occurs, the uplink rate differentiation multiple between UEs at different privilege levels is increased. Therefore, it is recommended that the UL_UNLIMITED_RATE_SW option of the gNBFlexExpDiffCfg.EquityLevelAlgoSw parameter be deselected for UEs with flexible experience differentiation in effect when the UL_NON_CON_SCH_SW of the NRDUCellAlgoSwitch.UllInconsecutiveSchSwitch parameter is selected. This ensures significant differentiation in uplink user experience.
High-frequency TDD	None	None	None	None

6.5.3.3 Hardware

For details, see [4.3.3 Hardware](#).

6.5.3.4 Cells

When the **NRDUCell.DuplexMode** parameter is set to **CELL_SUL**, the **NRDUCell.SubcarrierSpacing** parameter is set to **15KHZ**, and the **NRDUCellSulRelation.SulType** parameter is set to **FIX_SUL**, the

UL_NON_CON_SCH_SW option of the NRDUCellAlgoSwitch.UlInconsecutiveSchSwitch parameter cannot be selected.

6.5.3.5 Others

None

6.5.4 Operation and Maintenance

6.5.4.1 Data Configuration

6.5.4.1.1 Data Preparation

Table 6-9 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-9 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink Inconsecutive Scheduling Switch	NRDUCellAlgoSwitch.UlInconsecutiveSchSwitch	UL_NON_CON_SCH_SW	Set the UL_NON_CON_SCH_SW option of this parameter to its recommended value.
PUSCH Algorithm Extension Switch	NRDUCellPusch.PuschAlgoExtSwitch	UL_SCH_GAIN_OPT_SW	You are advised to select this option.
PUSCH Performance Switch	NRDUCellPusch.PuschPerformanceSwitch	UL_NONCONSEC_SCH_OPT_SW	You are advised to select this option.

Parameter Name	Parameter ID	Option	Setting Notes
Uplink PUSCH Algorithm Switch	NRDUCellPusch. <i>ULPuschAlgoSwitch</i>	UL_1RB_SCH_SWI TCH	When uplink scheduling on a per RB basis is in effect, the average uplink user-perceived rate decreases. Therefore, this function is recommended only when fewer than four RBs are required in scheduling (for example, when the network slice group with the TTI-based control policy is configured for a cell and the number of RBs corresponding to NRDUCellRes.<i>ULRBMaxRatio</i> is less than 4). In other scenarios, it is recommended that this option be deselected. Before you select this option, learn the resultant network impacts by referring to 6.5.2.2 Impacts. If this option needs to be selected, the PUSCH_PC_1RB_SW option of the NRDUCellULPcConfig.<i>ULPwrCtrlAlgoSwitch</i> parameter also needs to be selected.

Parameter Name	Parameter ID	Option	Setting Notes
PUSCH Performance Switch	NRDUCellPusch.PuschPerformanceSwitch	UL_1RB_SCH_ENH_SW	You are advised to select this option only when fewer than four RBs are required in scheduling (for example, when the network slice group with the TTI-based control policy is configured for a cell and the number of RBs corresponding to NRDUCellRes.UIRbMaxRatio is less than 4) and uplink scheduling on a per RB basis is enabled. You are advised to deselect this option in other scenarios.
PUSCH Performance Switch	NRDUCellPusch.PuschPerformanceSwitch	FAR_UE_MIN_RB_CONTROL_SW	You are advised to select this option only when fewer than four RBs are required in scheduling (for example, when the network slice group with the TTI-based control policy is configured for a cell and the number of RBs corresponding to NRDUCellRes.UIRbMaxRatio is less than 4) and uplink scheduling on a per RB basis is enabled. You are advised to deselect this option in other scenarios.

Parameter Name	Parameter ID	Option	Setting Notes
PUSCH Performance Switch	NRDUCellPusch.<i>PuschPerformanceSwitch</i>	PUSCH_DMRS_FDM_AND_1RB_OPT_SW	Uplink scheduling on a per RB basis (controlled by the UL_1RB_SCH_SWITCH option of NRDUCellPusch.<i>UlPuschAlgoSwitch</i>) decreases the average uplink UE throughput and is not recommended on live networks. Therefore, you are also advised to deselect the PUSCH_DMRS_FDM_AND_1RB_OPT_SW option.

6.5.4.1.2 Using MML Commands

Before using MML commands, refer to [6.5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for uplink non-consecutive scheduling
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, UlInconsecutiveSchSwitch=UL_NON_CON_SCH_SW-1;
//Turning on the switch for uplink scheduling gain optimization
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschAlgoExtSwitch=UL_SCH_GAIN_OPT_SW-1;
//Turning on the switch for uplink non-consecutive scheduling optimization
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL_NONCONSEC_SCH_OPT_SW-1;
//Disabling uplink scheduling on a per RB basis (because it is good practice to enable this function only in special scenarios detailed in the data preparation section and disable this function in other scenarios)
MOD NRDUCELLPUSCH: NrDuCellId=0, UlPuschAlgoSwitch=UL_1RB_SCH_SWITCH-0;
//Disabling enhancement of uplink scheduling on a per RB basis (because it is good practice to enable this function only in special scenarios detailed in the data preparation section and disable this function in other scenarios)
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL_1RB_SCH_ENH_SW-0;
//Disabling minimum RB control for cell edge UEs (because it is good practice to enable this function only in special scenarios detailed in the data preparation section and disable this function in other scenarios)
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=FAR_UE_MIN_RB_CONTROL_SW-0;
//Disabling the function controlled by PUSCH_DMRS_FDM_AND_1RB_OPT_SW (which is a recommended operation considering that the uplink scheduling on a per RB basis function is not recommended)
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=PUSCH_DMRS_FDM_AND_1RB_OPT_SW-0;
```

Deactivation Command Examples

```
//Turning off the switch for uplink non-consecutive scheduling  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, UlinconsecutiveSchSwitch=UL_NON_CON_SCH_SW-0;  
//Turning off the switch for uplink scheduling gain optimization  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschAlgoExtSwitch=UL_SCH_GAIN_OPT_SW-0;  
//Turning off the switch for uplink non-consecutive scheduling optimization  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL_NONCONSEC_SCH_OPT_SW-0;
```

6.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.5.4.2 Activation Verification

Uplink Non-Consecutive Scheduling

If the average uplink UE throughput (**User Uplink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

Uplink Non-Consecutive Scheduling Optimization

If the uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**) decreases after this function is enabled, this function has taken effect.

6.5.4.3 Network Monitoring

The indicators listed in [6.5.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

6.6 Uplink Frequency Selective Scheduling and Non-Frequency-Selective Scheduling

6.6.1 Principles

Uplink RBs can be allocated using frequency selective or non-frequency-selective scheduling. The gNodeB adaptively selects an RB allocation mode based on the number of UEs to be scheduled and PRB usage in a cell. When the cell has a few UEs to be scheduled and low PRB usage, the gNodeB selects frequency selective scheduling. Conversely, when the cell has many UEs to be scheduled and high PRB usage, the gNodeB selects non-frequency-selective scheduling.

The gNodeB can calculate the PRB usage of a cell based on the available RBs for the PUSCH or across the uplink full bandwidth. If the **FREQ_SEL_SCH_PRB_CALC_OPT_SW** option of the **NRDUCellPusch.PuschPerformanceSwitch** parameter is selected, the calculation is based on the available RBs for the PUSCH. If this option is deselected, the calculation is based on the available RBs across the uplink full bandwidth.

During uplink RB allocation, the scheduler checks whether the uplink PRB usage of a UE exceeds the threshold specified by the **NRDUCellPusch.AbnUePrbUsageThld**

parameter and whether the proportion of padding exceeds the threshold specified by the **NRDUCellPusch.AbnormalUePaddingRatioThld** parameter. If both conditions are met, the scheduler identifies the UE as a UE with abnormal BSR and dynamically adjusts the amount of data to be transmitted for this UE based on the **NRDUCellPusch.AbnUeDataVolume** and **NRDUCellPusch.AbnUeDataVolAdjPadRateThld** parameters.

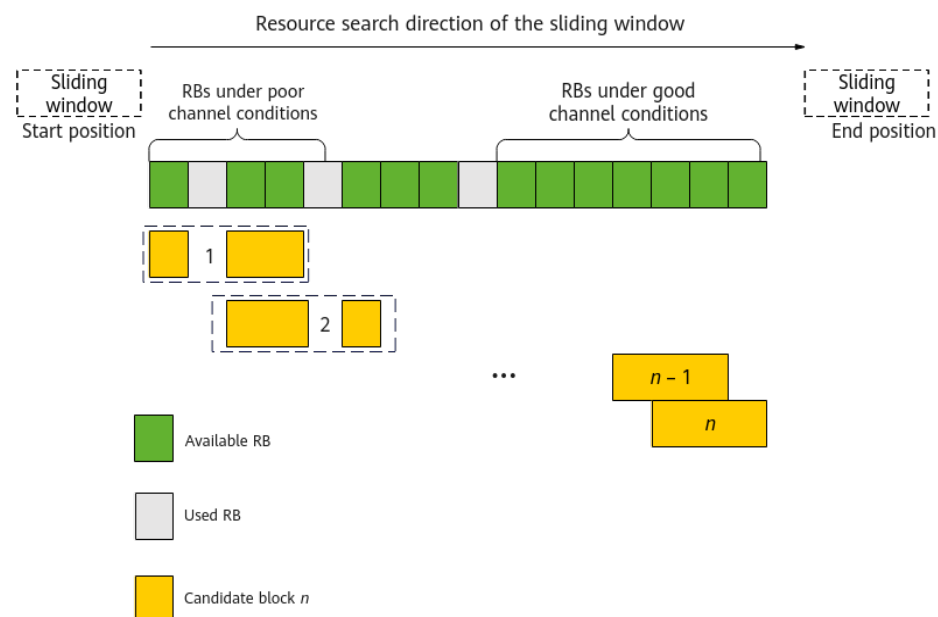
When the uplink PRB usage of a UE is greater than the PRB usage threshold for identifying abnormal UEs (configured using **NRDUCellPusch.AbnUePrbUsageThld**), the number of BSRs containing control channel data is measured for the UE in a period of time. If a UE reports BSRs containing control channel data but no control channel data exists in actual data transmission, and the number of such BSRs exceeds a specified proportion of all measured BSRs, this UE is identified as a UE with abnormal BSRs and its scheduling priority is lowered.

- Frequency selective scheduling

In frequency selective scheduling, the gNodeB selects the most suitable frequency resources for a UE based on channel quality differences derived from SRS measurements across uplink frequency bands. The uplink scheduler sets a sliding window based on the number of RBs required by each UE and searches for resource combinations from the lower end to the upper end of the available frequency band using this sliding window. It then selects the resource combination that yields the maximum benefits. [Figure 6-6](#) provides an example where each UE requires three RBs and the sliding window spans three RBs. The uplink scheduler searches for resource combinations from the lower end to the upper end of the frequency band using the sliding window. From the resultant resource combinations, that is, candidate blocks 1 to n , the uplink scheduler selects the resource combination that yields the maximum benefits and allocates it to the UE.

If uplink frequency selective scheduling optimization is enabled, the gNodeB calculates the expected benefits of a candidate block based on both the uplink spectral efficiency and SINR corresponding to the candidate block. If this function is disabled, the calculation is based on the uplink spectral efficiency. Uplink frequency selective scheduling optimization is controlled by the **UL_FREQ_SEL_SCH_OPT_SW** option of the **NRDUCellPusch.PuschPerformanceSwitch** parameter.

Figure 6-6 Resource allocation for uplink frequency selective scheduling



To fully utilize PUSCH frequency-domain resources and further improve system performance, decoupling uplink non-consecutive scheduling from frequency selective scheduling is introduced. This function is controlled by the **NONCON_FREQ_SEL_SCH_DECOUP_SW** option of the **NRDUCellPusch.PuschPerformanceSwitch** parameter.

- If this option is selected, the gNodeB chooses between consecutive or non-consecutive scheduling based on cell load, regardless of whether frequency selective scheduling is in effect. The selection logic is as follows: (1) In a lightly loaded cell, the gNodeB compares the spectral efficiency in consecutive scheduling with that in non-consecutive scheduling and automatically selects the more efficient one. (2) In a heavily loaded cell, the gNodeB performs non-consecutive scheduling for UEs that meet non-contiguous resource allocation conditions but consecutive scheduling for UEs if they do not meet non-contiguous resource allocation conditions.
- If this option is deselected and uplink non-consecutive scheduling is enabled (by selecting the **UL_NON_CON_SCH_SW** option of the **NRDUCellAlgoSwitch.UlinconsecutiveSchSwitch** parameter), the gNodeB chooses between consecutive or non-consecutive scheduling for UEs that meet non-contiguous resource allocation conditions based on whether frequency selective scheduling is in effect, as described below:
 - If frequency selective scheduling is in effect, the gNodeB selects non-consecutive or consecutive scheduling based on spectral efficiency.
 - If frequency selective scheduling does not take effect, the gNodeB selects non-consecutive scheduling.

 NOTE

Non-contiguous resource allocation conditions: UEs support uplink non-contiguous resource allocation and dynamic switching between contiguous and non-contiguous resource allocation. (This support is indicated by the *almostContiguousCP-OFDM-UL*, *ra-Type0-PUSCH*, and *dynamicSwitchRA-Type0-1-PUSCH* IEs.) For details, see [Table 6-7](#) in [6.5 Uplink Consecutive Scheduling and Non-Consecutive Scheduling](#).

- Non-frequency selective scheduling

In non-frequency-selective scheduling, the gNodeB allocates RBs to a UE based on its required number of RBs by searching from the start position of the available system bandwidth, in order from low to high frequency.

6.6.2 Network Analysis

6.6.2.1 Benefits

After uplink frequency selective scheduling takes effect in light-load scenarios, inter-cell interference decreases and the average uplink UE throughput (**User Uplink Average Throughput (DU)**) increases.

6.6.2.2 Impacts

The impacts between this function and other functions are described in [6.6.3.2.3 Function Impacts](#).

6.6.3 Requirements

6.6.3.1 Licenses

There are no license requirements for basic functions.

6.6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.6.3.2.1 Prerequisite Functions

None

6.6.3.2.2 Mutually Exclusive Functions

None

6.6.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Avoidance of RedCap uplink scheduling	NRDUCellRedCap.UIAvoidPol set to LEVEL1 or LEVEL2	<i>Stable Video Surveillance Experience</i>	When enabled together with avoidance of RedCap uplink scheduling, uplink frequency selective scheduling will produce reduced benefits for non-RedCap UEs. This is because non-RedCap UEs are scheduled from frequency-domain resources other than those on the BWP1s activated for RedCap UEs.

6.6.3.3 Hardware

For details, see [4.3.3 Hardware](#).

6.6.3.4 Others

None

6.6.4 Operation and Maintenance

6.6.4.1 Data Configuration

6.6.4.1.1 Data Preparation

[Table 6-10](#) describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

Table 6-10 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
PUSCH Performance Switch	NRDUCellPusch.PuschPerformanceSwitch	FREQ_SEL_SCH_PRB_CALC_OPT_SW	You are advised to select this option.
PUSCH Performance Switch	NRDUCellPusch.PuschPerformanceSwitch	UL_FREQ_SEL_SCH_H_OPT_SW	You are advised to select this option.

Parameter Name	Parameter ID	Option	Setting Notes
PUSCH Performance Switch	NRDUCellPusch.<i>PuschPerformanceSwitch</i>	NONCON_FREQ_SEL_SCH_DECOUP_SW	You are advised to select this option.
Abnormal UE PRB Usage Threshold	NRDUCellPusch.<i>AbnUePrbUsageThld</i>	None	It is recommended that Huawei engineers configure the threshold based on network requirements.
Abnormal UE Padding Ratio Threshold	NRDUCellPusch.<i>AbnormalUePaddingRatioThld</i>	None	It is recommended that Huawei engineers configure the threshold based on network requirements.
Abn UE Data Volume Adjust Padding Rate Thld	NRDUCellPusch.<i>AbnUeDataVolAdjPadRateThld</i>	None	It is recommended that Huawei engineers configure the threshold based on network requirements.
Abnormal UE Data Volume	NRDUCellPusch.<i>AbnUeDataVolume</i>	None	It is recommended that Huawei engineers configure the value based on network requirements.

6.6.4.1.2 Using MML Commands

Before using MML commands, refer to [6.6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for PRB calculation optimization in frequency selective scheduling
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=FREQ_SEL_SCH_PRB_CALC_OPT_SW-1;
//Turning on the switch for uplink frequency selective scheduling optimization
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL_FREQ_SEL_SCH_OPT_SW-1;
//Turning on the switch for decoupling uplink non-consecutive scheduling from frequency selective
scheduling
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=NONCON_FREQ_SEL_SCH_DECOUP_SW-1;
//Configuring the PRB usage threshold for identifying abnormal UEs
MOD NRDUCELLPUSCH: NrDuCellId=0, AbnUePrbUsageThld=10;
//Configuring the padding proportion threshold for identifying abnormal UEs
MOD NRDUCELLPUSCH: NrDuCellId=0, AbnormalUePaddingRatioThld=9;
//Configuring the padding rate threshold for adjusting the amount of data to be transmitted by abnormal
UEs
MOD NRDUCELLPUSCH: NrDuCellId=0, AbnUeDataVolAdjPadRateThld=50;
//Configuring the initial amount of data to be transmitted by abnormal UEs
MOD NRDUCELLPUSCH: NrDuCellId=0, AbnUeDataVolume=1;
```

Deactivation Command Examples

```
//Turning off the switch for PRB calculation optimization in frequency selective scheduling
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=FREQ_SEL_SCH_PRB_CALC_OPT_SW-0;
//Turning off the switch for uplink frequency selective scheduling optimization
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL_FREQ_SEL_SCH_OPT_SW-0;
//Turning off the switch for decoupling uplink non-consecutive scheduling from frequency selective
scheduling
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=NONCON_FREQ_SEL_SCH_DECOUP_SW-0;
```

6.6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.6.4.2 Activation Verification

In light load scenarios, if the average uplink UE throughput (**User Uplink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

6.6.4.3 Network Monitoring

The indicators listed in [6.6.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

6.7 Uplink Interference Randomization-based Scheduling

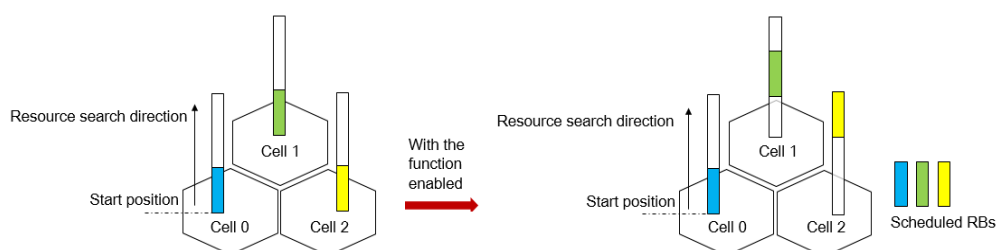
6.7.1 Principles

Uplink interference randomization-based scheduling staggers resources that are allocated to adjacent cells as far away from each other as possible in multi-cell

networking scenarios. This reduces the uplink interference from UEs in neighboring cells to the local cell and improves uplink system capacity and cell edge UE throughput when the network is moderately or lightly loaded. When the BWP of a UE is the same as the cell bandwidth, this function is enabled by setting **NRDUCellAlgoSwitch.UlIntrfRandomSchSwitch** to **ON**. When the BWP for a UE differs from the cell bandwidth, this function is enabled by setting **NRDUCellAlgoSwitch.UlIntrfRandomSchSwitch** to **ON** and selecting the **INTRF_RAND_AND_CELL_BW_OPT_SW** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter. This function is supported only in low frequency bands, not in high frequency bands. Uplink interference randomization-based scheduling is not recommended in scenarios where the NR system suffers interference from digital broadcasting.

Figure 6-7 shows how uplink interference randomization-based scheduling works.

Figure 6-7 Uplink interference randomization-based scheduling



Before this function is enabled, the RB allocation start positions are not staggered between adjacent cells. Instead, uplink RBs are scheduled at the same start position in the frequency band. The probability of RB position overlapping is high and the local cell experiences a large amount of uplink interference from UEs in neighboring cells.

After this function is enabled, the RB allocation start positions are staggered between adjacent cells based on the PCI. Uplink RBs are scheduled from different start positions in the frequency band, reducing uplink interference from UEs in neighboring cells to the local cell.

6.7.2 Network Analysis

6.7.2.1 Benefits

In medium- and light-load scenarios, the throughput of UEs at the cell edge increases. A smaller inter-site distance indicates stronger interference from neighboring cells and higher benefits delivered by this function.

In heavy load scenarios, this function may deliver no benefits. However, it will never generate negative results.

NOTE

$$\text{Cell load} = \text{N.PRB.UL.Used.Avg} / \text{N.PRB.UL.Avail.Avg}$$

6.7.2.2 Impacts

The impacts between this function and other functions are described in [6.7.3.2.3 Function Impacts](#).

6.7.3 Requirements

6.7.3.1 Licenses

There are no license requirements for basic functions.

6.7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.7.3.2.1 Prerequisite Functions

None

6.7.3.2.2 Mutually Exclusive Functions

None

6.7.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	For hyper cells, uplink interference randomization-based scheduling is supported only between TRPs within a hyper cell.
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	Uplink interference randomization-based scheduling will not take effect if all of the following conditions are met: (1) The HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter is selected. (2) Independent resource reservation applies in non-slicing scenarios. (3) The NRDUCellRes.UlMinRbStartPos parameter is set to 0 .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE FDD and NR Uplink Spectrum Sharing	LTE_NR_UL_SPECTRUM_SHARING_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	<i>LTE FDD and NR SUL Dynamic Spectrum Sharing</i>	The number of uplink RBs available for NR is reduced in SUL cells, causing the benefits of interference randomization to decrease.
Low-frequency TDD	NR DU cell resource management between network slices	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.NetworkSliceAlgoSwitch parameter	<i>Network Slicing</i>	Uplink interference randomization-based scheduling is not performed within the minimum RB quota if all of the following conditions are met: (1) Both the NR DU cell resource management between network slices function and the uplink interference randomization-based scheduling function are enabled. (2) Both the minimum RB quota and its start position are configured.
Low-frequency TDD	Avoidance of RedCap uplink scheduling	NRDUCellRedCap.UIAvoidPol set to LEVEL1 or LEVEL2	<i>Stable Video Surveillance Experience</i>	When enabled together with avoidance of RedCap uplink scheduling, uplink interference randomization-based scheduling will produce reduced benefits for non-RedCap UEs. This is because uplink scheduling for non-RedCap UEs avoids using the frequency-domain resources on the BWP1 activated for RedCap UEs.

6.7.3.3 Hardware

For details, see [4.3.3 Hardware](#).

6.7.3.4 Others

None

6.7.4 Operation and Maintenance

6.7.4.1 Data Configuration

6.7.4.1.1 Data Preparation

Table 6-11 describes the parameters used for function activation. No parameters are involved in function optimization.

Table 6-11 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink Interference Random Schedule Switch	NRDUCellAlgoSwitch.UlIntrfRandomSchSwitch	None	It is recommended that this parameter be set to ON for cells with medium or low load.
Uplink PUSCH Algorithm Switch	NRDUCellPusch.UlPuschAlgoSwitch	INTRF_RAND_AND_CELL_BW_OPT_SW	When the BWP for a UE is different from the cell bandwidth, uplink interference randomization-based scheduling can be enabled only if NRDUCellAlgoSwitch.UlIntrfRandomSchSwitch is set to ON and the INTRF_RAND_AND_CELL_BW_OPT_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter is selected.

6.7.4.1.2 Using MML Commands

Before using MML commands, refer to **6.7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink interference randomization-based scheduling when the BWP for a UE is the same as the cell bandwidth  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, UlintrfRandomSchSwitch=ON;  
//Enabling uplink interference randomization-based scheduling when the BWP for a UE is different from the cell bandwidth  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, UlintrfRandomSchSwitch=ON;  
MOD NRDUCELLPUSCH: NrDuCellId=0, UlpuschAlgoSwitch=INTRF_RAND_AND_CELL_BW_OPT_SW-1;
```

Optimization Command Examples

Optimization command examples are not involved.

Deactivation Command Examples

```
//Disabling uplink interference randomization-based scheduling  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, UlintrfRandomSchSwitch=OFF;
```

6.7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.7.4.2 Activation Verification

If the average uplink UE throughput (**User Uplink Average Throughput (DU)**) increases after uplink interference randomization-based scheduling is enabled, this function has taken effect.

6.7.4.3 Network Monitoring

Monitor **User Uplink Average Throughput (DU)** to obtain the average uplink UE throughput.

6.8 Uplink Waveform Adaptation

The NR system supports CP-OFDM and DFT-s-OFDM waveforms.

- CP-OFDM: cyclic prefix-OFDM. CP-OFDM is characterized by flexible resource allocation and large frequency diversity gain due to the use of non-contiguous frequency-domain resources. However, this resource allocation mode can produce a large maximum power reduction (MPR) value, which limits the maximum transmit power of UEs and therefore affects the throughput at the coverage edge.
- DFT-s-OFDM: discrete Fourier transform-spread OFDM. DFT-s-OFDM features a small MPR value, high transmit power of UEs, and high throughput at the coverage edge. However, it requires continuous frequency-domain resources.

NOTE

For details about the MPR value, see section "6.2.2 UE maximum output power reduction" in 3GPP TS 38.101.

Uplink waveform adaptation supports adaptation between the CP-OFDM and DFT-s-OFDM waveforms based on the radio environment (such as coverage) of UEs to improve user experience.

This feature includes RRC-reconfiguration-based dynamic waveform switching and DCI-based dynamic waveform switching. The differences between the two functions are as follows:

- RRC-reconfiguration-based dynamic waveform switching: The gNodeB instructs a UE to switch waveforms through an RRCReconfiguration message. For details about this function in low-frequency TDD, see *Turbo Uplink (Low-Frequency TDD)*. For details about this function in high-frequency TDD, see *Ultra Uplink (High-Frequency TDD)*.
- DCI-based dynamic waveform switching: The gNodeB instructs a UE to switch waveforms through DCI. For details, see [6.8.1 DCI-based Dynamic Waveform Switching](#).

DCI-based dynamic waveform switching does not require RRC connection reconfiguration, accelerating the switching speed and stabilizing user experience. Therefore, this function is recommended if the UE capability meets its requirements. For details, see [6.8.1.3.5 Others](#). In the case that both RRC-reconfiguration-based dynamic waveform switching and DCI-based dynamic waveform switching are enabled, if the UE capability meets the requirements of DCI-based dynamic waveform switching, DCI-based dynamic waveform switching takes effect; otherwise, RRC-reconfiguration-based dynamic waveform switching takes effect.

6.8.1 DCI-based Dynamic Waveform Switching

6.8.1.1 Principles

DCI-based dynamic waveform switching is controlled by the **DYNAMIC_WAVEFORM_SWITCHING_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter.

Before this function is enabled, the network side instructs a UE to always use the CP-OFDM waveform.

After this function is enabled, the network side instructs a UE to use the CP-OFDM or DFT-s-OFDM waveform through DCI based on the radio environment and the threshold (specified by the **NRDUCellPusch.SinrThldforWaveformSel** parameter). If a UE's uplink SINR is less than the threshold (the UE is in a poor radio environment, for example, at the coverage edge) and rank 1 is used, the UE uses the DFT-s-OFDM waveform. In other cases, the UE uses the CP-OFDM waveform.

NOTE

- To prevent ping-pong waveform switching, the gNodeB adds a hysteresis range of ± 2 dB to the threshold. That is, waveform switching is performed only when the difference between the UE SINR and the threshold exceeds 2 dB.
- If a UE's rank is not 1, the UE does not support the DFT-s-OFDM waveform and therefore only the CP-OFDM waveform can be used.

This document describes DCI-based dynamic waveform switching in the following scenarios:

- 3900 and 5900 series base stations (macro base stations): low-frequency TDD RF modules working in a TX/RX mode lower than 32T32R. For details about this function in 32T32R mode or higher, see *Massive MIMO Uplink Boosting (Low-Frequency TDD)*.
- DBS3900 LampSite and DBS5900 LampSite: low-frequency TDD, with distributed massive MIMO disabled. For details about this function with distributed massive MIMO enabled, see *Massive MIMO Uplink Boosting (Low-Frequency TDD)*.

 NOTE

For details about DCI-based dynamic waveform switching in high-frequency TDD, see *Ultra Uplink (High-Frequency TDD)*.

6.8.1.2 Network Analysis

6.8.1.2.1 Benefits

CEUs achieve higher throughput with DFT-S-OFDM than with CP-OFDM under the same scheduled bandwidth.

The function may produce reduced benefits or no benefits but has no negative results in the following scenarios:

- The UE's rank is not 1.
- The UE's uplink transmit power is limited by the SAR standard.
- The MPR values vary with terminal vendors.

 NOTE

Specific absorption rate (SAR) is a measure of the rate at which energy is absorbed per unit mass by a human body when exposed to an electromagnetic field. A larger value indicates a greater impact on the human body. To ensure safety, the transmit power of a UE must be lower than a specified SAR when the UE is close to the human body.

6.8.1.2.2 Impacts

The impacts between this function and other functions are described in [Function Impacts](#).

The single-UE throughput may not increase or even decrease if (1) the scheduled bandwidth decreases after the waveform mode is switched from CP-OFDM to DFT-s-OFDM and (2) frequency-domain resources are not contiguous. The specific negative results depend on the changes in (1) scheduled bandwidth and power spectral density and (2) increase in UE transmit power after the waveform mode is switched from CP-OFDM to DFT-s-OFDM.

6.8.1.3 Requirements

6.8.1.3.1 Licenses

None

6.8.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

Prerequisite Functions

None

Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DMRS configuration type	NRDUCellPusch.UIDmrsType	<i>Uplink Scheduling</i>	When NRDUCellPusch.UIDmrsType is set to TYPE2 , DCI-based dynamic waveform switching cannot be enabled.
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	When NRDUCell.HighSpeedFlag is set to HIGH_SPEED , DCI-based dynamic waveform switching cannot be enabled.

Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RRC-Reconfiguration-based Dynamic Waveform Switching	UL_WAVEFORM_ADAPT_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	In the case that both RRC-reconfiguration-based dynamic waveform switching and DCI-based dynamic waveform switching are enabled, if the UE capability meets the requirements of DCI-based dynamic waveform switching, DCI-based dynamic waveform switching takes effect; otherwise, RRC-reconfiguration-based dynamic waveform switching takes effect.

6.8.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

- 3900 and 5900 series base stations (macro base stations): All NR TDD-capable RF modules and NR FDD-capable RF modules that work in 4T4R mode or lower support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- DBS3900 LampSite and DBS5900 LampSite: All NR TDD- and NR FDD-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.8.1.3.4 Cells

DCI-based dynamic waveform switching cannot be enabled in SDL cells (with **NRDUCell.DuplexMode** set to **CELL_SDL**).

DCI-based dynamic waveform switching can be enabled only when **NRDUCell.NrDuCellNetworkingMode** is set to **NORMAL_CELL**.

6.8.1.3.5 Others

DCI-based dynamic waveform switching requires support from UEs. Whether a UE supports this function in a band depends on the number of carriers configured in the band and the UE's support for the **dynamicWaveformSwitch-r18** and **dynamicWaveformSwitchIntraCA-r18** fields.

- If only one carrier is configured in a band, the UE supports this function as long as it supports the **dynamicWaveformSwitch-r18** field.
- If multiple carriers are configured in a band, the UE supports this function only when the value of the **dynamicWaveformSwitchIntraCA-r18** field in the UE capability information is greater than or equal to the number of carriers configured in the band.

For details about the **dynamicWaveformSwitch-r18** and **dynamicWaveformSwitchIntraCA-r18** fields, see section 4.2.7.2 "BandNR parameters" in 3GPP TS 38.306 V18.2.0.

 NOTE

If a UE supports different bands and carriers are set up in these bands, whether DCI-based dynamic waveform switching can take effect for a band is determined independently of other bands. For example, if a UE meets the conditions for this function to take effect in band A but does not meet the conditions for this function to take effect in band B, this function can take effect in band A but not in band B.

6.8.1.4 Operation and Maintenance

6.8.1.4.1 Data Configuration

Data Preparation

[Table 6-12](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-12 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
UL Coverage Algorithm Switch	NRDUCellChnCovAlgo.UlCoverageAlgoSwitch	DYNAMIC_WAVEFORM_SWITCHING_SW	To enable DCI-based dynamic waveform switching, select this option.
SINR Threshold for Waveform Selection	NRDUCellPusch.SinrThldforWaveformSel	None	Set this parameter based on the network plan if DCI-based dynamic waveform switching is required.

Using MML Commands

Before using MML commands, refer to [6.8.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling DCI-based dynamic waveform switching
MOD NRDUCCELLCHNCOVALGO: NrDuCellId=0,
UlCoverageAlgoSwitch=DYNAMIC_WAVEFORM_SWITCHING_SW-1;
//Setting the SINR threshold for waveform selection
MOD NRDUCCELLPUSCH: NrDuCellId=0, SinrThldforWaveformSel=32;
```

Deactivation Command Examples

```
//Disabling DCI-based dynamic waveform switching  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0,  
ULCoverageAlgoSwitch=DYNAMIC_WAVEFORM_SWITCHING_SW-0;
```

Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.8.1.4.2 Activation Verification

- Step 1** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**.
- Step 2** In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **NR > Application Layer > Uu Interface Trace**.
- Step 3** In a cell that serves UEs supporting DCI-based dynamic waveform switching, check whether **dynamicTransformPrecoderFieldPresenceDCI-0-1-r18** is contained in at least one RRCReconfiguration message sent by the gNodeB to a UE. If yes, DCI-based dynamic waveform switching has taken effect in the cell.

----End

6.8.1.4.3 Network Monitoring

After this function is enabled, you can monitor its running status by checking the increase in the uplink user-perceived rate of CEUs. The increase is calculated using the following formula: $(N.ThpVol.UL.Edge - N.ThpVol.UE.UL.SmallPkt.Edge) / N.ThpTime.UE.UL.RmvSmallPkt.Edge$.

6.9 Adaptive HARQ-ACK Feedback on Downlink Transmission (High-Frequency TDD)

6.9.1 Principles

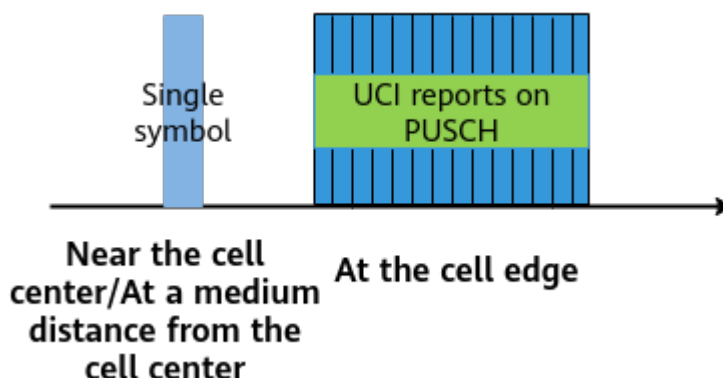
Adaptive HARQ-ACK feedback on downlink transmission is introduced to improve the uplink coverage for ACK/NACK feedback in high-frequency NR TDD scenarios. The **ACK_NACK_REPORTING_PUSCH_SW** option of the **NRDUCellAlgoSwitch.FullChannelCovEnhSwitch** parameter specifies whether to enable this function. After this function takes effect, the gNodeB checks the UE-reported signal quality to flexibly select a resource allocation mode for uplink ACK and NACK feedback to downlink HARQ, as shown in [Figure 6-8](#).

- When a UE is located near the cell center or at a medium distance from the cell center, the short PUCCH that spans one symbol is always used for ACK/NACK feedback. For details about the working principles of the PUCCH, see *Channel Management*.

- When a UE is located at the cell edge, the resource allocation mode is selected based on the setting of the **ACK_NACK_REPORTING_PUSCH_SW** option of the **NRDUCellAlgoSwitch.FullChannelCovEnhSwitch** parameter:
 - If this option is deselected, this function does not take effect, and the reporting mode is the same as that used before this option is deselected.
 - If this option is selected, the UE feeds back ACK/NACK over the PUSCH, and only one such UE can be scheduled at a time. In a slot with no PUSCH transmission scheduling, the UE feeds back ACK/NACK over the short PUCCH. If the **ACK_NACK_REPORTING_PUSCH_SW** option of the **NRDUCellAlgoSwitch.FullChannelCovEnhSwitch** parameter is deselected after ACK/NACK feedback reporting over the PUSCH has taken effect for a UE, this function does not become ineffective immediately and is processed as follows:
 - If the UE remains at the cell edge after this option is deselected, ACK/NACK feedback is still reported together with the PUSCH and only one UE is scheduled at a time.
 - If the UE moves to a location near the cell center or at a medium distance from the cell center after this option is deselected, the principle is the same as that used when the UE is located near the cell center or at a medium distance from the cell center.

When adaptive HARQ-ACK feedback on downlink transmission (controlled by the **ACK_NACK_REPORTING_PUSCH_SW** option of the **NRDUCellAlgoSwitch.FullChannelCovEnhSwitch** parameter) is enabled, ACK/NACK feedback is reported over the PUSCH and occupies PUSCH beam resources. To ensure user experience of UEs with common uplink data transmission, the proportion of PUSCH beam resources occupied by ACK/NACK feedback on the PUSCH must not exceed the HARQ-ACK scheduling guarantee proportion threshold specified by the **NRDUCellPusch.GtdHarqAckSchRatioThld** parameter.

- If the proportion of uplink beam resources occupied by ACK/NACK feedback over the PUSCH is less than the value of this parameter, UEs with HARQ-ACK transmitted with data on the PUSCH are preferentially scheduled to obtain PUSCH beam resources for ACK/NACK feedback.
- When the proportion of uplink beam resources occupied by ACK/NACK feedback over the PUSCH is greater than or equal to the value of this parameter, UEs with uplink data transmission are preferentially scheduled to obtain PUSCH beam resources for uplink data transmission. Beam resources for ACK/NACK feedback can be allocated only when PUSCH beam resources are idle.

Figure 6-8 Adaptive HARQ-ACK feedback on downlink transmission

It is recommended that this function be enabled for a cell with a medium or heavy load in NSA networking.

6.9.2 Network Analysis

6.9.2.1 Benefits

After adaptive HARQ-ACK feedback on downlink transmission takes effect for a cell with a medium or heavy load, UEs can transmit ACK/NACK (for downlink transmission) over the PUSCH when PUCCH resources are limited, thereby increasing the average downlink cell throughput.

6.9.2.2 Impacts

The impacts between this function and other functions are described in [6.9.3.2.3 Function Impacts](#).

In a cell with a medium or heavy load, adaptive HARQ-ACK feedback on downlink transmission increases the uplink coverage capability for UEs to feed back ACK/NACK to downlink HARQ. This improves the downlink coverage capability of the cell, and increases the proportion of CEUs in the downlink, the proportion of downlink low MCS indexes, and the proportion of QPSK for the PUSCH. As PUSCH resources for uplink data transmission are occupied (with the proportion dependent on the setting of the **NRDUCellPusch.GtdHarqAckSchRatioThld** parameter) by UEs with this function, the uplink PRB usage decreases, and the average uplink UE throughput and average uplink cell throughput may decrease.

6.9.3 Requirements

6.9.3.1 Licenses

There are no license requirements for basic functions.

6.9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.9.3.2.1 Prerequisite Functions

None

6.9.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	If adaptive HARQ-ACK feedback on downlink transmission is enabled, mmWave PUSCH MU-MIMO cannot be enabled.

6.9.3.2.3 Function Impacts

None

6.9.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

6.9.3.4 Cells

If the **NRDUCell.SlotAssignment** parameter is set to **2_3_DSUUU** for a cell, the **ACK_NACK_REPORTING_PUSCH_SW** option of the

NRDUCellAlgoSwitch.FullChannelCovEnhSwitch parameter must be deselected for the cell.

6.9.3.5 Others

None

6.9.4 Operation and Maintenance

6.9.4.1 Data Configuration

6.9.4.1.1 Data Preparation

[Table 6-13](#) and [Table 6-14](#) describe the parameters used for function activation and optimization, respectively.

Table 6-13 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Full Channel Coverage Enhancement Switch	NRDUCellAlgoSwitch.FullChannelCovEnhSwitch	ACK_NACK_REPORTING_PUSCH_SW	It is recommended that this option be selected for a cell with medium or heavy load in NSA networking.

Table 6-14 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
Guaranteed HARQ-ACK Sch Ratio Thld	NRDUCellPusch.GtdHarqAckSchRatioThld	Use the default value.

6.9.4.1.2 Using MML Commands

Before using MML commands, refer to [6.9.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling adaptive HARQ-ACK feedback on downlink transmission  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
FullChannelCovEnhSwitch=ACK_NACK_REPORTING_PUSCH_SW-1;
```

Optimization Command Examples

```
//Modifying the HARQ-ACK scheduling guarantee proportion threshold  
MOD NRDUCELLPUSCH: NrDuCellId=0, GtdHarqAckSchRatioThld=20;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling adaptive HARQ-ACK feedback on downlink transmission  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
FullChannelCovEnhSwitch=ACK_NACK_REPORTING_PUSCH_SW-0;
```

6.9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.9.4.2 Activation Verification

Use **N.Thp.DL.Samp.Index0** to **N.Thp.DL.Samp.Index9** to measure the proportion of samples in each throughput range. For example: Proportion of samples in the downlink throughput range 0 = **N.Thp.DL.Samp.Index0**/SUM(**N.Thp.DL.Samp.Index0** to **N.Thp.DL.Samp.Index9**). If the proportion of samples in the downlink low-throughput ranges increase after adaptive HARQ-ACK feedback on downlink transmission is enabled, this function has taken effect.

6.9.4.3 Network Monitoring

After this function is enabled, monitor the following:

- Scheduling frequency of an MCS index: **N.ChMeas.PUSCH.MCS.0** to **N.ChMeas.PUSCH.MCS.31**
- Proportion of QPSK for UL-SCH = **N.UL.SCH.QPSK.TB** / (**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**)
- Uplink RB usage: **Uplink Resource Block Utilizing Rate (DU)**
- Average uplink UE throughput: **User Uplink Average Throughput (DU)**
- Average uplink cell throughput: **Cell Uplink Average Throughput (DU)**
- Average downlink cell throughput: **Cell Downlink Average Throughput (DU)**
- Average downlink UE throughput: **User Downlink Average Throughput (DU)**
- Number of samples in a downlink low-throughput range: **N.Thp.DL.Samp.Index0** to **N.Thp.DL.Samp.Index9**

6.10 Uplink Scheduling Enhancement

6.10.1 Principles

In low-frequency scenarios, the gNodeB determines the number of RBs for a UE based on the average measured SINR. If the actual SINR in scheduling is lower than the average measured SINR, RBs allocated to the UE will be fewer than expected. As a result, UE data needs to be scheduled multiple times, increasing the UE data transmission delay. To address this issue, the uplink scheduling enhancement function is introduced. This function is controlled by the **UL_SCHEDULE_ENH_SW** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter. After uplink scheduling enhancement is enabled, the gNodeB determines the number of RBs for UEs based on the actual SINR in scheduling, ensuring more accurate RB resource allocation and shorter UE data transmission delay.

In addition, after uplink scheduling enhancement is enabled, processing varies as follows with the setting of the **UL_INTRF_RANDOMIZED_SCH_ENH_SW** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter:

- If this option is selected, the start positions of scheduling for adjacent cells are fixed strictly based on their PCIs, and the cells search for RBs from different start positions and allocate RBs to UEs. This reduces inter-cell interference.
- If this option is deselected, all cells search for RBs from the same start position and allocate RBs.

6.10.2 Network Analysis

6.10.2.1 Benefits

In low-frequency scenarios, if uplink scheduling enhancement is enabled and uplink MU-MIMO (controlled by the **UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch.MuMimoSwitch** parameter) is disabled, the number of RBs allocated by the gNodeB to UEs increases, and the uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**) increases.

6.10.2.2 Impacts

The impacts between this function and other functions are described in [6.10.3.2.3 Function Impacts](#).

In low-frequency cells, if uplink scheduling enhancement is enabled and uplink MU-MIMO is disabled (with the **UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch.MuMimoSwitch** parameter deselected), UEs are allocated an increased number of resources. However, the MCS indexes of UEs at the cell edge or a medium distance from the cell center may decrease because their power is limited.

6.10.3 Requirements

6.10.3.1 Licenses

There are no license requirements for basic functions.

6.10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.10.3.2.1 Prerequisite Functions

None

6.10.3.2.2 Mutually Exclusive Functions

None

6.10.3.2.3 Function Impacts

None

6.10.3.3 Hardware

For details, see [4.3.3 Hardware](#).

6.10.3.4 Others

None

6.10.4 Operation and Maintenance

6.10.4.1 Data Configuration

6.10.4.1.1 Data Preparation

[Table 6-15](#) describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

Table 6-15 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink PUSCH Algorithm Switch	NRDUCellPusch.UlPuschAlgoSwitch	UL_SCHEDULE_E NH_SW	You are advised to select this option.

Parameter Name	Parameter ID	Option	Setting Notes
Uplink PUSCH Algorithm Switch	NRDUCeLLPusch.UlPuschAlgoSwitch	UL_INTRF_RANDOMIZED_SCH_ENH_SW	You are advised to select this option.

6.10.4.1.2 Using MML Commands

Before using MML commands, refer to [6.10.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink scheduling enhancement
MOD NRDUCeLLPUSCH: NrDuCellId=0, UlPuschAlgoSwitch=UL_SCHEDULE_ENH_SW-1;
//Enabling uplink interference randomization-based scheduling enhancement
MOD NRDUCeLLPUSCH: NrDuCellId=0, UlPuschAlgoSwitch=UL_INTRF_RANDOMIZED_SCH_ENH_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink scheduling enhancement
MOD NRDUCeLLPUSCH: NrDuCellId=0, UlPuschAlgoSwitch=UL_SCHEDULE_ENH_SW-0;
//Disabling uplink interference randomization-based scheduling enhancement
MOD NRDUCeLLPUSCH: NrDuCellId=0, UlPuschAlgoSwitch=UL_INTRF_RANDOMIZED_SCH_ENH_SW-0;
```

6.10.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.10.4.2 Activation Verification

If the uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**) increases after this function is enabled, this function has taken effect.

6.10.4.3 Network Monitoring

The indicators listed in [6.10.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

6.11 PUCCH Symbol Sharing with the PUSCH

6.11.1 Principles

In high-frequency scenarios, PUCCH symbol sharing with the PUSCH is introduced to increase available PUSCH time-domain resources. This function is controlled by the **PUSCH_SHR_PUCCH_SYMBOL_RES_SW** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter and is supported only in high frequency bands. When PUCCH symbol sharing with the PUSCH is enabled, the PUSCH can share PUCCH symbol resources.

6.11.2 Network Analysis

6.11.2.1 Benefits

After this function is enabled in high-frequency NR TDD scenarios, the average uplink cell throughput increases significantly if the number of UEs is small and there are UEs performing uplink large-packet services in the cell. For example, the uplink cell throughput can increase by about 33% at most in high-frequency NR TDD single-UE peak rate scenarios.

6.11.2.2 Impacts

The impacts between this function and other functions are described in [6.11.3.2.3 Function Impacts](#).

6.11.3 Requirements

6.11.3.1 Licenses

There are no license requirements for basic functions.

6.11.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.11.3.2.1 Prerequisite Functions

None

6.11.3.2.2 Mutually Exclusive Functions

None

6.11.3.2.3 Function Impacts

None

6.11.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

6.11.3.4 Others

None

6.11.4 Operation and Maintenance

6.11.4.1 Data Configuration

6.11.4.1.1 Data Preparation

Table 6-16 describes the parameters used for function activation. No parameters are involved in function optimization.

Table 6-16 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink PUSCH Algorithm Switch	NRDUCellPusch. UlPuschAlgoSwitch	PUSCH_SHR_PUC CH_SYMBOL_RES_ SW	You are advised to select this option in high-frequency scenarios.

6.11.4.1.2 Using MML Commands

Before using MML commands, refer to [6.11.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for PUCCH symbol sharing with the PUSCH  
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPuschAlgoSwitch=PUSCH_SHR_PUCCH_SYMBOL_RES_SW-1;
```

Optimization Command Examples

Optimization command examples are not involved.

Deactivation Command Examples

```
//Turning off the switch for PUCCH symbol sharing with the PUSCH  
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPuschAlgoSwitch=PUSCH_SHR_PUCCH_SYMBOL_RES_SW-0;
```

6.11.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.11.4.2 Activation Verification

When the number of UEs is small and there are UEs performing uplink large-packet services in a cell, if the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

6.11.4.3 Network Monitoring

The indicators listed in [6.11.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

6.12 PUSCH and PUCCH RB Adaptation

6.12.1 Principles

In low frequency bands, PUSCH and PUCCH RB adaptation is introduced to increase the number of available PUSCH RBs. This function dynamically checks

whether the PUCCH RBs are used in each TTI. If not, the RBs that are allocated to the PUCCH but are not used can be dynamically allocated to the PUSCH for UEs to send large-packet service data, thereby improving resource utilization. This function is controlled by the **PUSCH_AND_PUCCH_RB_ADAPT_SW** option of the **NRDUCellPusch.PuschAlgoExtSwitch** parameter. When this option is selected, RBs are adaptively allocated to the PUSCH and PUCCH. When this option is deselected, RBs are fixedly allocated to the PUSCH and PUCCH.

6.12.2 Network Analysis

6.12.2.1 Benefits

In low frequency bands, this function increases the number of available PUSCH RBs, thereby increasing the uplink user-perceived rate of UEs running uplink large-packet services and the average uplink cell throughput.

6.12.2.2 Impacts

The impacts between this function and other functions are described in [6.12.3.2.3 Function Impacts](#).

This function may have the following network impacts:

- The uplink user-perceived rate increases.
- The IBLER for uplink scheduling fluctuates.
- The average uplink MCS index fluctuates.
- The downlink user-perceived rate fluctuates.
- The **Uplink Resource Block Utilizing Rate (DU)** fluctuates.
- The **N.PRB.UL.PUSCH.Avail.Avg** increases.
- The uplink cell throughput increases.
- The **N.ChMeas.MIMO.UL.Pair.PR**B fluctuates.
- The uplink RBLER fluctuates.

6.12.3 Requirements

6.12.3.1 Licenses

There are no license requirements for basic functions.

6.12.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.12.3.2.1 Prerequisite Functions

None

6.12.3.2.2 Mutually Exclusive Functions

None

6.12.3.2.3 Function Impacts

None

6.12.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

6.12.3.4 Others

None

6.12.4 Operation and Maintenance

6.12.4.1 Data Configuration

6.12.4.1.1 Data Preparation

[Table 6-17](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-17 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
PUSCH Algorithm Extension Switch	NRDUCellPusch.PuschAlgoExtSwitch	PUSCH_AND_PUCCH_RB_ADAPT_SW	You are advised to select this option in low-frequency scenarios.

6.12.4.1.2 Using MML Commands

Before using MML commands, refer to [6.12.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for PUSCH and PUCCH RB adaptation  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschAlgoExtSwitch=PUSCH_AND_PUCCH_RB_ADAPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch for PUSCH and PUCCH RB adaptation  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschAlgoExtSwitch=PUSCH_AND_PUCCH_RB_ADAPT_SW-0;
```

6.12.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.12.4.2 Activation Verification

In a cell with a small number of UEs, some of which are running uplink large-packet services, if the **N.PR.B.UL.PUSCH.Avail.Avg** increases after this function is enabled, this function has taken effect.

6.12.4.3 Network Monitoring

The indicators listed in [6.12.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

6.13 Uplink Scheduling of Multi-SLIV UEs in the Same Slot

6.13.1 Principles

The `NRDUCellPusch.SchMultiSlivTypeUeSwitch` parameter specifies whether uplink scheduling of multi-SLIV UEs is supported in the same slot. This parameter is set to **ON** by default, indicating that scheduling of multi-SLIV UEs is supported. If this parameter is set to **OFF**, scheduling of only single-SLIV UEs is supported. There are multi-SLIV UEs scheduled in the same slot when SRS carrier switching, super uplink, or UL and DL Decoupling is enabled.

6.13.2 Network Analysis

6.13.2.1 Benefits

Scheduling of multi-SLIV UEs in the same slot is supported to increase the user-perceived rate.

6.13.2.2 Impacts

The impacts between this function and other functions are described in [6.13.3.2.3 Function Impacts](#).

6.13.3 Requirements

6.13.3.1 Licenses

There are no license requirements for basic functions.

6.13.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.13.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS carrier switching	SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoSwitch parameter	<i>Carrier Aggregation</i>	When SRS carrier switching is enabled, uplink scheduling of multi-SLIV UEs in the same slot is supported.
Low-frequency TDD	Super uplink	SUPER_UL_TDM_SCH_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	When super uplink is enabled, uplink scheduling of multi-SLIV UEs in the same slot is supported.
Low-frequency TDD	UL and DL Decoupling	NRDUCellAlgoSwitch.ULDLDecouplingSwitch	<i>UL and DL Decoupling</i>	When UL and DL Decoupling is enabled, uplink scheduling of multi-SLIV UEs in the same slot is supported.

6.13.3.2.2 Mutually Exclusive Functions

None

6.13.3.2.3 Function Impacts

None

6.13.3.3 Hardware

For details, see [4.3.3 Hardware](#).

6.13.3.4 Others

None

6.13.4 Operation and Maintenance

6.13.4.1 Data Configuration

6.13.4.1.1 Data Preparation

Table 6-18 describes the parameters used for function activation. No parameters are involved in function optimization.

Table 6-18 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Schedule Multiple SLIV Type UE Switch	NRDUCellPusch.SchMultiSlivTypeUeSwitch	None	Use the default value.

6.13.4.1.2 Using MML Commands

Before using MML commands, refer to **6.13.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the multi-SLIV UE scheduling switch  
MOD NRDUCELLPUSCH: NRDuCellId=0, SchMultiSlivTypeUeSwitch=ON;
```

Optimization Command Examples

Optimization command examples are not involved.

Deactivation Command Examples

```
//Turning off the multi-SLIV UE scheduling switch  
MOD NRDUCELLPUSCH: NRDuCellId=0, SchMultiSlivTypeUeSwitch=OFF;
```

6.13.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.13.4.2 Activation Verification

In scenarios where there are multi-SLIV UEs in the same slot, if the average uplink UE throughput (**User Uplink Average Throughput (DU)**) increases after

scheduling of multi-SLIV UEs in the same slot is enabled, this function has taken effect.

6.13.4.3 Network Monitoring

The indicators listed in [6.13.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

6.14 Uplink DCI Extension-based Scheduling (Low-Frequency TDD)

6.14.1 Principles

As more UEs are scheduled in a heavily loaded cell, not all RBs in the cell can be allocated due to limited uplink control channel element (CCE) resources. In this case, the downlink data volume decreases due to delayed uplink TCP ACK feedback. The uplink DCI extension-based scheduling function is introduced to increase the uplink RB usage in a cell. This function uses additional downlink slots to transmit DCI, improving the utilization of uplink RB resources and average uplink cell throughput. This function also prevents the decrease in the downlink data volume caused by delayed uplink TCP ACK feedback, increasing the average downlink cell throughput. The uplink DCI extension-based scheduling function is controlled by the **DCI_EXTEND_UL_CONGEST_SW** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter.

The enabled uplink DCI extension-based scheduling function can only take effect when the uplink and downlink RBs are sufficient while the uplink CCEs are insufficient in the cell. Uplink RBs are sufficient when the uplink RB usage is less than or equal to the value of the **NRDUCellPusch.UlDciExtRbUsageThld** parameter. After this function takes effect, the **NRDUCellPusch.ExtSlotDciAllocUePct** parameter specifies the percentage of UEs for which DCI for uplink scheduling can be transmitted in more slots. In heavy-load scenarios where more traffic volume occurs in the uplink than in the downlink, the **NRDUCellPdcch.HeavyLoadUlCceAdjPolicy** parameter can be configured to increase the percentage of available uplink CCEs in slots where CCEs can be allocated for both the uplink and downlink.

After uplink DCI extension-based scheduling is enabled, if uplink DCI extension-based scheduling enhancement (controlled by the **UL_SCH_DCI_EXT_SW** option of the **NRDUCellUlSch.UlSchAlgoSwitch** parameter) is enabled, uplink DCI extension-based scheduling can take effect only when all of the following conditions are met.

- The number of UEs in the local cell is greater than the UE number threshold for uplink DCI extension-based scheduling (specified by the **NRDUCellUlSch.UlSchDciExtUsrNumThld** parameter).
- In the evaluation of whether uplink DCI extension-based scheduling takes effect, if the gNodeB has been transmitting DCI in more downlink slots, the ratio of the number of UEs to the UE number threshold (specified by **NRDUCellUlSch.UlSchDciExtUsrNumThld**) must be greater than the **NRDUCellUlSch.UlSchDciExtExitThld** parameter value.

Uplink DCI extension-based scheduling can be applied to low- and high-speed cells. This document describes only uplink DCI extension-based scheduling in low-speed cells (with **NRDUCell.HighSpeedFlag** set to **LOW_SPEED**). For details about uplink DCI extension-based scheduling in high-speed cells (with **NRDUCell.HighSpeedFlag** set to **HIGH_SPEED**), see *High Speed Mobility*.

6.14.2 Network Analysis

6.14.2.1 Benefits

With uplink DCI extension-based scheduling enabled in heavy-load scenarios, the number of scheduled UEs in the uplink, uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**), and average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) increase. In addition, for MU-MIMO scenarios, the proportion of UEs paired in uplink MU-MIMO increases; for TCP services, the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases because uplink response packets are sent more promptly.

In heavy-load scenarios, enabling uplink DCI extension-based scheduling enhancement ensures the proportion of uplink DCI extension-based scheduling taking effect in heavy load scenarios and effectively reduces the ping-pong effect of this function.

6.14.2.2 Impacts

The impacts between this function and other functions are described in [6.14.3.2.3 Function Impacts](#).

The impacts on the network are as follows:

- According to section "UE procedure for reporting control information" in 3GPP TS 38.213, K1 must be greater than K2. (K1 is used to determine the HARQ time sequence for downlink data transmission, and K2 is used to determine the uplink scheduling time sequence). After the uplink DCI extension-based scheduling function is enabled, the DCI is sent in multiple downlink slots. As a result, the values of K1 and K2 increase, while the downlink retransmission delay increases as the value of K1 increases.
- After uplink DCI extension-based scheduling takes effect, the proportion of UEs paired in uplink MU-MIMO increases. For details, see the impact descriptions in *MIMO (TDD)*.
- After uplink DCI extension-based scheduling takes effect, the following impacts arise: (1) For TCP services, uplink response packets are sent more promptly, and more PDCP packets are transmitted. When PDSCH PRB resources or PDCCH resources are insufficient, the waiting delay increases, which may cause the downlink user-plane delay to increase and the **User Downlink Average Throughput (DU)** to decrease. (2) For UDP services, the number of available downlink CCEs decreases. When PDCCH resources are insufficient, the **Cell Downlink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** decrease.
- After uplink DCI extension-based scheduling takes effect, the number of UEs that are successfully scheduled in the uplink increases, and uplink scheduling is accelerated. As a result, UEs may enter idle mode earlier, and the average number of UEs in RRC_CONNECTED mode may decrease.

- If the percentage of required uplink CCEs is greater than that specified by the **NRDUCellPdcch.UlMaxCcePct** parameter and the downlink data volume is not limited by uplink TCP ACKs before uplink DCI extension-based scheduling is enabled, the **Cell Downlink Average Throughput (DU)** may decrease after uplink DCI extension-based scheduling takes effect.
- After the uplink DCI extension-based scheduling function is enabled, the DCI is sent in multiple downlink slots, which increases the CPU usage of the baseband processing unit.

6.14.3 Requirements

6.14.3.1 Licenses

There are no license requirements for basic functions.

6.14.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.14.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH uplink-to-downlink CCE ratio adaptation	UL_DL_CCE_RATIO_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Channel Management</i>	Uplink DCI extension-based scheduling requires that PDCCH uplink-to-downlink CCE ratio adaptation be enabled.

6.14.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Uplink DCI extension-based scheduling cannot be enabled for combined cells.
Low-frequency TDD	Pattern-free PDCCH rate matching activation rate increase	PDCCH_NO_PAT_RM_RATIO_INC_SW option of the NRDUCellservExp.SpdTurboAlgoSwitch parameter	<i>Downlink Scheduling</i>	"Pattern-free PDCCH rate matching activation rate increase" and "uplink DCI extension-based scheduling" are mutually exclusive, indicating that they cannot be both enabled.
Low-frequency TDD	Time-domain power saving BWP	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	If the TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter is selected, the DCI_EXTEND_UL_CONGEST_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter must be deselected.
Low-frequency TDD	Combining noise suppression	RF_COMBINING_NOISE_SUPPR_SW option of the NRDUCellULSch.RfCombSwitch parameter	None	Uplink DCI extension-based scheduling and combining noise suppression cannot be both enabled.

6.14.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dynamic uplink slot aggregation	REDCAP_UL_COVERAGE_ENH_SW option of the NRDUCellFeatureSw.UeCapBasebandFunSw parameter	<i>RedCap</i>	For an NR TDD cell, uplink DCI extension-based scheduling does not take effect on RedCap UEs with dynamic uplink slot aggregation in effect.

6.14.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.14.3.4 Others

- **NRDUCell.DlBandwidth** is set to **CELL_BW_40M**, **CELL_BW_50M**, **CELL_BW_60M**, **CELL_BW_70M**, **CELL_BW_80M**, **CELL_BW_90M**, or **CELL_BW_100M**.
- **NRCell.FrequencyBand** and **NRDUCell.FrequencyBand** are set to **N38**, **N40**, **N41**, **N77**, **N78**, or **N79**. **NRDUCell.SlotAssignment** is set to **4_1_DDDSU**, **8_2_DDDDDDDSUU**, or **8_2_DDDSUUDDDD**.

- When **NRDUCell.HighSpeedFlag** is set to **LOW_SPEED**, **NRDUCell.NrDuCellNetworkingMode** must be set to **NORMAL_CELL** or **FUSION_CELL**.
- When **NRDUCell.HighSpeedFlag** is set to **HIGH_SPEED**, **NRDUCell.NrDuCellNetworkingMode** must be set to **NORMAL_CELL** or **HYPER_CELL**.

6.14.4 Operation and Maintenance

6.14.4.1 Data Configuration

6.14.4.1.1 Data Preparation

[Table 6-19](#) and [Table 6-20](#) describe the parameters used for function activation and optimization, respectively.

Table 6-19 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink PUSCH Algorithm Switch	NRDUCellPusch.UlPuschAlgoSwitch	DCI_EXTEND_UL_CONGEST_SW	Select this option in uplink CCE overuse scenarios.
UL Scheduling Algorithm Switch	NRDUCellUISch.UlSchAlgoSwitch	UL_SCH_DCI_EXT_SW	Select this option in uplink CCE overuse scenarios.
UL Schedule DCI Ext User Number Threshold	NRDUCellUISch.UlSchDCIExtUsrNumThld	None	Use the recommended value.
UL Schedule DCI Extension Exit Threshold	NRDUCellUISch.UlSchDCIExtExitThld	None	Use the recommended value.

Table 6-20 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
UL DCI Extension RB Usage Threshold	NRDUCellPusch.UlDciExtRbUsageThld	Use the recommended value.
Extended Slot DCI Alloc UE Percentage	NRDUCellPusch.ExtSlotDciAllocUePct	Use the recommended value.
Heavy Load Uplink CCE Adjustment Policy	NRDUCellPdcch.HeavyLoadUICceAdjPolicy	It is recommended that this parameter be set to RATIO_OPT when both uplink load and downlink load are heavy. In other scenarios, the value RATIO_FIXED is recommended.

6.14.4.1.2 Using MML Commands

Before using MML commands, refer to [6.14.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink DCI extension-based scheduling
MOD NRDUCCELLPUSCH: NrDuCellId=0, UlPuschAlgoSwitch=DCI_EXTEND_UL_CONGEST_SW-1;
//Enabling uplink DCI extension-based scheduling enhancement and setting the thresholds for uplink DCI
extension-based scheduling to take effect
MOD NRDUCCELLULSCH: NrDuCellId=0, UlSchAlgoSwitch=UL_SCH_DCI_EXT_SW-1;
MOD NRDUCCELLULSCH: NrDuCellId=0, UlSchDciExtUsrNumThld=200;
MOD NRDUCCELLULSCH: NrDuCellId=0, UlSchDciExtExitThld=80;
```

Optimization Command Examples

```
//Configuring the RB usage threshold for uplink DCI extension-based scheduling to take effect
MOD NRDUCCELLPUSCH: NrDuCellId=0, UlDciExtRbUsageThld=60;
//Configuring the percentage of UEs for which DCI for uplink scheduling can be transmitted in more slots
MOD NRDUCCELLPUSCH: NrDuCellId=0, ExtSlotDciAllocUePct=50;
//Configuring the percentage of available uplink CCEs in slots where CCEs can be allocated for both the
uplink and downlink
MOD NRDUCCELLPDCCH: NrDuCellId=0, HeavyLoadUICceAdjPolicy=RATIO_OPT;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink DCI extension-based scheduling  
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPuschAlgoSwitch=DCI_EXTEND_UL_CONGEST_SW-0;  
//Disabling uplink DCI extension-based scheduling enhancement  
MOD NRDUCELLULSCH: NrDuCellId=0, UISchAlgoSwitch=UL_SCH_DCI_EXT_SW-0;
```

6.14.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.14.4.2 Activation Verification

If the value of **N.User.Schedule.Ul.Sum** increases after uplink DCI extension-based scheduling is enabled, this function has taken effect.

6.14.4.3 Network Monitoring

Monitor the following to obtain the total number of scheduled UEs in the uplink, average uplink and downlink cell throughputs, average downlink UE throughput, and uplink PRB usage.

- Total number of scheduled UEs in the uplink in a cell:
N.User.Schedule.Ul.Sum
- Average downlink cell throughput: **Cell Downlink Average Throughput (DU)**
- Average uplink cell throughput: **Cell Uplink Average Throughput (DU)**
- Uplink PRB usage: **Uplink Resource Block Utilizing Rate (DU)**
- Average downlink UE throughput: **User Downlink Average Throughput (DU)**

7 MCS Selection

The signal to interference plus noise ratio (SINR) indicates the uplink channel quality of a UE. Due to the impact of channel variations on signals, the SINR measured by the gNodeB sometimes varies significantly from the SINR at the time of scheduling. If the measured SINR deviates from the actual channel quality, the IBLER cannot close in on its optimal value. Therefore, the gNodeB must adjust the SINR at the moment of measurement, and then determines the MCS for uplink scheduling based on the SINR adjustment result.

SINR Adjustment

The initial value of uplink SINR adjustment is specified by the **NRDUCellPusch.InitUISinrAdjust** parameter. During SINR adjustment, the scheduler calculates the SINR adjustment based on the uplink target IBLER.

- When the IBLER is higher than the uplink target IBLER, the SINR adjustment is adjusted downwards.
- When the IBLER is lower than the uplink target IBLER, the SINR adjustment is adjusted upwards.

The uplink scheduler adds up the SINR at the measurement point and the SINR adjustment to obtain the adjusted SINR. The adjusted SINR corresponds to the current channel quality of a UE.

Table 7-1 lists optimization functions for SINR adjustment.

Table 7-1 Optimization functions for SINR adjustment

Function Name	Low-Frequency NR TDD	High-Frequency NR TDD	Reference
Uplink target IBLER adaptation	√	√	7.1 Uplink Target IBLER Adaptation

Function Name	Low-Frequency NR TDD	High-Frequency NR TDD	Reference
Uplink outer-loop optimization for scheduling with lower MCS indexes	-	√	7.2 Uplink Outer-Loop Optimization for Scheduling with Lower MCS Indexes
Uplink power reduction optimization	√	√	7.3 Uplink Power Reduction Optimization
Average SINR optimization in uplink scheduling	√	√	7.4 Average SINR Optimization in Uplink Scheduling
Power-difference-based uplink SINR adjustment	-	√	7.7 Power-Difference-based Uplink SINR Adjustment

 NOTE

In the preceding table, "√" indicates that the function is supported in the specified RAT, while "-" indicates that the function is not supported.

MCS Finalization

The scheduler compares the adjusted SINR with the gNodeB demodulation performance and selects an appropriate MCS for uplink transmission.

During periodic SRS resource allocation in low-frequency NR TDD, if the configured usage value is only **antennaSwitching**, SRS antenna port switching is triggered when PUSCH scheduling is performed in the adjacent symbol that follows the SRS symbol. According to 3GPP TS 38.101 V15.9.0, a transition period of 15 μs is allowed when the antenna ports between two uplink symbols change. The specific transition period depends on the UE capability. During the transition period, signals may be lost or distorted. The uplink MCS index of a UE can be reduced to mitigate the impact during the transition period. The **NRDUCellUAmc.PuschMcsAdjVal** parameter specifies the maximum decrease in the MCS index, while the actual MCS index decrease is adjusted by the gNodeB based on the current MCS index of the UE.

 NOTE

For details about the **usage** field, see descriptions of **SRS-Config** in section 6.3.2 "Radio resource control information elements" of 3GPP TS 38.331 V15.6.0.

Table 7-2 lists optimization functions for MCS selection.

Table 7-2 Optimization functions for MCS selection

Function Name	Low-Frequency NR TDD	High-Frequency NR TDD	Reference
Uplink maximum modulation order optimization	√	-	7.5 Uplink Maximum Modulation Order Optimization
MCS selection for uplink scheduling on a per RB basis	√	-	7.6 MCS Selection for Uplink Scheduling on a per RB Basis
Uplink PAMC	√	√	7.8 Uplink PAMC
Uplink dual filtering	√	-	7.9 Uplink Dual Filtering
Configuration of upper limit to MCS indexes for uplink completion messages	√	√	7.10 Configuration of Upper Limit to MCS Indexes for Uplink Completion Messages

 NOTE

In the preceding table, "√" indicates that the function is supported in the specified RAT, while "-" indicates that the function is not supported.

7.1 Uplink Target IBLER Adaptation

7.1.1 Principles

On live networks, the requirements for IBLER vary with scenarios. As such, the uplink target IBLER must be adapted to the UE channel variations to increase

uplink throughput of UEs. Uplink target IBLER adaptation can take effect if the **UL_IBLER_ADAPT_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter is selected. (However, for services with high requirements on delay, such as VoNR services, uplink target IBLER adaptation does not take effect and the uplink target IBLER is fixed at 1%.)

- When this option is deselected, the uplink target IBLER is specified by the **NRDUCellPusch.UITargetIbler** parameter (default value: 10%). When the MCS index reaches the highest value, the uplink target IBLER is fixed at 1%.
- When this option is selected, the target value is determined based on how much the SINR is fluctuating. A larger SINR fluctuation indicates a larger target IBLER value.

7.1.2 Network Analysis

7.1.2.1 Benefits

After this function is enabled, the throughput of UEs at the cell edge may increase when the channel quality fluctuates significantly.

7.1.2.2 Impacts

The impacts between this function and other functions are described in [7.1.3.2.3 Function Impacts](#).

After this function is enabled, the network IBLER may increase. In NSA networking, when this function is in effect and PDCCH CCE resources are limited, the number of contention resolution failures (= **N.RA.Contention.Msg3 – N.RA.Contention.Resolution.Succ**) may increase. As a result, the contention-based random access success rate decreases.

7.1.3 Requirements

7.1.3.1 Licenses

There are no license requirements for basic functions.

7.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.1.3.2.1 Prerequisite Functions

None

7.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink Subframe MCS Adjustment Switch	UL_SF_MCS_ADJ_SW option of the NRDUCellUplinkMcUplinkMCSAlgoSw parameter	None	When uplink target IBLER adaptation is enabled, the switch for uplink subframe-level MCS adjustment must be turned off.
High-frequency TDD	None	None	None	None

7.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Probability-based uplink MCS selection	UL_PBMCSSEL_SW option of the NRDUCellUplinkMcUplinkPerformanceSw parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	When both probability-based uplink MCS selection and uplink target IBLER adaptation are enabled, probability-based uplink MCS selection preferentially takes effect in a cell.
Low-frequency TDD	Uplink precise MCS optimization	UL_PRECISE_MCS_OPT_SW option of the NRDUCellUplinkMcUplinkPerformanceSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	Uplink precise MCS optimization preferentially takes effect when it is enabled together with uplink target IBLER adaptation.
Low-frequency TDD	Probability-based uplink MU-MIMO MCS selection	UL_MU_PBMCSSEL_SW option of the NRDUCellUplinkMcUplinkPerformanceSw parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	Probability-based uplink MU-MIMO MCS selection preferentially takes effect when it is enabled together with uplink target IBLER adaptation.

7.1.3.3 Hardware

For details, see [4.3.3 Hardware](#).

7.1.3.4 Others

None

7.1.4 Operation and Maintenance

7.1.4.1 Data Configuration

7.1.4.1.1 Data Preparation

Table 7-3 describes the parameters used for function activation. No parameters are involved in function optimization.

Table 7-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Adaptive Edge Experience Enhance Switch	NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch	UL_IBLER_ADAPT_SW	Select the UL_IBLER_ADAPT_SW option of this parameter in weak coverage scenarios where channel quality fluctuates significantly.

7.1.4.1.2 Using MML Commands

Before using MML commands, refer to **7.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink IBLER adaptation  
MOD NRDUCCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=UL_IBLER_ADAPT_SW-1;
```

Optimization Command Examples

Optimization command examples are not involved.

Deactivation Command Examples

```
//Disabling uplink IBLER adaptation  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=UL_IBLER_ADAPT_SW-0;
```

7.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.1.4.2 Activation Verification

If the average uplink UE throughput (**User Uplink Average Throughput (DU)**) increases after uplink target IBLER adaptation is enabled, this function has taken effect.

7.1.4.3 Network Monitoring

The indicators listed in [7.1.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.2 Uplink Outer-Loop Optimization for Scheduling with Lower MCS Indexes

7.2.1 Principles

For services of some scheduling types (such as SR-triggered scheduling and preallocation), the gNodeB decreases the MCS index for scheduling. As a result, the BLER of these services is lower than that of other services. To balance the BLERs of different types of services and increase the uplink user-perceived rate, uplink outer-loop optimization for scheduling with lower MCS indexes is introduced. This function is controlled by the **DEC_MCS_SCHEDUL_OL_OPT_SW** option of the **NRDUCellULAmc.UlAmcPerformanceSw** parameter. If this option is selected, the gNodeB adjusts the SINR adjustment for such services to prevent a high SINR adjustment from affecting the BLERs of other scheduled services.

7.2.2 Network Analysis

7.2.2.1 Benefits

After this function takes effect, the SINR adjustment value is adjusted for services scheduled with lower MCS indexes to balance the BLERs of different types of services. As a result, the uplink initial BLER may decrease and the UE data transmission delay is shortened.

7.2.2.2 Impacts

The impacts between this function and other functions are described in [7.2.3.2.3 Function Impacts](#).

7.2.3 Requirements

7.2.3.1 Licenses

There are no license requirements for basic functions.

7.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.2.3.2.1 Prerequisite Functions

None

7.2.3.2.2 Mutually Exclusive Functions

None

7.2.3.2.3 Function Impacts

None

7.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

7.2.3.4 Others

None

7.2.4 Operation and Maintenance

7.2.4.1 Data Configuration

7.2.4.1.1 Data Preparation

[Table 7-4](#) describes the parameters used for function activation. No parameters are involved in function optimization.

Table 7-4 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink AMC Performance Switch	NRDUCeLLULAmc. ULAmcPerforman ceSw	DEC_MCS_SCHED UL_OL_OPT_SW	You are advised to select this option when the uplink PRB usage is greater than 30% in high-frequency NR TDD scenarios.

7.2.4.1.2 Using MML Commands

Before using MML commands, refer to [7.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink outer-loop optimization for scheduling with lower MCS indexes (for high-frequency TDD only)
MOD NRDUCeLLULAMC: NrDuCellId=0, ULAmcPerformanceSw=DEC_MCS_SCHEDUL_OL_OPT_SW-1;
```

Optimization Command Examples

Optimization command examples are not involved.

Deactivation Command Examples

```
//Disabling uplink outer-loop optimization for scheduling with lower MCS indexes (for high-frequency TDD only)
MOD NRDUCeLLULAMC: NrDuCellId=0, ULAmcPerformanceSw=DEC_MCS_SCHEDUL_OL_OPT_SW-0;
```

7.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.2.4.2 Activation Verification

If the average uplink UE throughput (**User Uplink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

7.2.4.3 Network Monitoring

The indicators listed in [7.2.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.3 Uplink Power Reduction Optimization

7.3.1 Principles

Power reduction methods for UEs are defined in section 6.2.2 "UE maximum output power reduction" in 3GPP TS 38.101-1 V16.2.0 and 3GPP TS 38.101-2 V16.2.0. The power reduction value varies with the modulation scheme, waveform, and RB allocation. When a UE switches from a low-order modulation scheme to a high-order one (from 64QAM to 256QAM in a low-frequency cell or from QPSK to 16QAM or from 16QAM to 64QAM in a high-frequency cell), the uplink BLER may be high for the high-order modulation scheme as a result of power reduction.

To resolve this issue, uplink power reduction optimization is introduced. This function is controlled by the **UL_256QAM_MPR_OPT_SW** option of the **NRDUCellUIMc.UIMcAlgoSw** parameter. After this function takes effect:

- In low-frequency cells, when UEs switch from 64QAM to 256QAM, this function evaluates channel quality based on the power reduction value (specified by the **NRDUCellUIMc.PowerReductionValue256Qam** parameter) and makes corresponding adaptations to scheduling. In addition, this function reduces the uplink target IBLER for the UEs during the switching, with this target value specified by the **NRDUCellUIMc.PowerReductionIblerTarget** parameter.
- In high-frequency cells, when UEs switch from QPSK to 16QAM or from 16QAM to 64QAM, this function evaluates channel quality based on the adaptively obtained power reduction value and makes corresponding adaptations to scheduling. In addition, this function reduces the uplink target IBLER for the UEs during the switching.

7.3.2 Network Analysis

7.3.2.1 Benefits

After the uplink power reduction optimization function is enabled, the uplink IBLER decreases, and the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) and average uplink UE throughput (**User Uplink Average Throughput (DU)**) increase.

7.3.2.2 Impacts

The impacts between this function and other functions are described in [7.3.3.2.3 Function Impacts](#).

Uplink power reduction optimization has the following impacts on low-frequency cells:

- The proportion of 256QAM for UL-SCH decreases.
- The proportion of 64QAM for UL-SCH increases.
- The average MCS index for uplink scheduling decreases.
- In SA networking, the proportion of uplink transmissions using rank 2 for UEs increases. This proportion is calculated as follows:
Proportion of uplink transmissions using rank 2 for UEs = $\frac{N.PUSCH.Tb_{UL.Rank2}}{N.PUSCH.Tb_{UL.Rank1} + N.PUSCH.Tb_{UL.Rank2}}$
- A smaller difference between the configured power reduction value for 256QAM (specified by **NRDUCellULAmc.PowerReductionValue256Qam**) and the actual power reduction value of UEs results in this function delivering higher benefits. A larger difference results in not only smaller benefits, but possibly negative results.

Uplink power reduction optimization has the following impacts on high-frequency cells:

- The proportion of 64QAM for UL-SCH decreases.
- The proportion of QPSK for UL-SCH increases.
- The proportion of 16QAM for UL-SCH fluctuates.
- The average MCS index for uplink scheduling decreases.

7.3.3 Requirements

7.3.3.1 Licenses

There are no license requirements for basic functions.

7.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.3.2.1 Prerequisite Functions

None

7.3.3.2.2 Mutually Exclusive Functions

None

7.3.3.2.3 Function Impacts

None

7.3.3.3 Hardware

For details, see [4.3.3 Hardware](#).

7.3.3.4 Others

None

7.3.4 Operation and Maintenance

7.3.4.1 Data Configuration

7.3.4.1.1 Data Preparation

[Table 7-5](#) and [Table 7-6](#) describe the parameters used for function activation and optimization, respectively.

Table 7-5 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink AMC Algorithm Switch	NRDUCellULAmc.UlAmcAlgoSw	UL_256QAM_MPR_OPT_SW	Use the recommended value.

Table 7-6 Parameters used for optimization (only in low-frequency scenarios)

Parameter Name	Parameter ID	Setting Notes
Power Reduction Value for 256QAM	NRDUCellUlamc.PowerReductionValue256Qam	Use the default value. A small difference between this parameter's value and the actual power reduction value of a UE results in significant positive impact of this function, while a large difference results in little positive impact or may even render negative results.
Power Reduction IBLER Target	NRDUCellUlamc.PowerReductionIblerTarget	Use the default value. A large parameter value results in a high uplink IBLER and a high average MCS index. A small value of this parameter results in a low uplink IBLER and low average MCS index.

7.3.4.1.2 Using MML Commands

Before using MML commands, refer to [7.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink power reduction optimization  
MOD NRDUCELLULAMC: NrDuCellId=0, UlaMcAlgoSw=UL_256QAM_MPR_OPT_SW-1;
```

Optimization Command Examples

```
//Configuring the power reduction value for 256QAM in low-frequency scenarios  
MOD NRDUCELLULAMC: NrDuCellId=0, PowerReductionValue256Qam=3;  
//Configuring the target IBLER for power reduction in low-frequency scenarios  
MOD NRDUCELLULAMC: NrDuCellId=0, PowerReductionIblerTarget=2;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink power reduction optimization
MOD NRDUCELLULAMC: NrDuCellId=0, ULAmcAlgoSw=UL_256QAM_MPR_OPT_SW-0;
```

7.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.3.4.2 Activation Verification

Uplink power reduction optimization has taken effect in low-frequency cells if the proportion of 256QAM for UL-SCH decreases and that of 64QAM for UL-SCH increases after this function is enabled. These proportions are calculated as follows:

- Proportion of 256QAM for UL-SCH = $\frac{N.UL.SCH.256QAM.TB}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)}$
- Proportion of 64QAM for UL-SCH = $\frac{N.UL.SCH.64QAM.TB}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)}$

Uplink power reduction optimization has taken effect in high-frequency cells if the proportion of 64QAM for UL-SCH decreases and that of QPSK for UL-SCH increases after this function is enabled. These proportions are calculated as follows:

- Proportion of 64QAM for UL-SCH = $\frac{N.UL.SCH.64QAM.TB}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)}$
- Proportion of QPSK for UL-SCH = $\frac{N.UL.SCH.QPSK.TB}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)}$

7.3.4.3 Network Monitoring

Monitor **Cell Uplink Average Throughput (DU)** to obtain the average uplink cell throughput.

Monitor **User Uplink Average Throughput (DU)** to obtain the average uplink UE throughput.

Monitor **N.ChMeas.PUSCH.MCS.0** to **N.ChMeas.PUSCH.MCS.31** to obtain the number of times an MCS index is selected for uplink scheduling.

7.4 Average SINR Optimization in Uplink Scheduling

7.4.1 Principles

When the gNodeB determines the RB resource size based on the average measured SINR but does not enable lower threshold protection for the minimum RB scheduling granularity during calculation of the average measured SINR, the SINR is relatively high and the allocated RBs are fewer than expected. As a result, UE data needs to be scheduled multiple times, increasing the UE data transmission delay. Average SINR optimization in uplink scheduling (controlled by the **UL_SCH_AVG_SINR_OPT_SW** option of the **NRDUCellUIMc.UlAmcAlgoSw** parameter) is introduced to enable lower threshold protection for the minimum RB scheduling granularity during calculation of the average measured SINR, ensuring more accurate RB resource allocation and shorter UE data transmission delay.

7.4.2 Network Analysis

7.4.2.1 Benefits

After average SINR optimization in uplink scheduling is enabled, lower threshold protection for the minimum RB scheduling granularity is performed during calculation of the average measured SINR. This ensures more accurate RB resource allocation and shorter UE data transmission delay for small-packet-service UEs with limited uplink power and improves the average uplink UE throughput (**User Uplink Average Throughput (DU)**).

7.4.2.2 Impacts

The impacts between this function and other functions are described in [7.4.3.2.3 Function Impacts](#).

After average SINR optimization in uplink scheduling is enabled, a lower threshold is applied to the protection of the minimum RB scheduling granularity during the calculation of the average measured SINR. This increases the number of RBs allocated to UEs that run uplink small-packet services with limited power. As a result, the average PSD for such UEs decreases, leading to a reduction in the average MCS index for uplink scheduling.

7.4.3 Requirements

7.4.3.1 Licenses

There are no license requirements for basic functions.

7.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.4.3.2.1 Prerequisite Functions

None

7.4.3.2.2 Mutually Exclusive Functions

None

7.4.3.2.3 Function Impacts

None

7.4.3.3 Hardware

For details, see [4.3.3 Hardware](#).

7.4.3.4 Others

None

7.4.4 Operation and Maintenance

7.4.4.1 Data Configuration

7.4.4.1.1 Data Preparation

[Table 7-7](#) describes the parameters used for function activation. No parameters are involved in function optimization.

Table 7-7 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink AMC Algorithm Switch	NRDUCellUlamc.UlAmcAlgoSw	UL_SCH_AVG_SINR_OPT_SW	This option is selected by default for newly deployed cells. It is good practice to select this option for existing cells.

7.4.4.1.2 Using MML Commands

Before using MML commands, refer to [7.4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for average SINR optimization in uplink scheduling  
MOD NRDUCELLULAMC: NrdCellId=0, ULAmcAlgoSw=UL_SCH_AVG_SINR_OPT_SW-1;
```

Optimization Command Examples

Optimization command examples are not involved.

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch for average SINR optimization in uplink scheduling  
MOD NRDUCELLULAMC: NrdCellId=0, ULAmcAlgoSw=UL_SCH_AVG_SINR_OPT_SW-0;
```

7.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.4.2 Activation Verification

If the average uplink UE throughput (**User Uplink Average Throughput (DU)**) increases after average SINR optimization in uplink scheduling is enabled, this function has taken effect.

7.4.4.3 Network Monitoring

The indicators listed in [7.4.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.5 Uplink Maximum Modulation Order Optimization

7.5.1 Principles

The uplink maximum modulation order optimization function is introduced in low frequency bands to increase uplink spectral efficiency of UEs. This function is controlled by the **UL_MAX_MODULATION_ORDER_OPT_SW** option of the **NRDUCellULAmc.UlAmcAlgoSw** parameter:

- If this option is selected,
 - When the modulation order supported by a UE is higher than that indicated by the **supportedModulationOrderUL** field, the uplink maximum modulation order selected for the UE can be higher than that indicated by this field, thereby improving uplink spectral efficiency for the UE.
 - When the modulation order supported by a UE is not higher than that indicated by the **supportedModulationOrderUL** field, the uplink maximum modulation order selected for the UE cannot be higher than that indicated by this field.
- If this option is deselected, the uplink maximum modulation order selected for the UE cannot be higher than that indicated by the **supportedModulationOrderUL** field.

NOTE

For details about the **supportedModulationOrderUL** field, see the descriptions of the *FeatureSetUplinkPerCC* IE in section 6.3.3 "UE capability information elements" of 3GPP TS 38.331 V15.4.0.

7.5.2 Network Analysis

7.5.2.1 Benefits

After this function is enabled in low frequency bands, uplink spectral efficiency increases when the modulation order supported by a UE is higher than that indicated by **supportedModulationOrderUL**.

7.5.2.2 Impacts

The impacts between this function and other functions are described in [7.5.3.2.3 Function Impacts](#).

7.5.3 Requirements

7.5.3.1 Licenses

There are no license requirements for basic functions.

7.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.5.3.2.1 Prerequisite Functions

None

7.5.3.2.2 Mutually Exclusive Functions

None

7.5.3.2.3 Function Impacts

None

7.5.3.3 Hardware

For details, see [4.3.3 Hardware](#).

7.5.3.4 Others

None

7.5.4 Operation and Maintenance

7.5.4.1 Data Configuration

7.5.4.1.1 Data Preparation

[Table 7-8](#) describes the parameters used for function activation. No parameters are involved in function optimization.

Table 7-8 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink AMC Algorithm Switch	NRDUCellUIAmc. <i>UIAmcAlgoSw</i>	UL_MAX_MODULATION_ORDER_OPT_SW	You are advised to select this option.

7.5.4.1.2 Using MML Commands

Before using MML commands, refer to [7.5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for uplink maximum modulation order optimization (recommended when the modulation order supported by a UE is higher than that indicated by the supportedModulationOrderUL field)
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcAlgoSw=UL_MAX_MODULATION_ORDER_OPT_SW-1;
```

Optimization Command Examples

Optimization command examples are not involved.

Deactivation Command Examples

```
//Turning off the switch for uplink maximum modulation order optimization
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcAlgoSw=UL_MAX_MODULATION_ORDER_OPT_SW-0;
```

7.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.5.4.2 Activation Verification

None

7.5.4.3 Network Monitoring

The indicators listed in [7.5.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.6 MCS Selection for Uplink Scheduling on a per RB Basis

7.6.1 Principles

MCS selection for uplink scheduling on a per RB basis is introduced in low frequency bands to improve the uplink PRB usage. This function is controlled by the **UL_1RB_MCS_SEL_SW** option of the **NRDUCellUIAmc.UIAmcPerformanceSw** parameter. When uplink scheduling on a per RB basis is enabled:

- If MCS selection for uplink scheduling on a per RB basis is enabled, the MCS for such scheduling is selected in line with the principle of reducing the BLER. This improves the uplink PRB usage and reduces the uplink BLER.
- If MCS selection for uplink scheduling on a per RB basis is disabled, the MCS for such scheduling is selected in line with the principle of reducing the resource usage. This leads to high uplink BLER.

7.6.2 Network Analysis

7.6.2.1 Benefits

If the function switch is turned on, the uplink BLER decreases and the uplink PRB usage increases.

7.6.2.2 Impacts

The impacts between this function and other functions are described in [7.6.3.2.3 Function Impacts](#).

7.6.3 Requirements

7.6.3.1 Licenses

There are no license requirements for basic functions.

7.6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.6.3.2.1 Prerequisite Functions

None

7.6.3.2.2 Mutually Exclusive Functions

None

7.6.3.2.3 Function Impacts

None

7.6.3.3 Hardware

For details, see [4.3.3 Hardware](#).

7.6.3.4 Others

None

7.6.4 Operation and Maintenance

7.6.4.1 Data Configuration

7.6.4.1.1 Data Preparation

Table 7-9 describes the parameters used for function activation. No parameters are involved in function optimization.

Table 7-9 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink AMC Performance Switch	NRDUCellUIMc.UlAmcPerformanceSw	UL_1RB_MCS_SEL_SW	You are advised to select this option only in special scenarios where fewer than four RBs are required in scheduling (for example, the network slice group with the TTI-based control policy is configured for a cell and the number of RBs corresponding to NRDUCellRes.UlRbMaxRatio is less than 4), and when uplink scheduling on a per RB basis is also enabled. You are advised to deselect this option in other scenarios.

7.6.4.1.2 Using MML Commands

Before using MML commands, refer to **7.6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Disabling MCS selection for uplink scheduling on a per RB basis (because it is good practice to enable this function only in special scenarios detailed in the data preparation section and disable this function in other scenarios)  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcPerformanceSw=UL_1RB_MCS_SEL_SW-0;
```

Optimization Command Examples

Optimization command examples are not involved.

Deactivation Command Examples

Deactivation command examples are not involved.

7.6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.6.4.2 Activation Verification

None

7.6.4.3 Network Monitoring

For details, see [7.6.2 Network Analysis](#).

7.7 Power-Difference-based Uplink SINR Adjustment

7.7.1 Principles

Power-difference-based uplink SINR adjustment is introduced to improve uplink spectral efficiency in high-frequency NR TDD scenarios. This function is controlled by the **PWR_DIFF_BASED_UL_SINR_ADJ_SW** option of the **NRDUCellUIAmc.UIAmcAlgoSw** parameter.

- When this function is enabled, the gNodeB keeps information about the difference between the SRS and PUSCH power reduction values based on the reported power headroom. During conversion of the SRS SINR into the PUSCH SINR, the gNodeB adjusts the PUSCH SINR based on the dynamically obtained difference between the SRS and PUSCH power reduction values. This improves the PUSCH AMC accuracy and uplink spectral efficiency.
- When this function is disabled, the gNodeB does not adjust the PUSCH SINR based on the difference between the SRS and PUSCH power reduction values.

7.7.2 Network Analysis

7.7.2.1 Benefits

After this function takes effect in high-frequency NR TDD scenarios (with the uplink PRB usage being greater than 30%) where the small-packet and large-packet hybrid burst traffic model is used, the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) and average uplink UE throughput (**User Uplink Average Throughput (DU)**) increase. In other scenarios, this function may not deliver any benefits.

7.7.2.2 Impacts

The impacts between this function and other functions are described in [7.7.3.2.3 Function Impacts](#).

7.7.3 Requirements

7.7.3.1 Licenses

There are no license requirements for basic functions.

7.7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.7.3.2.1 Prerequisite Functions

None

7.7.3.2.2 Mutually Exclusive Functions

None

7.7.3.2.3 Function Impacts

None

7.7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

7.7.3.4 Others

None

7.7.4 Operation and Maintenance

7.7.4.1 Data Configuration

7.7.4.1.1 Data Preparation

Table 7-10 describes the parameters used for function activation. No parameters are involved in function optimization.

Table 7-10 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink AMC Algorithm Switch	NRDUCellUAmc. UAmcAlgoSw	PWR_DIFF_BASED _UL_SINR_ADJ_S W	You are advised to select this option when the uplink PRB usage is greater than 30% in high-frequency NR TDD scenarios.

7.7.4.1.2 Using MML Commands

Before using MML commands, refer to **7.7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling power-difference-based uplink SINR adjustment (for high-frequency TDD only)  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcAlgoSw=PWR_DIFF_BASED_UL_SINR_ADJ_SW-1;
```

Optimization Command Examples

Optimization command examples are not involved.

Deactivation Command Examples

```
//Disabling power-difference-based uplink SINR adjustment (for high-frequency TDD only)  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcAlgoSw=PWR_DIFF_BASED_UL_SINR_ADJ_SW-0;
```

7.7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.7.4.2 Activation Verification

In scenarios where the uplink PRB usage is greater than 30% and the small-packet and large-packet hybrid burst traffic model is used, if the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

7.7.4.3 Network Monitoring

The indicators listed in [7.7.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.8 Uplink PAMC

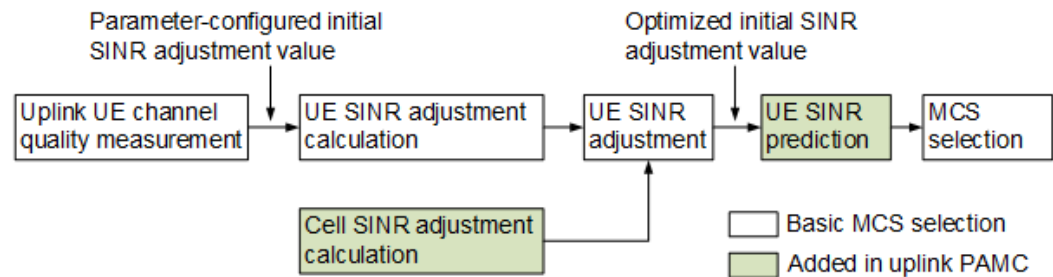
7.8.1 Principles

Basic MCS selection adjusts the SINR and determines the MCS based on how the IBLER converges for UEs. For UEs that have just accessed a network, a fixed initial SINR adjustment value (specified by **NRDUCellPusch.InitULSinrAdjust**) is used. Therefore, the SINR adjustment value does not immediately approach the expected value if the UEs are scheduled infrequently because of low traffic volume. This slow change in the SINR adjustment value results in a failure to accurately indicate uplink channel quality, affecting uplink throughput of UEs. In addition, the amount by which the UE channel fluctuates also affects the accuracy of SINR adjustment calculation in basic MCS selection. In this context, the uplink perceptively adaptive modulation and coding scheme (PAMC) function is introduced. This function is supported in both low and high frequency bands.

The uplink PAMC function is enabled using the **UL_PAMC_SWITCH** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter.

Based on basic MCS selection, the uplink PAMC function adds cell SINR adjustment calculation for newly admitted UEs to obtain a more accurate initial SINR adjustment value. In addition, this function adds the UE SINR prediction process. **Figure 7-1** shows how the uplink PAMC function works.

Figure 7-1 Uplink PAMC principles



Cell SINR adjustment calculation: Measures and integrates statistics on how IBLER converges for all UEs in a cell to obtain the initial SINR adjustment value of UEs, accelerating the convergence process for newly admitted UEs.

UE SINR prediction: Predicts the SINR adjustment amount at the next scheduling moment based on the adjusted SINR and how the UE channel fluctuates, ensuring more accurate MCS selection and higher throughput.

7.8.2 Network Analysis

7.8.2.1 Benefits

You are advised not to enable this function in the current version because this function needs to be optimized. Enabling this function in scenarios with significant uplink signal quality fluctuation decreases the average uplink cell throughput and average uplink user-perceived rate. Enabling this function in other scenarios does not bring negative results.

7.8.2.2 Impacts

The impacts between this function and other functions are described in [7.8.3.2.3 Function Impacts](#).

After uplink PAMC is enabled, the uplink target IBLER changes adaptively based on channel quality. A low target IBLER value is used for CCUs and a high target IBLER value is used for CEUs, improving cell throughput. A large number of CCUs lead to a decrease in the cell uplink IBLER. A large number of CEUs lead to an increase in the cell uplink IBLER.

7.8.3 Requirements

7.8.3.1 Licenses

There are no license requirements for basic functions.

7.8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.8.3.2.1 Prerequisite Functions

None

7.8.3.2.2 Mutually Exclusive Functions

None

7.8.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Spatial channel estimation under uplink weak coverage	UL_WEAK_COV_SPACE_CHN_ESTI_SW option of the NRDUCellChn CovAlgo.UlCoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	When both "uplink PAMC" and "spatial channel estimation under uplink weak coverage" are enabled, the MCS indexes selected for UEs are inaccurate. As a result, the average uplink throughput of CEUs may decrease.
Low-frequency TDD	Probability-based uplink MCS selection	UL_PBMCSSEL_SW option of the NRDUCellUplinkMc.UlMcPerformanceSw parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	When both uplink PAMC and probability-based uplink MCS selection are enabled, probability-based uplink MCS selection preferentially takes effect in a cell.
Low-frequency TDD	Dynamic uplink slot aggregation	REDCAP_UL_COVERAGE_ENH_SW option of the NRDUCellFeatureSw.UeCapBasedFunSw parameter	<i>RedCap</i>	Uplink PAMC does not take effect on RedCap UEs with dynamic uplink slot aggregation in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink precise MCS optimization	UL_PRECISE_MCS_OPT_SW option of the NRDUCellAiAlgo.AiAmcAlgoSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	Uplink precise MCS optimization preferentially takes effect when it is enabled together with uplink PAMC.
Low-frequency TDD	Probability-based uplink MU-MIMO MCS selection	UL_MU_PBMCSEL_SW option of the NRDUCellUIMc.UlAmcPerformanceSw parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	Probability-based uplink MU-MIMO MCS selection preferentially takes effect when it is enabled together with uplink PAMC.
Low-frequency TDD	Uplink 4-layer transmission for a single FWA user	UL_SU_FOUR_LAYER_SW option of the NRDUCellUIRank.UlRankAlgoSw parameter	<i>FWA</i>	If uplink 4-layer transmission for a single FWA user has taken effect, uplink PAMC does not take effect for the user.
High-frequency TDD	None	None	None	None

7.8.3.3 Hardware

For details about the hardware requirements of other functions, see [4.3.3 Hardware](#).

7.8.3.4 Others

None

7.8.4 Operation and Maintenance

7.8.4.1 Data Configuration

7.8.4.1.1 Data Preparation

Table 7-11 describes the parameters used for function activation. No parameters are involved in function optimization.

Table 7-11 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink PUSCH Algorithm Switch	NRDUCellPusch.UlPuschAlgoSwitch	UL_PAMC_SWITCH H	You are advised to deselect this option.

7.8.4.1.2 Using MML Commands

Before using MML commands, refer to [7.8.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink PAMC  
MOD NRDUCELLPUSCH: NrDuCellId=0, UlPuschAlgoSwitch=UL_PAMC_SWITCH-1;
```

Optimization Command Examples

Optimization command examples are not involved.

Deactivation Command Examples

```
//Disabling uplink PAMC  
MOD NRDUCELLPUSCH: NrDuCellId=0, UlPuschAlgoSwitch=UL_PAMC_SWITCH-0;
```

7.8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.8.4.2 Activation Verification

If the value of **N.ThpVol.UL.Cell** (uplink traffic volume received at the MAC layer in a cell) increases after the uplink PAMC function is enabled, this function has taken effect.

7.8.4.3 Network Monitoring

Monitor the **N.ThpVol.UL.Cell** counter to obtain the uplink traffic volume received at the MAC layer in a cell.

7.9 Uplink Dual Filtering

7.9.1 Principles

Sounding reference signal (SRS) is the reference signal used in the uplink. The time required by the gNodeB to measure the SRS SINR on the full cell bandwidth is positively correlated with the SRS period: $\text{Time} = \text{Number of RBs corresponding to the configured SRS bandwidth} / \text{Number of RBs corresponding to the SRS frequency hopping subband bandwidth} \times \text{SRS period}$. When the SRS period is long and the SRS frequency hopping subband bandwidth is narrow, more than one second is required to measure the SRS SINR on the full cell bandwidth. For example, assume that for an NR TDD cell, the configured SRS bandwidth corresponds to 272 RBs, the SRS period is 80 ms, and the SRS frequency hopping subband bandwidth corresponds to 16 RBs. The time required for measuring the SRS SINR on the full cell bandwidth is 1360 ms ($272/16 \times 80 = 1360$).

NOTE

- For details about the number of RBs corresponding to the SRS bandwidth, see C_{SRS} in Table 6.4.1.4.3-1: SRS bandwidth configuration in 3GPP TS 38.211 V15.5.0.
- For details about the number of RBs corresponding to the SRS frequency hopping subband bandwidth, see m_{SRS} corresponding to B_{SRS} in Table 6.4.1.4.3-1: SRS bandwidth configuration in 3GPP TS 38.211 V15.5.0.
- The SRS period is specified by the **NRDUCellSrs.SrsPeriod** parameter.
- For more information about SRS, see *Channel Management*.

For the gNodeB to accommodate more UEs in an NR TDD cell with a medium or heavy load, some UEs are allocated a long SRS period and a narrow SRS frequency hopping subband. In this case, the timeliness of the SRS SINR reported by the gNodeB cannot adapt to radio environment changes, affecting the MCS index selection and scheduling for uplink services and decreasing the uplink throughput.

As such, the uplink dual filtering function is introduced. This function is controlled by the **UL_SRS_DMRS_DUAL_FILTER_SW** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter. Before this function is enabled, the gNodeB performs filtering based only on the SRS SINR measurements. After this function is enabled, the gNodeB supports filtering based on both DMRS SINR and SRS SINR measurements (dual filtering for short) to obtain as many real-time SINR measurements as possible, thereby increasing average uplink cell throughput (**Cell Uplink Average Throughput (DU)**). This function is supported only in low frequency bands, not in high frequency bands.

7.9.2 Network Analysis

7.9.2.1 Benefits

You are advised not to enable this function in the current version because this function needs to be optimized.

In medium- and heavy-load eMBB scenarios in NR TDD: When the time required by the gNodeB to measure the SRS SINR on the full bandwidth is greater than or equal to a specified threshold (**NRDUCellUplinkDualFilterThld**) and there is interference, after uplink dual filtering is enabled and takes effect, the average uplink UE throughput (**User Uplink Average Throughput (DU)**) and average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) decrease; when the time required for measuring the SRS SINR on the full bandwidth is less than the specified threshold, enabling this function does not bring negative results.

7.9.2.2 Impacts

The impacts between this function and other functions are described in [7.9.3.2.3 Function Impacts](#).

After uplink dual filtering is enabled, the uplink MCS index may decrease but the CCE allocation failure rate may increase.

7.9.3 Requirements

7.9.3.1 Licenses

There are no license requirements for basic functions.

7.9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.9.3.2.1 Prerequisite Functions

None

7.9.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink MCS index selection optimization based on narrowband SRS	UL_SRS_FREQ_HOPPING_MCS_OPT_SW option of the NRDUCellHighSpeedMob.HighSpeedAlgoOptSw parameter	<i>High Speed Mobility</i>	"Uplink MCS index selection optimization based on narrowband SRS" and "uplink dual filtering" are mutually exclusive.
Low-frequency TDD	High-reliability uplink MCS selection	UL_HIGH_RELIAB_MCS_SEL_SW option of the NRDUCellUplinkMcPerformanceSw parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	High-reliability uplink MCS selection can be enabled only when uplink dual filtering is disabled.

7.9.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Spatial channel estimation under uplink weak coverage	UL_WEAK_COV_SPACE_CHN_ESTI_SW option of the NRDUCellChn CovAlgo.UlCoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	If both uplink dual filtering and spatial channel estimation under uplink weak coverage are enabled, the MCS indexes selected for UEs are inaccurate. As a result, the average uplink throughput of CEUs may decrease.
Low-frequency TDD	Dual-Filtering Threshold	NRDUCellUplinkMcDualFilterThld	<i>Uplink Scheduling</i>	After uplink dual filtering is enabled, it does not take effect unless the time required by the gNodeB to measure the SRS SINR on the full bandwidth is greater than or equal to the threshold specified by NRDUCellUplinkMcDualFilterThld .

7.9.3.3 Hardware

For details about the hardware requirements of other functions, see [4.3.3 Hardware](#).

7.9.3.4 Others

None

7.9.4 Operation and Maintenance

7.9.4.1 Data Configuration

7.9.4.1.1 Data Preparation

Table 7-12 describes the parameters used for function activation. No parameters are involved in function optimization.

Table 7-12 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink PUSCH Algorithm Switch	NRDUCellPusch.UlPuschAlgoSwitch	UL_SRS_DMRS_DUAL_FILTER_SW	You are advised to deselect this option.

7.9.4.1.2 Using MML Commands

Before using MML commands, refer to **7.9.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling the dual filtering function  
MOD NRDUCELLPUSCH: NrDuCellId=0, UlPuschAlgoSwitch=UL_SRS_DMRS_DUAL_FILTER_SW-1;
```

Optimization Command Examples

Optimization command examples are not involved.

Deactivation Command Examples

```
//Disabling the dual filtering function  
MOD NRDUCELLPUSCH: NrDuCellId=0, UlPuschAlgoSwitch=UL_SRS_DMRS_DUAL_FILTER_SW-0;
```

7.9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.9.4.2 Activation Verification

If the throughput of uplink data received at the MAC layer in a cell (**N.ThpVol.UL.Cell/N.ThpTime.UL.Cell**) increases after this function is enabled, this function has taken effect.

7.9.4.3 Network Monitoring

Monitor **N.ThpVol.UL.Cell** and **N.ThpTime.UL.Cell** to obtain the throughput of uplink data received at the MAC layer in a cell.

Monitor **Cell Uplink Average Throughput (DU)** to obtain the average uplink cell throughput.

Monitor **User Uplink Average Throughput (DU)** to obtain the average uplink UE throughput.

7.10 Configuration of Upper Limit to MCS Indexes for Uplink Completion Messages

7.10.1 Principles

Configuration of upper limit to MCS indexes for uplink completion messages is a function introduced to improve the reliability in the initial access phase of UEs. This function is controlled by the **NRDUCellUISch.CmpMsgMcsUpperLimit** parameter, which specifies the upper limit to the MCS indexes for Msg5 and RRC reconfiguration complete messages during UE access.

- When the Msg5 power control and MCS selection optimization function (controlled by the **MSG5_PWR_CTRL_MCS_SEL_OPT_SW** option of the **NRDUCellUISch.UISchAlgoSwitch** parameter) is enabled, the **NRDUCellUISch.CmpMsgMcsUpperLimit** setting determines the upper limit to the MCS indexes for Msg5 and RRC reconfiguration complete messages.
- When the Msg5 power control and MCS selection optimization function is disabled, the **NRDUCellUISch.CmpMsgMcsUpperLimit** setting determines the upper limit to the MCS indexes for RRC reconfiguration complete messages only. That is, configuration of upper limit to MCS indexes for uplink completion messages does not take effect on Msg5.

7.10.2 Network Analysis

7.10.2.1 Benefits

A larger upper limit to the MCS indexes for uplink completion messages may result in a higher MCS index for Msg5 and RRC reconfiguration completion messages, which decreases the UE access success rate. Conversely, a smaller upper

limit may result in a lower MCS index for Msg5 and RRC reconfiguration completion messages, which increases the UE access success rate.

7.10.2.2 Impacts

The impacts between this function and other functions are described in [7.10.3.2.3 Function Impacts](#).

If the upper limit to the MCS indexes for uplink completion messages is too low, the signaling may be segmented and require multiple transmission times, causing the UE access success rate to decrease.

7.10.3 Requirements

7.10.3.1 Licenses

There are no license requirements for basic functions.

7.10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.10.3.2.1 Prerequisite Functions

None

7.10.3.2.2 Mutually Exclusive Functions

None

7.10.3.2.3 Function Impacts

None

7.10.3.3 Hardware

For details, see [4.3.3 Hardware](#).

7.10.3.4 Cells

None

7.10.4 Operation and Maintenance

7.10.4.1 Data Configuration

7.10.4.1.1 Data Preparation

Table 7-13 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-13 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Complete Message MCS Upper Limit	NRDUCellULSch.CmpMsgMcsUpperLimit	None	The value 11 is recommended.

7.10.4.1.2 Using MML Commands

Before using MML commands, refer to **7.10.3 Requirements** and complete the parameter configurations for related functions based on the impact and dependency relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling configuration of upper limit to MCS indexes for uplink completion messages (using the setting of 11 as an example)  
MOD NRDUCELLULSCH: NrDuCellId=0, CmpMsgMcsUpperLimit=11;
```

Deactivation Command Examples

Deactivation command examples are not involved.

7.10.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.10.4.2 Activation Verification

If the UE access success rate changes after this function is enabled, this function has taken effect.

7.10.4.3 Network Monitoring

The indicators listed in [7.10.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

8 DMRS Resource Allocation

8.1 Principles

DMRS resource allocation is aimed at PUSCH demodulation. Section 6.4.1 "Reference signals" in 3GPP TS 38.211 V15.4.0 defines PUSCH DMRS, in terms of DMRS symbols in the time domain and DMRS configuration types in the frequency domain.

- DMRS symbol
 - Front-loaded DMRS
 - Number of front-loaded DMRS symbols: In accordance with specifications, front-loaded DMRS spans one or two symbols, which can be adaptively selected in the current version.
 - Number of front-loaded DMRS positions: One front-loaded DMRS position is supported.

NOTE

The number of DMRS positions is not equivalent to the number of DMRS symbols. The former indicates the number of available DMRS positions, whereas the latter indicates the number of symbols occupied by one DMRS position.

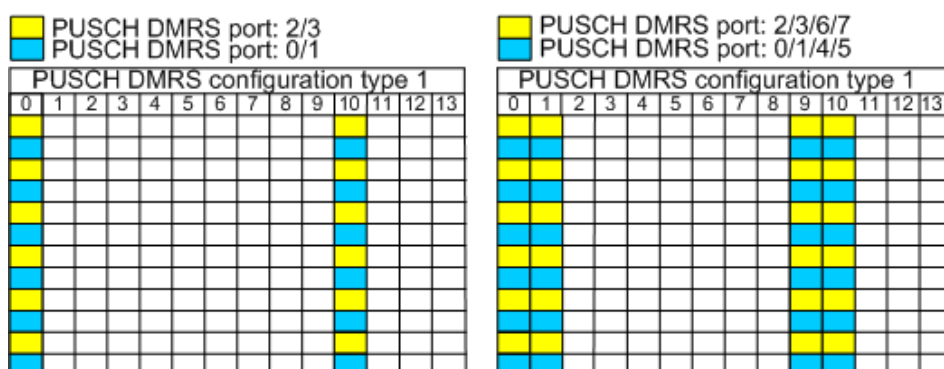
- In single-symbol DMRS scenarios, one DMRS position spans one symbol.
- In dual-symbol DMRS scenarios, one DMRS position spans two symbols.

- Additional DMRS
 - Number of additional DMRS symbols: Additional DMRS spans the same number of symbols as front-loaded DMRS.
 - When front-loaded DMRS spans one symbol, additional DMRS at each position spans one symbol.
 - When front-loaded DMRS spans two symbols, additional DMRS at each position spans two symbols.
 - Number of additional DMRS positions: **pos0**, **pos1**, **pos2**, and **pos3** are defined in specifications, but only **pos0**, **pos2**, and **pos1** are supported in the current version. The **NRDUCellPusch.UlAdditionalDmrsPos** parameter specifies the

additional DMRS symbol position. The time-domain positions of additional DMRS symbols in a slot are determined based on the PUSCH symbol configuration, and do not require special indication.

- When **NRDUCellPusch.UAdditionalDmrsPos** is set to **NOT_CONFIG**, no additional DMRS symbol is configured in the uplink.
 - When **NRDUCellPusch.UAdditionalDmrsPos** is set to **POS1**, a maximum of one additional DMRS position is allowed in the uplink.
 - When **NRDUCellPusch.UAdditionalDmrsPos** is set to **POS2**, a maximum of two additional DMRS positions are allowed in the uplink. (Setting the **NRDUCellPusch.UAdditionalDmrsPos** parameter to **POS2** is supported only in low frequency bands.)
 - DMRS configuration type
- The DMRS configuration type is specified by the **NRDUCellPusch.UIDmrsType** parameter, which can be set to **TYPE1** or **TYPE2**.
- **Figure 8-1** shows the DMRS when **NRDUCellPusch.UIDmrsType** is set to **TYPE1** (using DMRS of type B PUSCH as an example).

Figure 8-1 PUSCH DMRS configuration type 1

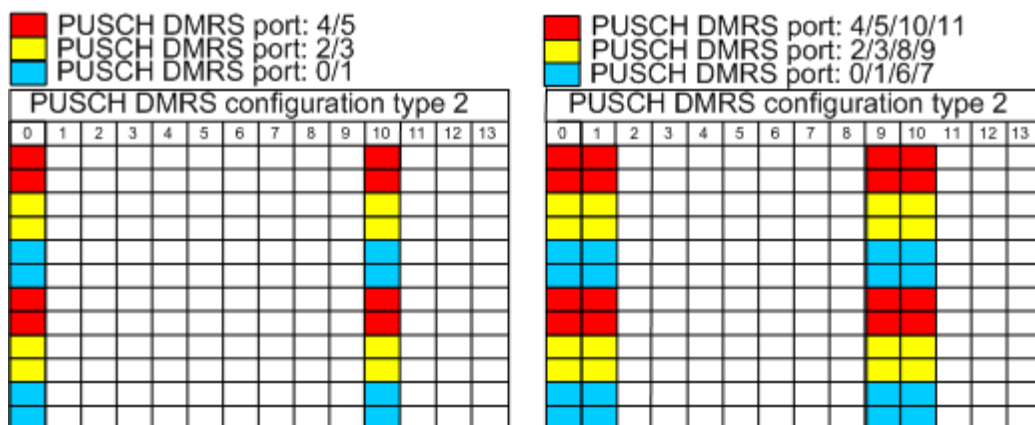


In DMRS configuration type 1, there are two CDM groups:

- The first group (CDM group 0) includes DMRS REs marked in blue in the preceding figure. The corresponding DMRS ports are as follows:
 - 0/1 (both front-loaded DMRS and additional DMRS span one symbol)
 - 0/1/4/5 (both front-loaded DMRS and additional DMRS span two symbols)
- The second group (CDM group 1) includes DMRS REs marked in yellow in the preceding figure. The corresponding DMRS ports are as follows:
 - 2/3 (both front-loaded DMRS and additional DMRS span one symbol)
 - 2/3/6/7 (both front-loaded DMRS and additional DMRS span two symbols)

- **Figure 8-2** shows the DMRS when **NRDUCellPusch.UIDmrsType** is set to **TYPE2** (using DMRS of type B PUSCH as an example).

Figure 8-2 PUSCH DMRS configuration type 2



In DMRS configuration type 2, there are three CDM groups:

- The first group (CDM group 0) includes DMRS REs marked in blue in the preceding figure. The corresponding DMRS ports are as follows:
 - 0/1 (both front-loaded DMRS and additional DMRS span one symbol)
 - 0/1/6/7 (both front-loaded DMRS and additional DMRS span two symbols)
- The second group (CDM group 1) includes DMRS REs marked in yellow in the preceding figure. The corresponding DMRS ports are as follows:
 - 2/3 (both front-loaded DMRS and additional DMRS span one symbol)
 - 2/3/8/9 (both front-loaded DMRS and additional DMRS span two symbols)
- The third group (CDM group 2) includes DMRS REs marked in red in the preceding figure. The corresponding DMRS ports are as follows:
 - 4/5 (both front-loaded DMRS and additional DMRS span one symbol)
 - 4/5/10/11 (both front-loaded DMRS and additional DMRS span two symbols)

NOTE

The effective values of the **NRDUCellPusch.UIAdditionalDmrsPos** and **NRDUCellPusch.UIDmrsType** parameters are determined by the UE capability. For example, if the **NRDUCellPusch.UIDmrsType** parameter is set to **TYPE2**, the effective value is **TYPE1** for UEs that support only type 1 or **TYPE2** for UEs that support type 2.

• FDM between PUSCH data and DMRS

After the DMRS port of a UE is determined, frequency-division multiplexing (FDM) between PUSCH data and DMRS can be performed in uplink scheduling. This function is controlled by the

NRDUCellPusch.PuschDmrsFdmSwitch parameter with the default value being **ON**. **NRDUCellPusch.PuschDmrsFdm256QamMcsThld** and **NRDUCellPusch.PuschDmrsFdm64QamMcsThld** are used to control the MCS index thresholds for FDM between PUSCH data and DMRS, for UEs using the 256QAM MCS index table and those using the 64QAM MCS index table, respectively. If the MCS index of a UE is greater than or equal to the configured MCS index threshold, FDM between PUSCH data and DMRS is allowed. If the MCS index of a UE is less than the configured MCS index threshold, FDM between PUSCH data and DMRS is disallowed.

8.2 Network Analysis

8.2.1 Benefits

DMRS resource allocation for UEs is aimed at PUSCH demodulation. DMRS resource overheads can be properly configured based on service scenarios to maximize system capacity.

Other sub-functions have the following impacts on the network:

- **FDM between PUSCH data and DMRS**

After this function is enabled, the cell supports FDM between PUSCH data and DMRS. If this function takes effect, the data rate of UEs near the cell center increases.

8.2.2 Impacts

The impacts between this function and other functions are described in [8.3.2.3 Function Impacts](#).

8.3 Requirements

8.3.1 Licenses

There are no license requirements for basic functions.

8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.2.1 Prerequisite Functions

None

8.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	DCI-based dynamic waveform switching	DYNAMIC_WAVEFORM_SWITCHING_SW option of the NRDUCellChnCfgAlgo.UICoverageAlgoSwitch parameter	<i>Ultra Uplink (High-Frequency TDD)</i>	When NRDUCellPusch.UIDmrsType is set to TYPE2 , DCI-based dynamic waveform switching cannot be enabled.
Low-frequency TDD	DCI-based dynamic waveform switching	DYNAMIC_WAVEFORM_SWITCHING_SW option of the NRDUCellChnCfgAlgo.UICoverageAlgoSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	When NRDUCellPusch.UIDmrsType is set to TYPE2 , DCI-based dynamic waveform switching cannot be enabled.
Low-frequency TDD	DCI-based dynamic waveform switching	DYNAMIC_WAVEFORM_SWITCHING_SW option of the NRDUCellChnCfgAlgo.UICoverageAlgoSwitch parameter	<i>Uplink Scheduling</i>	When NRDUCellPusch.UIDmrsType is set to TYPE2 , DCI-based dynamic waveform switching cannot be enabled.
High-frequency TDD	Window extension for DMRS-based channel estimation	DMRS_CHN_EST_WINDOW_EXT_SW option of the NRDUCellAlgoSwitch.FullChannelCovEnhSwitch parameter	None	If NRDUCellPusch.UIDmrsType is set to TYPE2 , window extension for DMRS-based channel estimation cannot be enabled.
Low-frequency TDD	PUSCH scheduling in self-contained slots	PUSCH_SCH_IN_SELF_CONTAINED_SLOT_SW option of the NRDUCellPusch.UIPuschAlgoSwitch parameter	None	If NRDUCellPusch.UIDmrsType is not set to TYPE1 , PUSCH scheduling in self-contained slots cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUSCH scheduling based on SRS interference avoidance	SRS_INTRF_AVOID_PUSCH_SCH_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	None	If the SRS_INTRF_AVOID_PUSCH_SCH_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter is selected, the NRDUCellPusch.UIDmrsType parameter cannot be set to TYPE2 .
Low-frequency TDD	PUSCH scheduling in self-contained slots	PUSCH_SCH_IN_S_SLOT_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	None	If NRDUCellPusch.UlAdditionalDmrsPos is not set to POS1 , PUSCH scheduling in self-contained slots cannot be enabled.

8.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	A high-speed cell supports up to two uplink additional DMRS positions. Therefore, NRDUCellPusch.UlAdditionalDmrsPos must be set to POS2. For NR TDD, if the maximum train speed is 300 km/h in the 2.6 GHz frequency band or 200 km/h in the 3.5 GHz frequency band, a high-speed cell can be configured with one uplink additional DMRS (by setting NRDUCellPusch.UlAdditionalDmrsPos to POS1).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DMRS interference reduction	DMRS_RESOURCE_ADJ_SW option of the NRDUCellChn CovAlgo.UICoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	- After frequency-division multiplexing between PUSCH data and DMRS is enabled, the uplink throughput of CCUs may decrease in cells with a high proportion of such UEs and high uplink PRB usage. Therefore, it is good practice to disable uplink DMRS interference reduction in such scenarios. - In other scenarios, DMRS interference reduction can be enabled after frequency-division multiplexing between PUSCH data and DMRS is enabled, but the padding proportion and accordingly the Cell Uplink Average Throughput (DU) will decrease.
Low-frequency TDD	Uplink DMRS adaptation	VOICE_DMRS_ADAPT_SW option of the NRDUCellDmrs.UIDmrsSwitch parameter	<i>VoNR</i>	Frequency-division multiplexing between PUSCH data and DMRS does not take effect for UEs with VoNR enabled and using an uplink slot with 2-symbol front-loaded DMRSs and 2-symbol additional DMRSs.

8.3.3 Hardware

For details, see [4.3.3 Hardware](#).

8.3.4 Others

None

8.4 Operation and Maintenance

8.4.1 Data Configuration

8.4.1.1 Data Preparation

Table 8-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink DMRS type	NRDUCellPusch. ULDmrsType	None	Use the default value.
Uplink Additional DMRS Position	NRDUCellPusch. ULAdditionalDmrsPos	None	In high-frequency scenarios: You are advised to set this parameter to POS1. In low-frequency scenarios: • Set this parameter to NOT_CONFIG in fixed-point scenarios (UE moving speed ≤ 10 km/h). • Set this parameter to POS1 in slow mobility scenarios ($10 \text{ km/h} < \text{UE moving speed} < 120 \text{ km/h}$). • Set this parameter to POS2 in high-speed mobility scenarios (UE moving speed $\geq 120 \text{ km/h}$).
PUSCH and DMRS FDM Switch	NRDUCellPusch.PuschDmrsFdmSwitch	None	Use the default value.
PUSCH and DMRS FDM 64QAM MCS Thld	NRDUCellPusch.PuschDmrsFdm64QamMcsThld	None	Use the default value.
PUSCH and DMRS FDM 256QAM MCS Thld	NRDUCellPusch.PuschDmrsFdm256QamMcsThld	None	Use the default value.

8.4.1.2 Using MML Commands

Before using MML commands, refer to [8.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Setting the uplink DMRS configuration type to type 1
MOD NRDUCELLPUSCH: NrDuCellId=0, UldMrsType=TYPE1;
//Configuring the uplink additional DMRS position in slow mobility scenarios
MOD NRDUCELLPUSCH: NrDuCellId=0, UIAdditionalDmrsPos=POS1;
//Turning on the switch for FDM between PUSCH data and DMRS
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschDmrsFdmSwitch=ON;
//Configuring the MCS threshold for frequency-division multiplexing between PUSCH data and DMRS for
UEs using the 64QAM MCS index table (using low-frequency TDD cells as an example)
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschDmrsFdm64QamMcsThld=18;
//Configuring the MCS threshold for frequency-division multiplexing between PUSCH data and DMRS for
UEs using the 256QAM MCS index table (using low-frequency TDD cells as an example)
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschDmrsFdm256QamMcsThld=12;
```

Deactivation Command Examples

```
//Turning off the switch for FDM between PUSCH data and DMRS
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschDmrsFdmSwitch=OFF;
```

8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.4.2 Activation Verification

This function is a basic function and takes effect by default. Activation verification is not involved.

8.4.3 Network Monitoring

The indicators listed in [8.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

9 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

10 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

11 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

12 Reference Documents

- 3GPP TS 38.214: "NR; Physical layer procedure for data"
- 3GPP TS 38.211: "NR; Physical channels and modulation"
- 3GPP TS 38.212: "NR; Multiplexing and channel coding"
- 3GPP TS 38.213: "NR; Physical layer procedures for control"
- 3GPP TS 38.306: "NR; User Equipment (UE) radio access capabilities"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.321: "NR; Medium Access Control (MAC) protocol specification"
- 3GPP TS 38.101-1: "NR; User Equipment (UE) radio transmission and reception"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- *QoS Management*
- *Channel Management*
- *FWA*
- *MIMO (TDD)*
- *High Speed Mobility*
- *UE Power Saving*
- *Carrier Aggregation*
- *VoNR*
- *Super Uplink*
- *Hyper Cell*
- *Cell Combination*
- *UL and DL Decoupling*
- *DRX*
- *CoMP*
- *URLLC*
- *Network Slicing*
- *Power Control*
- *LTE FDD and NR SUL Dynamic Spectrum Sharing*
- *EPC-based NSA Basics*

- *Turbo Uplink (Low-Frequency TDD)*
- *Massive MIMO Uplink Boosting (Low-Frequency TDD)*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*
- *Coordinated Interference Management (Low-Frequency TDD)*
- *mmWave Beam Management (High-Frequency TDD)*
- *RedCap*
- *Downlink Scheduling*
- *Service-differentiated Experience Guarantee*
- *Multi-Frequency Smart Aggregation*
- *Stable Video Surveillance Experience*
- *Ultra Uplink (High-Frequency TDD)*